



ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
(भारत सरकार का उद्यम)
GRID CONTROLLER OF INDIA LIMITED
(A Government of India Enterprise)



[formerly Power System Operation Corporation Limited (POSOCO)]
राष्ट्रीय भार प्रेषण केन्द्र / **National Load Despatch Centre**

कार्यालय : बी-9, प्रथम एवं द्वितीय तल, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराय, नई दिल्ली - 110016
Office : 1st and 2nd Floor, B-9, Qutab Institutional Area, Katwaria Sarai, New Delhi -110016
CIN : U40105DL2009GOI188682, Website : www.grid-india.in, E-mail : gridindiacc@grid-india.in, Tel.: 011- 42785855

संदर्भ संख्या : - GRID-INDIA/NLDC/MR/

दिनांक : 25.10.2023

सेवा में,

वितरण सूची के अनुसार

विषय:- निष्पादन रिपोर्ट – सितंबर 2023

महोदय,

आई०ई०जी०सी०-2010 के प्रावधान के अनुसार, सितंबर 2023 माह की अखिल भारतीय प्रणाली की निष्पादन रिपोर्ट संलग्न है।

धन्यवाद

भवदीय

(सुरजीत बनर्जी)

मुख्य महाप्रबन्धक (प्रणाली प्रचालन)

(रा०भा०प्रे०के०)

संलग्नक: मासिक प्रणाली रिपोर्ट

वितरण सूची Distribution List

- 1 सचिव, के.वि.नि.आ. तीसरा एवं चौथा तल, चंद्रलोक भवन, 36, जनपथ, नई दिल्ली-110001
Secretary, CERC, 3rd & 4th Floor, Chanderlok Building, 36, Janpath, New Delhi-110001
- 2 मुख्य अभियंता (जी एम), के.वि.प्रा., सेवा भवन, आर. के. पुरम, नई दिल्ली-110066
Chief Engineer (GM), CEA, Sewa Bhavan, R.K.Puram, New Delhi-110066
- 3 सदस्य सचिव, उ. क्षेत्र. वि. स., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Member Secretary, NRPC, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 4 सदस्य सचिव, प. क्षेत्र. वि. स., एफ-3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई - 400093
Member Secretary, WRPC, F-3, MIDC Area, Andheri (East), Mumbai-400093
- 5 सदस्य सचिव, द. क्षेत्र. वि. स., 29, रेस कोर्स क्रॉस रोड, बंगलूरु - 560009
Member Secretary, SRPC, 29, Race Course Cross Road, Bangalore-560009
- 6 सदस्य सचिव, पू. क्षेत्र. वि. स., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Member Secretary, ERPC, 14, Golf Club Road, Kolkata-700033
- 7 सदस्य सचिव, उ. पू. क्षेत्र. वि. स., मेघालय राज्य आवासीय वित्त सहकारी समिति लिमिटेड भवन, नॉग्रिम हिल्स, शिलोंग - 793003
Member Secretary, NERPC, Meghalaya State Housing Finance Cooperative Society Ltd. Building, Nongrim Hills, Shillong -793003
- 8 मुख्य अभियंता, राष्ट्रीय विद्युत समिति, एनआरपीसी भवन, तृतीय तल, 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Chief Engineer, National Power Committee, NRPC Building, 3rd floor, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 9 कार्यपालक निदेशक, पू. क्षेत्र. भा. प्रे. के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14, Golf Club Road, Kolkata-700033
- 10 कार्यपालक निदेशक, प. क्षेत्र. भा. प्रे. के., एफ-3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई - 400093
Executive Director, WRLDC, F-3, MIDC Area, Andheri (East), Mumbai-400093
- 11 कार्यपालक निदेशक, उ. क्षेत्र. भा. प्रे. के., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRPC, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 12 कार्यपालक निदेशक, द. क्षेत्र. भा. प्रे. के., 29, रेस कोर्स क्रॉस रोड, बंगलूरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009
- 13 मुख्य महाप्रबंधक, पू. क्षेत्र. भा. प्रे. के., डोंगतिह, लोअर नॉग्रह, लापलंग, शिलोंग- 793006
Chief General Manager, NERLDC, Dongtiah, Lower Nongrah, Laplang, Shillong-793006



ग्रिड-इंडिया
GRID-INDIA

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

NATIONAL LOAD DESPATCH CENTRE –राष्ट्रीय भार प्रेषण केन्द्र

मासिक ऑपरेशन रिपोर्ट

Monthly
Operation Report



सितंबर 2023

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
राष्ट्रीय भार प्रेषण केंद्र, नई दिल्ली

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

NATIONAL LOAD DESPATCH CENTRE, NEW DELHI



ग्रिड-इंडिया
GRID-INDIA

माह सितंबर 2023 के लिए प्रचालन निष्पादन रिपोर्ट
OPERATIONAL PERFORMANCE REPORT FOR THE
MONTH OF SEPTEMBER-2023

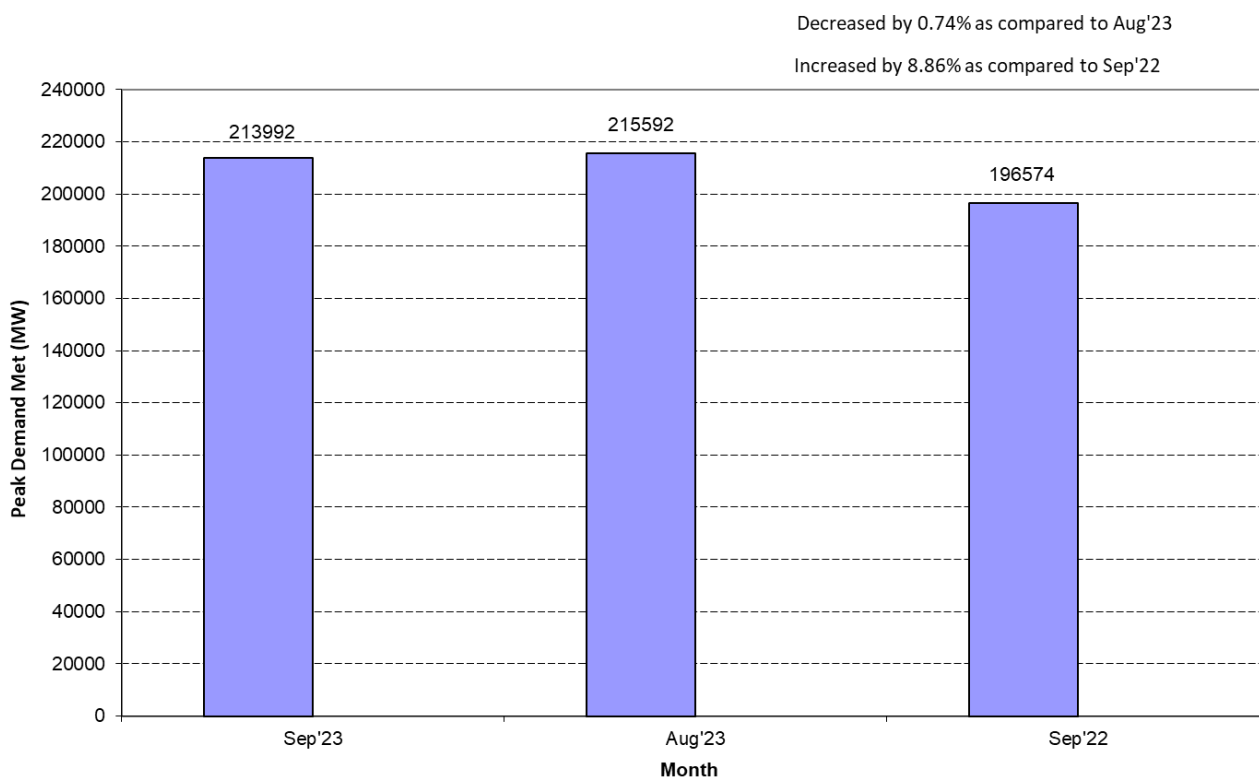
पृष्ठसूची CONTENTS

1	माह सितंबर 2023 की सारांश रिपोर्ट Summary of report for the month of SEPTEMBER 2023	1-5
2	क्षेत्रवार स्थापित विद्युत क्षमता Region wise Installed Capacity	6
3	राष्ट्रीय स्तर पर सांध्यकालीन शिखर मांगपूर्ति Peak demand met at National level	7-8
4	राष्ट्रीय स्तर पर विद्युत ऊर्जा खपत Energy consumption at National level	9-10
5	फ्रिक्वेंसी रूपरेखा Frequency profile	11-12
6	राष्ट्रीय स्तर पर जल विद्युत ऊर्जा उत्पादन Hydro generation at National level	13-14
7	राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन Wind generation at National level	15
8	राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन Solar generation at National level	16
9	राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति Daily maximum demand met at National level	17
10	01 सितंबर के माँग एवं उत्पादन प्लॉट(अधिकतम माँग पूर्ति)व माँग का डाटा (15 मिनट अंतराल) Demand and generation plots for 01-SEPTEMBER (Max Demand Met) and Demand data (15-minute interval)	18-22
11	ऊर्जा आपूर्ति की वास्तविक स्थिति Actual power supply position	23
12	ऊर्जा आवश्यकता एवं आपूर्ति की पिछले वर्ष माह सितंबर 2022 से तुलना Energy requirement and met comparison with SEPTEMBER 2022	24
13	शिखर माँग आवश्यकता एवं आपूर्ति की पिछले वर्ष माह सितंबर 2022 से तुलना Peak demand requirement and met comparison with SEPTEMBER 2022	25

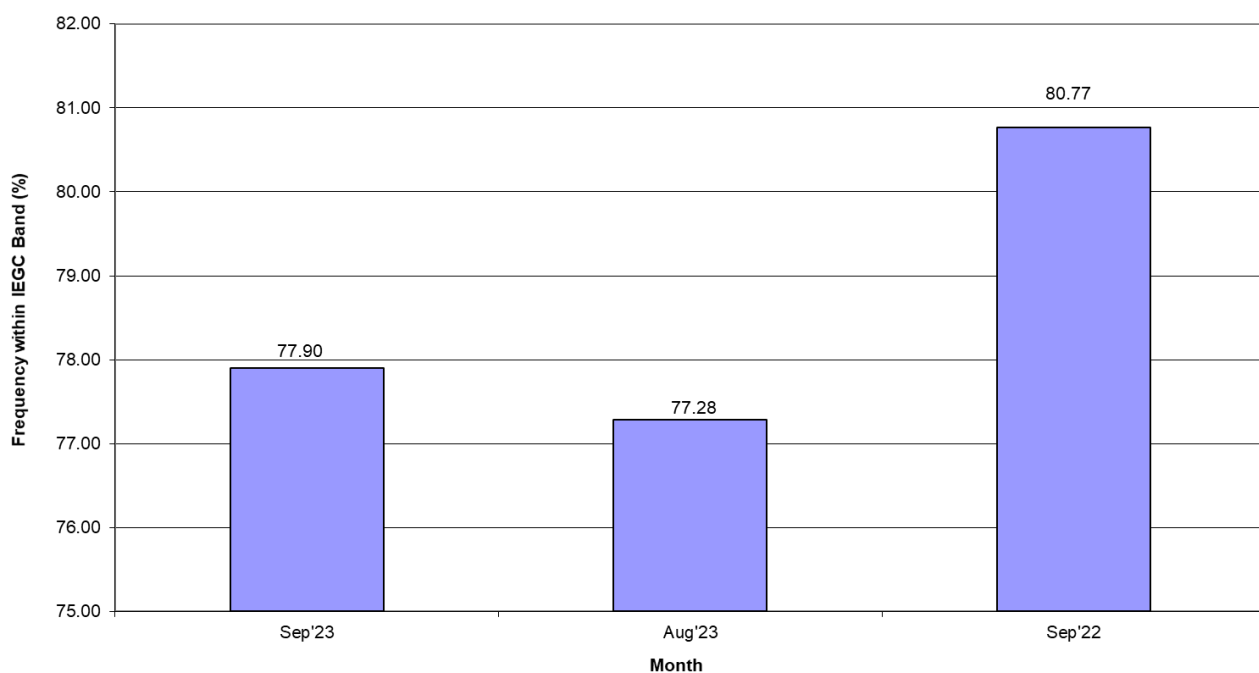
14	क्षेत्रीय स्तर पर शिड्यूल एवं ड्रावल Schedule & drawal at regional level	26
15	अंतर क्षेत्रीय विनिमय Inter-regional exchanges	27-36
16	अंतर राष्ट्रीय विद्युत विनिमय Inter-national exchanges	37-39
17	एसटीओए (द्विपक्षीय एवं सामूहिक) एवं यू आइ का ब्यौरा Bilateral and collective short-term open access & UI details	40
18	पावर मार्केट की सूचना Power market information	41-44
19	अक्षय ऊर्जा प्रमाण पत्र प्रक्रिया Renewable energy certificate mechanism	45-47
20	मुख्य घटनाएँ एवं डिस्टर्बेंस Important events/disturbances	48-60
21	नए कमीशंड पारेषण तत्व New elements commissioned	61
22	विभव रूपरेखा Voltage profile	62
23	सितंबर 2023 के दौरान रा.भा.प्रे.के.द्वारा निगरानी में तत्वों के आउटेज की सूची List of outage of elements monitored by NLDC for the month of SEPTEMBER 2023 (To be updated)	
24	अब तक उच्चतम (मांग व ऊर्जा) आँकड़े : माह सितंबर 2023 तक All Time Highest (Demand & Energy) figures: Till SEPTEMBER 2023	63
25	सितंबर 2023 के लिए सिस्टम रिलाइबिलिटी सूचकांक System Reliability Indices for SEPTEMBER 2023	64

1. SUMMARY OF REPORT FOR THE MONTH OF SEPTEMBER-2023

EVENING PEAK HOUR (at 2000hrs) DEMAND MET AT NATIONAL LEVEL (MW)



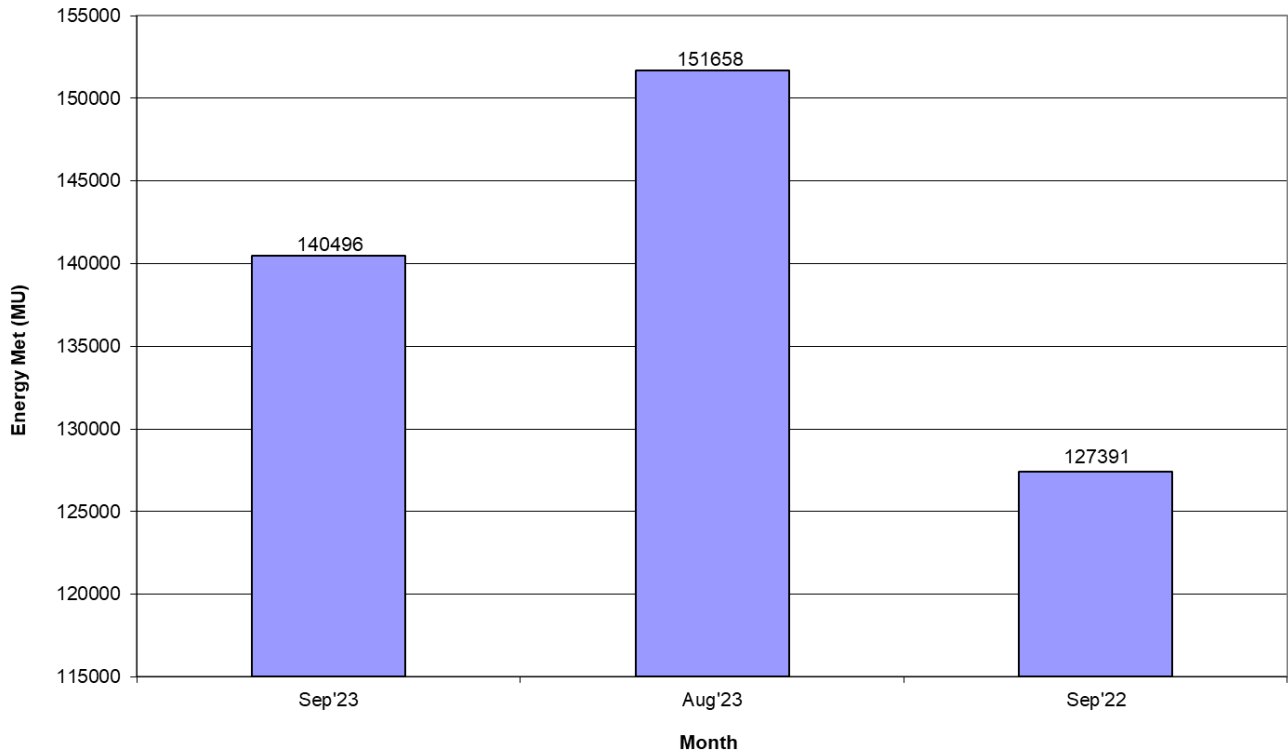
Frequency within IEGC Band



ENERGY MET AT NATIONAL LEVEL (MU)

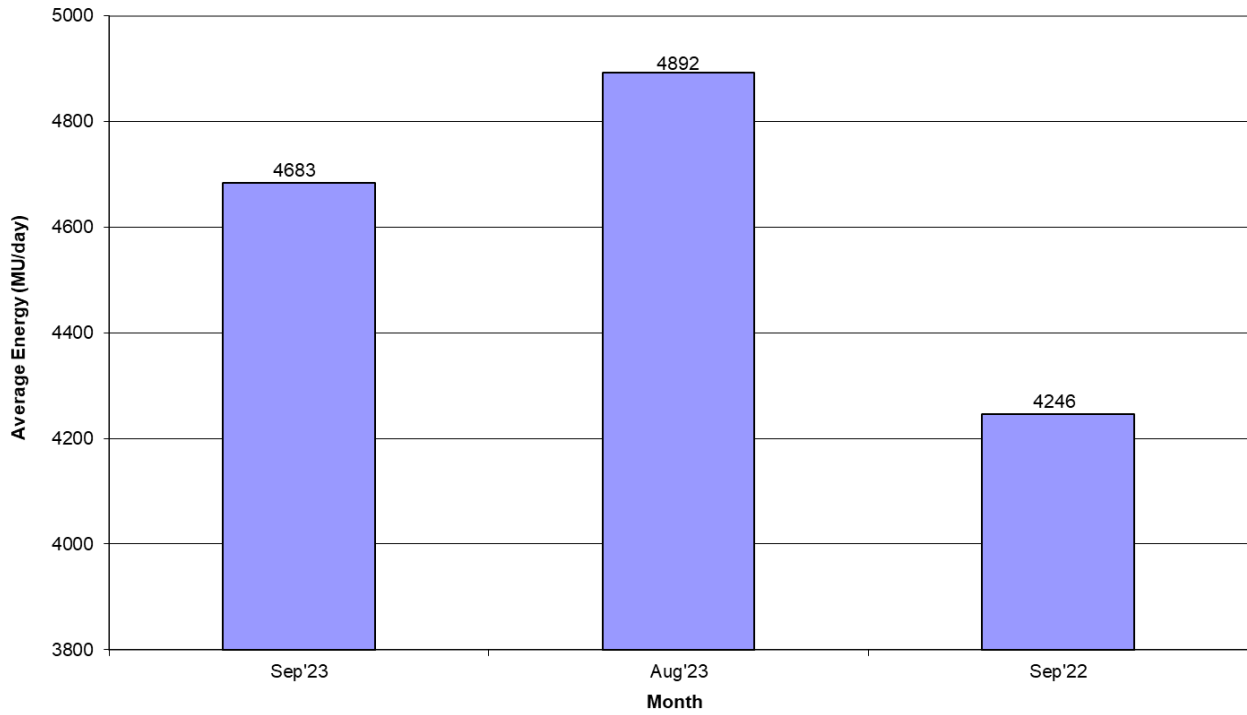
Decreased by 7.36% as compared to Aug'23

Increased by 10.29% as compared to Sep'22



AVERAGE ENERGY MET AT NATIONAL LEVEL (MU/Day)

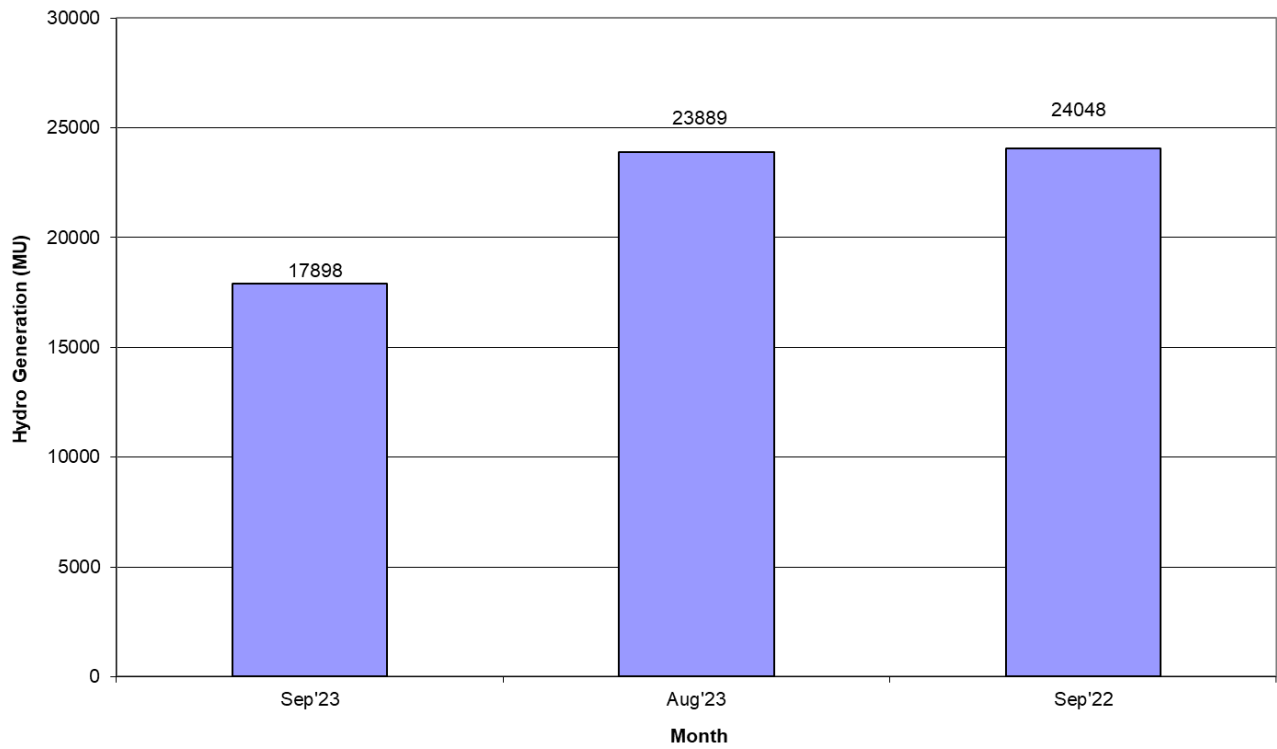
Increased by 10.29% as compared to Sep'22



HYDRO GENERATION AT NATIONAL LEVEL (MU)

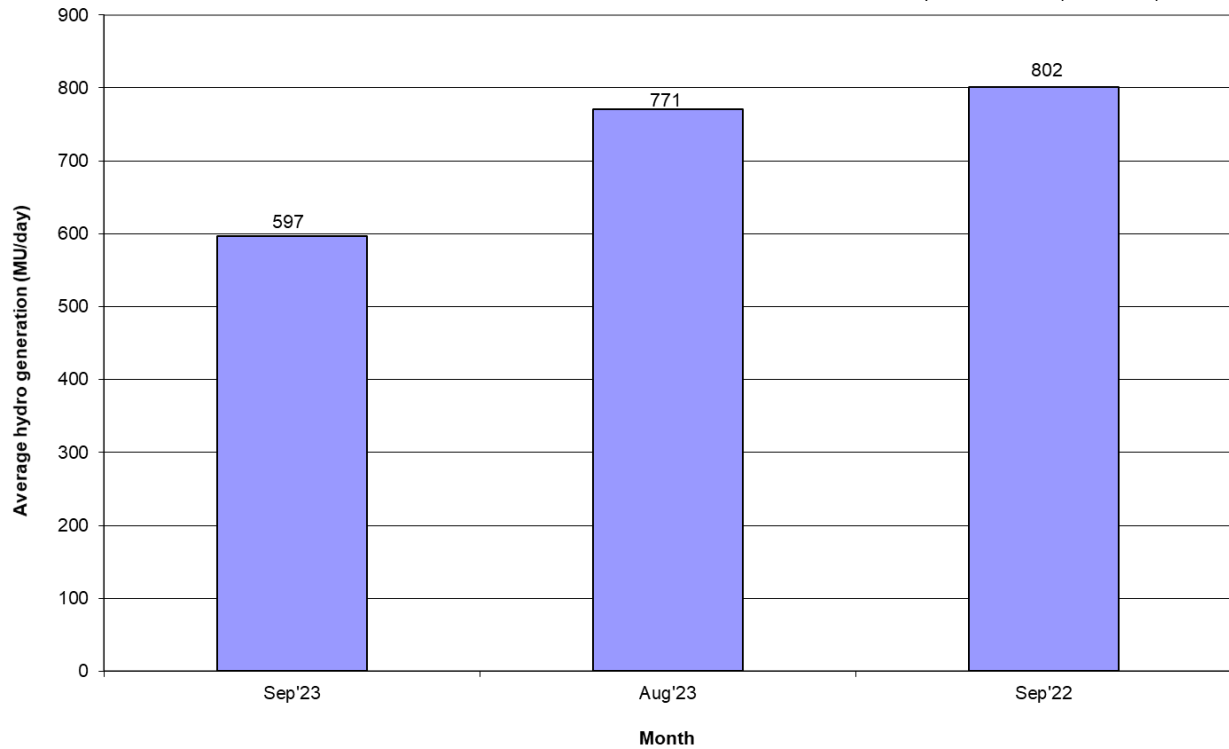
Decreased by 25.08% as compared to Aug'23

Decreased by 25.58% as compared to Sep'22



AVERAGE HYDRO GENERATION AT NATIONAL LEVEL (MU/Day)

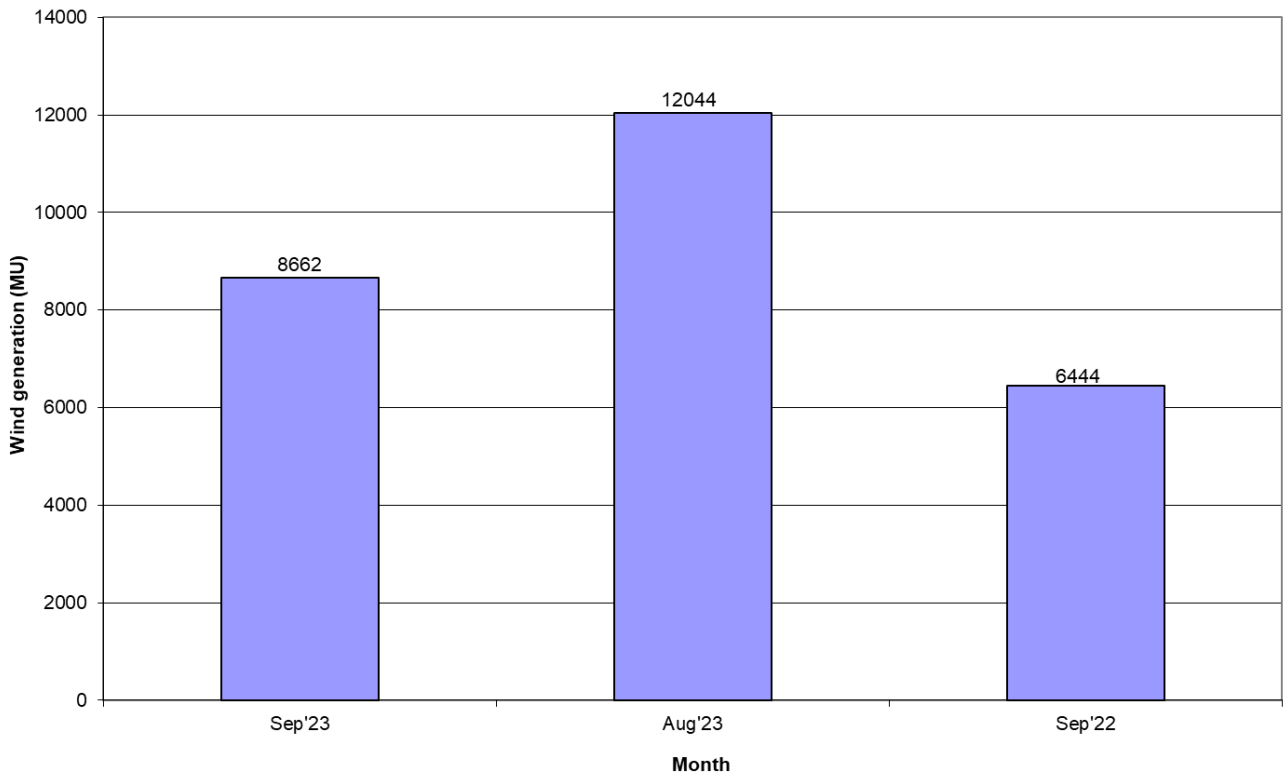
Decreased by 25.58% as compared to Sep'22



WIND GENERATION AT NATIONAL LEVEL (MU)

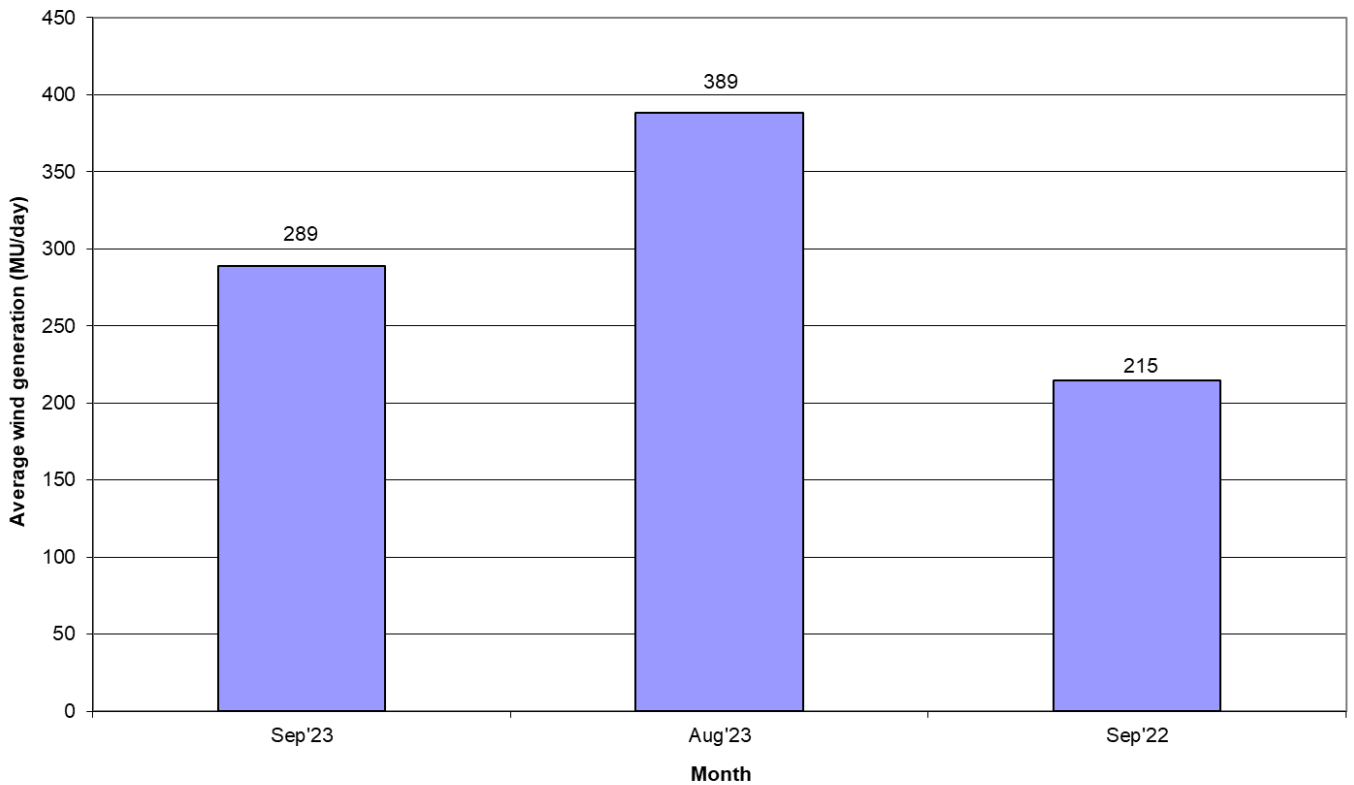
Decreased by 28.08% as compared to Aug'23

Increased by 34.42% as compared to Sep'22



AVERAGE WIND GENERATION AT NATIONAL LEVEL (MU/Day)

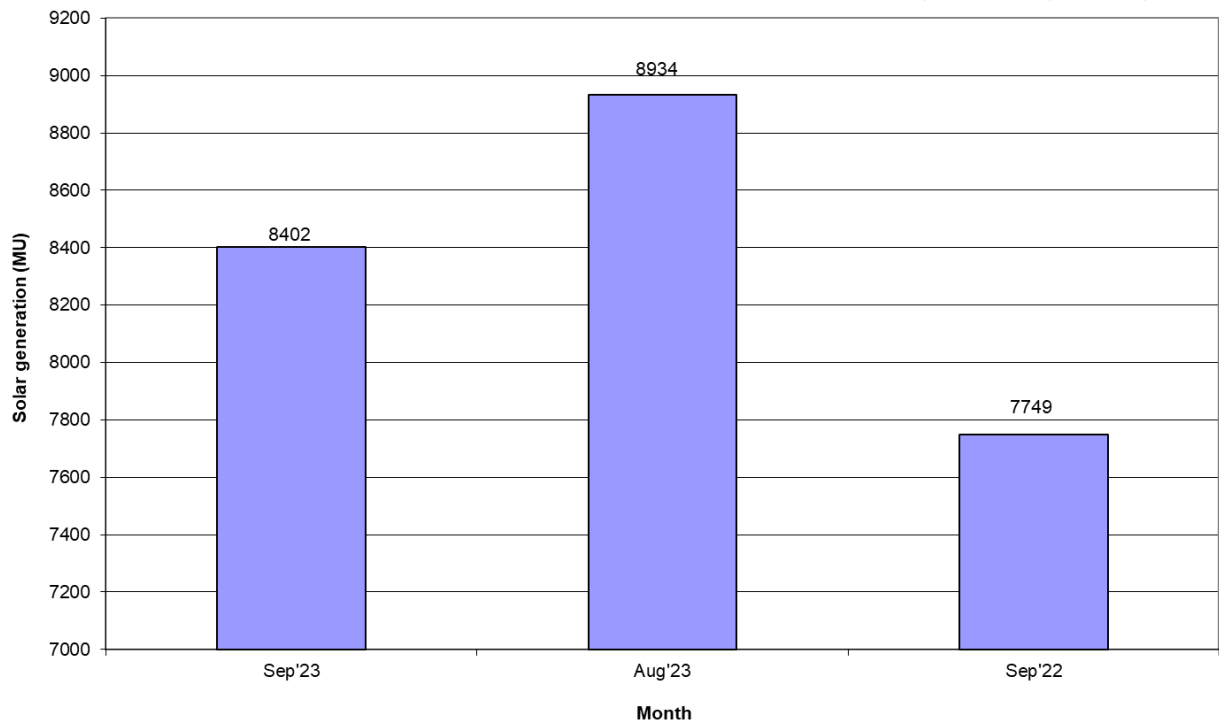
Increased by 34.42% as compared to Sep'22



SOLAR GENERATION AT NATIONAL LEVEL (MU)

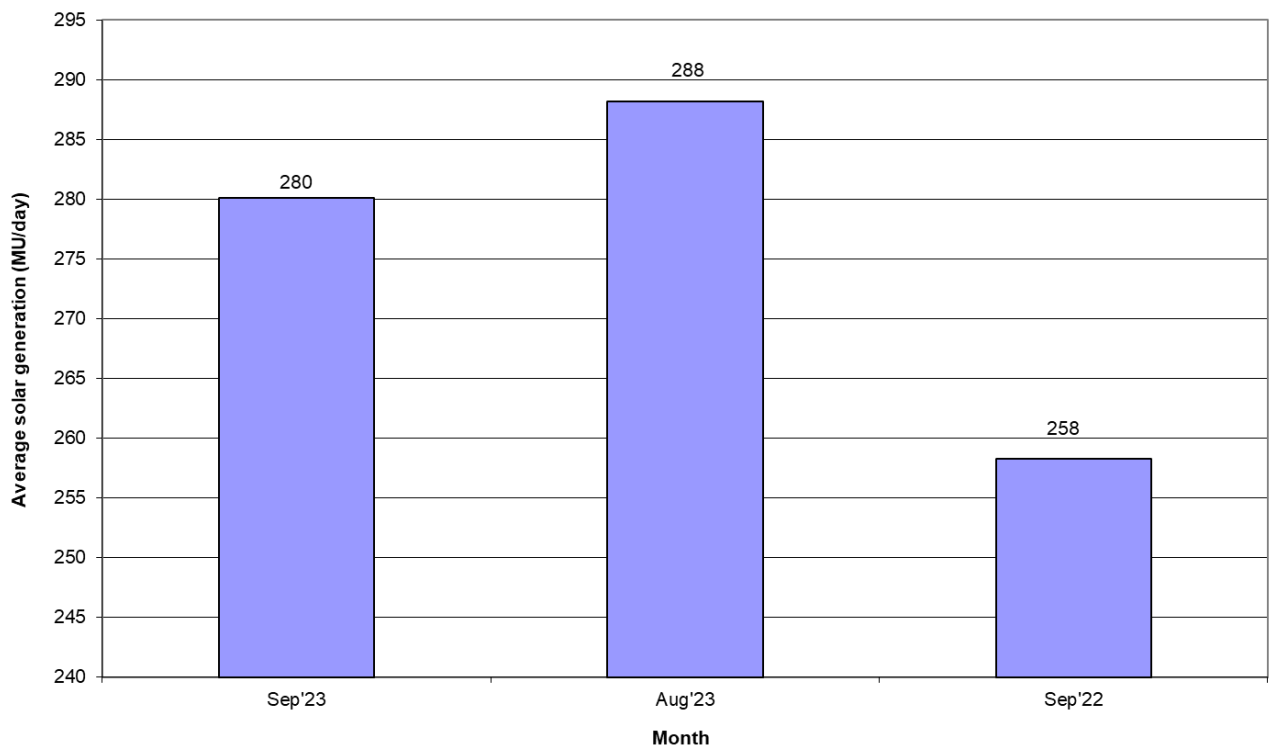
Decreased by 5.95% as compared to Aug'23

Increased by 8.44% as compared to Sep'22



AVERAGE SOLAR GENERATION AT NATIONAL LEVEL (MU/Day)

Increased by 8.44% as compared to Sep'22



2. ALL INDIA REGIONWISE INSTALLED CAPACITY

As on 30.09.2023
(All figures are in MW)

S No	Region	THERMAL					NUCLEAR	HYDRO	RES @ MNRE	GRAND TOTAL
		COAL	LIGNITE	GAS	DIESEL	TOTAL				
1	NR	55737	1580	5995	0	63312	1620	20752	34507	120191
2	WR	74513	1400	10806	0	86719	2540	7563	42553	139374
3	SR	47578	3640	6492	434	58144	3320	11827	52215	125506
4	ER	28448	0	80	0	28528	0	4764	1896	35188
5	NER	770	0	1665	36	2471	0	1944	574	4989
6	ISLANDS	0	0	0	120	120	0	0	38	158
	ALL INDIA	207045	6620	25038	589	239293	7480	46850	131783	425406

Remarks: Data from CEA website

3. राष्ट्रीय स्तर पर संध्याकालीन शिखर विद्युत मांग पूर्ति
EVENING PEAK HOUR DEMAND MET AT NATIONAL LEVEL

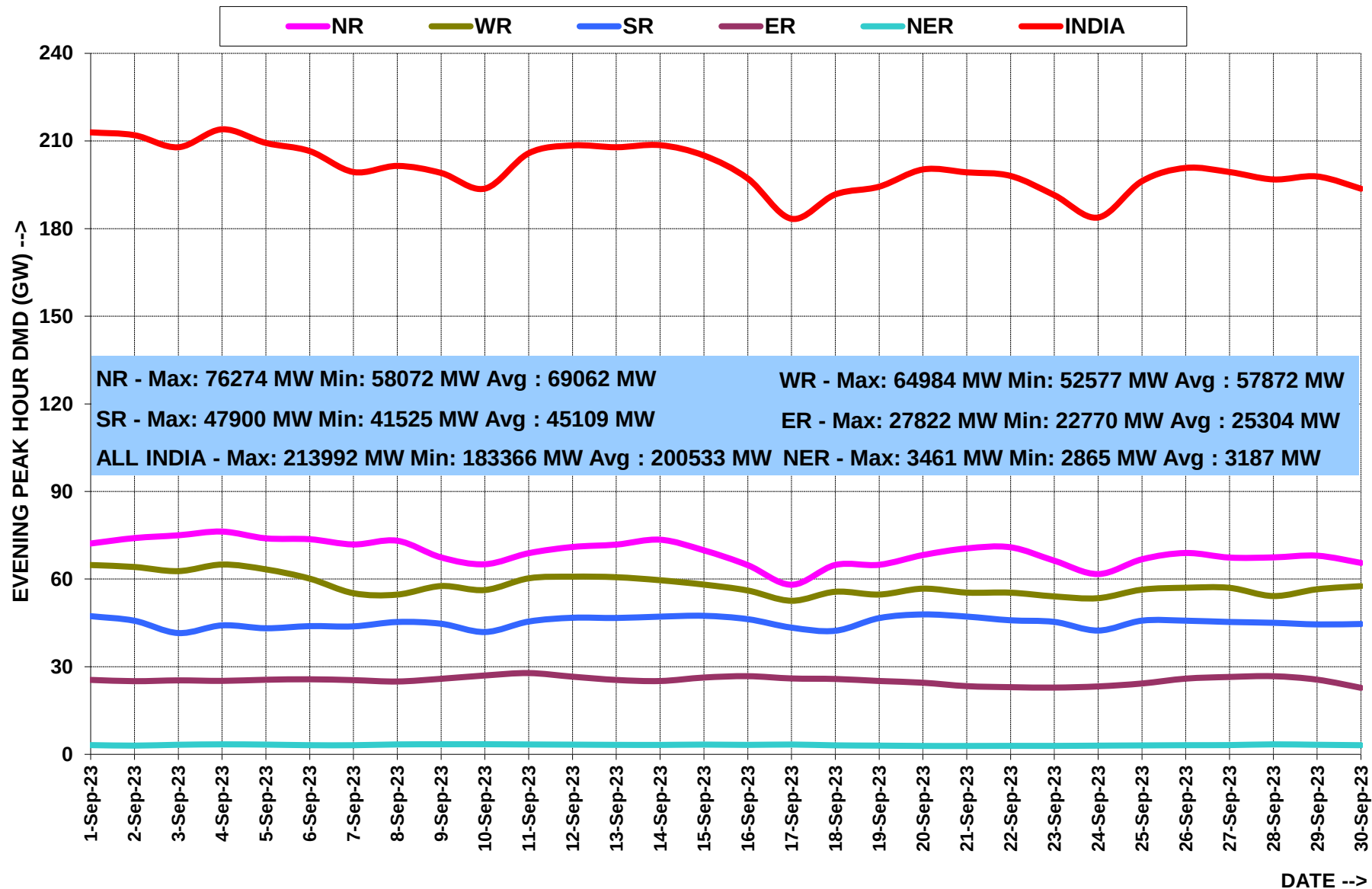
माह: सितंबर 2023 MONTH:- SEP 2023

सभी आंकड़े मेगावाट मे All figures in MW

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
1-Sep-23	72187	64813	47304	25488	3162	212954
2-Sep-23	74045	64151	45734	25049	2989	211968
3-Sep-23	74994	62732	41525	25324	3276	207851
4-Sep-23	76274	64984	44155	25155	3424	213992
5-Sep-23	73951	63325	43165	25544	3332	209317
6-Sep-23	73649	60160	43894	25696	3134	206533
7-Sep-23	71860	55165	43813	25403	3125	199366
8-Sep-23	73127	54689	45313	24935	3394	201458
9-Sep-23	67419	57619	44709	25861	3447	199055
10-Sep-23	65090	56261	41879	27001	3461	193692
11-Sep-23	68868	60268	45464	27822	3380	205802
12-Sep-23	70990	60828	46762	26587	3319	208486
13-Sep-23	71811	60646	46688	25471	3244	207860
14-Sep-23	73480	59609	47150	25100	3222	208561
15-Sep-23	69841	58116	47429	26297	3340	205023
16-Sep-23	64783	56093	46288	26759	3238	197161
17-Sep-23	58072	52577	43380	25979	3358	183366
18-Sep-23	64843	55671	42310	25805	3076	191705
19-Sep-23	64898	54712	46640	25113	2988	194351
20-Sep-23	68228	56726	47900	24520	2880	200254
21-Sep-23	70499	55380	47163	23348	2865	199255
22-Sep-23	70888	55353	45890	22997	2905	198033
23-Sep-23	66290	54068	45342	22854	2898	191452
24-Sep-23	61718	53465	42380	23253	2979	183795
25-Sep-23	66778	56381	45772	24251	3057	196239
26-Sep-23	68930	57036	45757	25923	3132	200778
27-Sep-23	67369	57007	45315	26497	3170	199358
28-Sep-23	67433	54199	45033	26744	3419	196828
29-Sep-23	68013	56522	44479	25569	3271	197854
30-Sep-23	65538	57592	44629	22770	3127	193656
उच्चतम MAXIMUM	76274	64984	47900	27822	3461	213992
निम्नतम MINIMUM	58072	52577	41525	22770	2865	183366
औसत AVERAGE	69062	57872	45109	25304	3187	200533
अब तक का उच्चतम All Time Max.	76274	64984	52366	28579	3510	215592
दिनांक Date	04.09.23	04.09.23	18.04.23	18.04.23	12.08.22	28.08.23

नोट : सभी आंकड़े 2000 बजे की मांगपूर्ति पर आधारित है ।
 Note: The above figures correspond to value at 2000 Hrs

EVENING PEAK HOUR DEMAND (2000hrs) MET DURING THE MONTH OF SEPTEMBER 2023



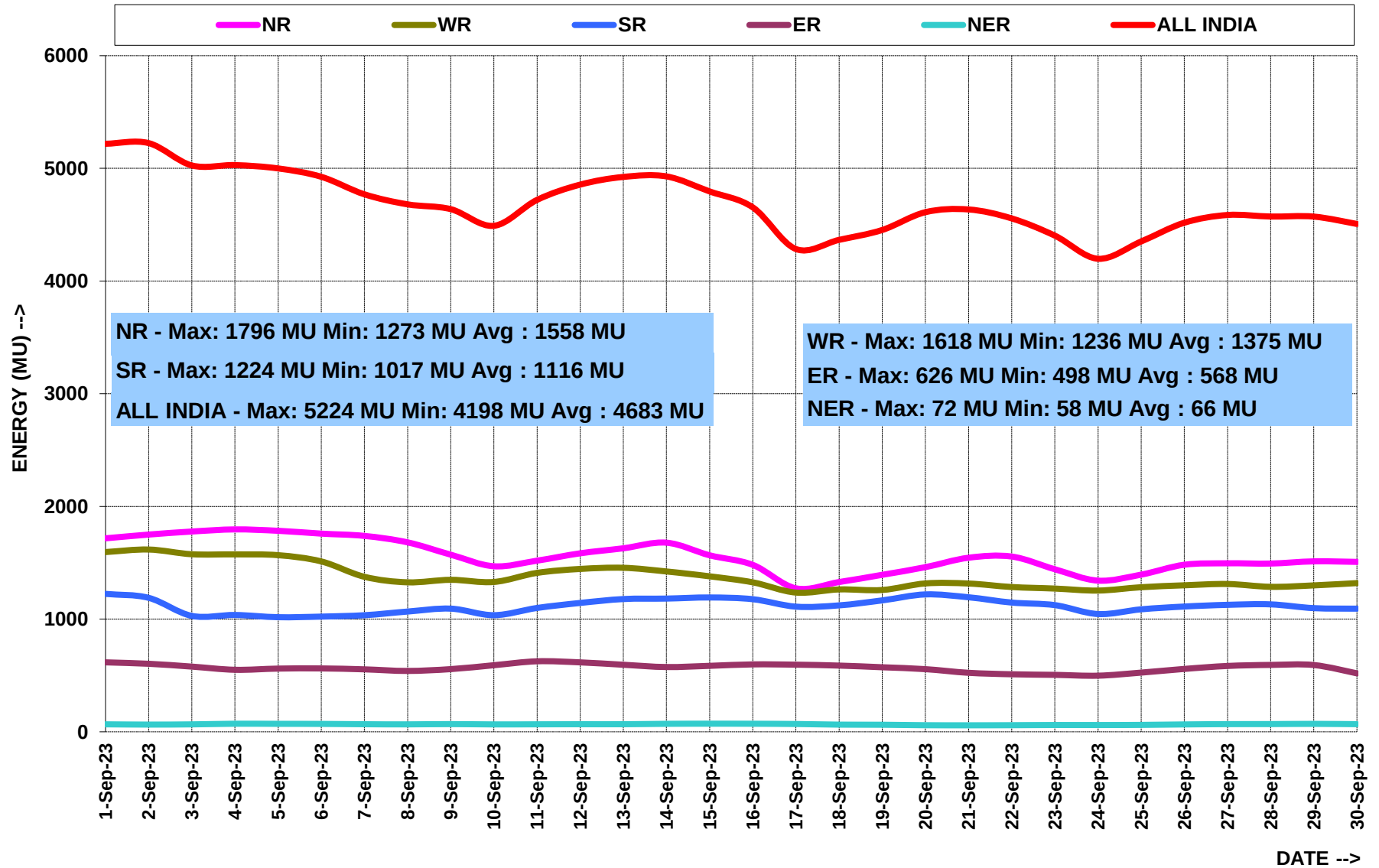
4. राष्ट्रीय स्तर पर विद्युत ऊर्जा आपूर्ति ENERGY MET AT NATIONAL LEVEL

माह: सितंबर 2023 MONTH:- SEP 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Sep-23	1717	1595	1224	616	66	5219
02-Sep-23	1750	1618	1189	604	64	5224
03-Sep-23	1777	1576	1029	578	66	5026
04-Sep-23	1796	1574	1037	550	72	5029
05-Sep-23	1784	1567	1017	561	71	5000
06-Sep-23	1759	1512	1022	562	70	4925
07-Sep-23	1739	1376	1035	554	67	4770
08-Sep-23	1681	1327	1067	540	66	4681
09-Sep-23	1572	1349	1093	556	68	4638
10-Sep-23	1469	1330	1035	591	66	4490
11-Sep-23	1519	1410	1099	626	66	4720
12-Sep-23	1584	1445	1144	616	67	4856
13-Sep-23	1629	1456	1178	595	67	4924
14-Sep-23	1678	1423	1182	575	71	4929
15-Sep-23	1566	1380	1192	585	72	4795
16-Sep-23	1481	1326	1177	598	71	4654
17-Sep-23	1273	1236	1111	596	69	4284
18-Sep-23	1329	1264	1122	588	64	4366
19-Sep-23	1393	1259	1166	573	62	4453
20-Sep-23	1461	1317	1219	556	58	4611
21-Sep-23	1545	1316	1194	524	58	4636
22-Sep-23	1555	1285	1147	510	59	4556
23-Sep-23	1442	1271	1124	505	60	4403
24-Sep-23	1342	1253	1045	498	60	4198
25-Sep-23	1395	1283	1087	525	61	4352
26-Sep-23	1483	1300	1112	558	65	4518
27-Sep-23	1495	1312	1127	584	68	4586
28-Sep-23	1493	1287	1131	593	69	4573
29-Sep-23	1513	1299	1097	593	70	4572
30-Sep-23	1508	1320	1093	519	68	4507
कुल TOTAL	46729	41263	33493	17030	1981	140496
उच्चतम MAXIMUM	1796	1618	1224	626	72	5224
निम्नतम MINIMUM	1273	1236	1017	498	58	4198
औसत AVERAGE	1558	1375	1116	568	66	4683
संचयी 2023-24 Cumulative 2023-24	262110	252136	208797	103423	10739	837206
अब तक का उच्चतम All Time Max.	1796	1618	1329	639	72	5224
दिनांक Date	04.09.23	02.09.23	19.04.23	08.06.23	15.09.23	02.09.23

ENERGY MET DURING THE MONTH OF SEPTEMBER' 2023



5. वर्ष 2023-24 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR YEAR 2023-24

राष्ट्रीय ग्रिड NATIONAL GRID								
फ्रिक्वेंसी रूपरेखा (Hz)			<49.9	49.9-50.05	>50.05	उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी
% समय	Apr-23	All India Grid	10.54	67.90	21.56	50.33	49.49	49.99
	May-23	All India Grid	9.83	68.48	21.69	50.43	49.48	49.99
	Jun-23	All India Grid	6.47	67.82	25.71	50.41	49.51	50.01
	Jul-23	All India Grid	4.60	74.96	20.44	50.23	49.76	50.01
	Aug-23	All India Grid	7.08	77.28	15.63	50.20	49.74	50.00
	Sep-23	All India Grid	5.26	77.90	16.84	50.21	49.77	50.00
	2023-24 (upto Sep)	All India Grid	7.30	72.41	20.29	50.43	49.48	50.00

5.1 सितंबर 2023 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR SEPTEMBER 2023

फ्रिक्वेंसी रूपरेखा (Hz)		<49.9	49.9-50.05	>50.05	उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी	एफ़.वी.आई.
% समय	01-Sep-23	20.34	70.29	9.38	50.12	49.52	49.96	0.11
	02-Sep-23	4.49	80.91	14.59	50.19	49.72	50.00	0.03
	03-Sep-23	1.66	77.14	21.20	50.26	49.85	50.02	0.03
	04-Sep-23	4.10	61.20	34.70	50.24	49.78	50.03	0.04
	05-Sep-23	1.12	78.88	20.00	50.20	49.79	50.01	0.03
	06-Sep-23	1.96	82.88	15.16	50.18	49.80	50.00	0.02
	07-Sep-23	2.18	80.08	17.74	50.25	49.85	50.01	0.03
	08-Sep-23	2.70	82.88	14.42	50.30	49.79	50.01	0.03
	09-Sep-23	5.22	83.33	11.45	50.18	49.81	49.99	0.03
	10-Sep-23	3.65	83.44	12.92	50.26	49.79	50.00	0.03
	11-Sep-23	1.15	82.37	16.48	50.23	49.85	50.01	0.02
	12-Sep-23	7.01	82.01	10.97	50.14	49.75	49.99	0.03
	13-Sep-23	5.71	85.86	8.44	50.23	49.77	49.99	0.03
	14-Sep-23	11.64	81.69	6.67	50.20	49.73	49.98	0.04
	15-Sep-23	5.83	75.96	18.21	50.15	49.74	50.00	0.03
	16-Sep-23	6.62	75.07	18.31	50.29	49.69	50.00	0.05
	17-Sep-23	6.08	65.38	28.54	50.23	49.77	50.01	0.04
	18-Sep-23	2.30	73.89	23.81	50.30	49.78	50.02	0.04
	19-Sep-23	5.46	75.31	19.22	50.22	49.76	50.01	0.04
	20-Sep-23	3.45	79.41	17.14	50.18	49.84	50.00	0.03
	21-Sep-23	2.33	72.62	25.06	50.24	49.82	50.02	0.03
	22-Sep-23	1.06	83.08	15.86	50.16	49.82	50.01	0.02
	23-Sep-23	4.61	80.39	15.00	50.23	49.83	50.00	0.03
	24-Sep-23	3.16	64.97	31.88	50.30	49.80	50.03	0.06
	25-Sep-23	3.89	87.40	8.72	50.15	49.84	49.99	0.02
	26-Sep-23	8.48	79.69	11.83	50.20	49.79	49.99	0.03
	27-Sep-23	12.53	75.88	11.59	50.23	49.73	49.98	0.05
	28-Sep-23	4.48	81.30	14.22	50.15	49.76	50.00	0.03
	29-Sep-23	7.33	77.13	15.54	50.20	49.75	50.00	0.04
	30-Sep-23	7.40	76.49	16.11	50.16	49.71	49.99	0.04

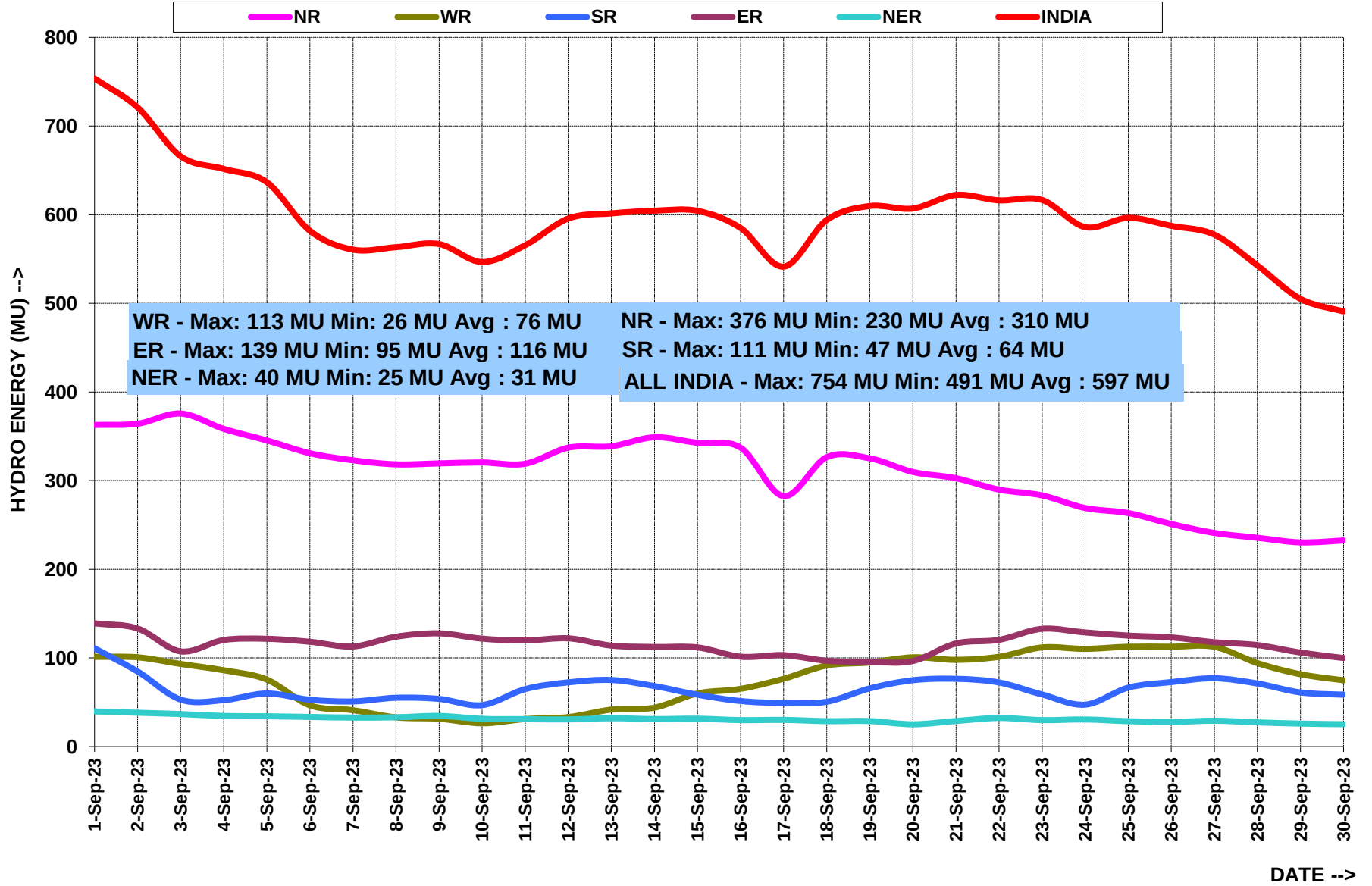
6.राष्ट्रीय स्तर पर जल विद्युत उत्पादन
HYDRO GENERATION AT NATIONAL LEVEL

माह: सितंबर 2023 MONTH:- SEP 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Sep-23	363	101	111	139	40	754
02-Sep-23	364	101	85	133	38	721
03-Sep-23	376	93	53	107	37	666
04-Sep-23	358	86	52	120	35	652
05-Sep-23	345	76	60	122	34	637
06-Sep-23	331	46	53	118	33	582
07-Sep-23	323	41	51	113	33	561
08-Sep-23	318	33	55	124	33	563
09-Sep-23	319	31	54	128	35	567
10-Sep-23	321	26	47	122	31	547
11-Sep-23	319	31	65	120	31	566
12-Sep-23	337	33	72	122	31	596
13-Sep-23	339	42	75	114	32	601
14-Sep-23	349	44	68	112	31	605
15-Sep-23	343	60	58	112	31	604
16-Sep-23	337	65	51	101	30	585
17-Sep-23	283	77	49	103	30	541
18-Sep-23	326	91	51	97	29	594
19-Sep-23	325	95	66	95	29	610
20-Sep-23	310	101	75	96	25	607
21-Sep-23	303	98	76	116	29	622
22-Sep-23	290	101	72	120	32	616
23-Sep-23	283	112	59	133	30	617
24-Sep-23	269	110	47	129	31	586
25-Sep-23	263	113	67	125	29	596
26-Sep-23	251	113	73	123	28	587
27-Sep-23	241	113	77	118	29	577
28-Sep-23	236	94	71	114	27	543
29-Sep-23	230	82	61	106	26	505
30-Sep-23	233	75	59	100	25	491
कुल TOTAL	9285	2283	1913	3484	932	17898
उच्चतम MAXIMUM	376	113	111	139	40	754
निम्नतम MINIMUM	230	26	47	95	25	491
औसत AVERAGE	310	76	64	116	31	597
संचयी 2023-24 Cumulative 2023-24	53982	9621	12352	17421	4262	97638
अब तक का उच्चतम All Time Max.	443	167	208	157	41	877
दिनांक Date	01.08.23	18.12.14	31.08.18	14.09.22	29.08.23	30.08.22

HYDRO ENERGY DURING THE MONTH OF SEPTEMBER' 2023



7. राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन
WIND GENERATION AT NATIONAL LEVEL
माह: सितंबर 2023 MONTH:- SEP 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Sep-23	25	47	54	-----	-----	125
02-Sep-23	47	61	122	-----	-----	230
03-Sep-23	51	83	136	-----	-----	269
04-Sep-23	30	106	228	-----	-----	364
05-Sep-23	10	88	198	-----	-----	297
06-Sep-23	13	100	195	-----	-----	309
07-Sep-23	10	190	225	-----	-----	425
08-Sep-23	4	207	230	-----	-----	441
09-Sep-23	8	212	245	-----	-----	465
10-Sep-23	21	231	249	-----	-----	501
11-Sep-23	30	214	224	-----	-----	468
12-Sep-23	34	183	190	-----	-----	407
13-Sep-23	32	134	184	-----	-----	350
14-Sep-23	16	80	192	-----	-----	287
15-Sep-23	9	89	179	-----	-----	278
16-Sep-23	15	159	199	-----	-----	373
17-Sep-23	19	194	151	-----	-----	365
18-Sep-23	32	150	138	-----	-----	319
19-Sep-23	19	121	141	-----	-----	281
20-Sep-23	10	114	117	-----	-----	241
21-Sep-23	11	81	92	-----	-----	184
22-Sep-23	15	70	64	-----	-----	149
23-Sep-23	17	73	99	-----	-----	189
24-Sep-23	16	67	150	-----	-----	232
25-Sep-23	14	46	115	-----	-----	176
26-Sep-23	8	22	121	-----	-----	150
27-Sep-23	4	27	139	-----	-----	170
28-Sep-23	4	40	147	-----	-----	191
29-Sep-23	9	46	149	-----	-----	205
30-Sep-23	14	54	154	-----	-----	223
कुल TOTAL	547	3288	4827	-----	-----	8662
उच्चतम MAXIMUM	51	231	249	-----	-----	501
निम्नतम MINIMUM	4	22	54	-----	-----	125
औसत AVERAGE	18	110	161	-----	-----	289
संचयी 2023-24 Cumulative 2023-24	5411	24178	25603	-----	-----	55191
अब तक का उच्चतम All Time Max.	86	273	298	-----	-----	611
दिनांक Date	07.08.23	07.08.23	21.07.23	-----	-----	02.08.23

***Source:** As reported by SLDCs. Limited visibility of embedded wind generator data.

8. राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन
SOLAR GENERATION AT NATIONAL LEVEL

माह: सितंबर 2023 MONTH:- SEP 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Sep-23	140	60	118	2.5	1.0	322
02-Sep-23	143	60	112	1.9	1.1	317
03-Sep-23	144	55	93	2.1	1.3	296
04-Sep-23	142	56	62	1.7	1.3	263
05-Sep-23	137	58	55	1.9	1.3	253
06-Sep-23	132	53	105	5.1	1.1	297
07-Sep-23	134	45	99	5.6	1.1	284
08-Sep-23	130	42	95	1.5	0.9	269
09-Sep-23	128	42	104	2.2	1.0	277
10-Sep-23	127	47	98	2.6	1.0	276
11-Sep-23	129	54	98	2.5	1.0	285
12-Sep-23	130	54	113	1.8	1.2	301
13-Sep-23	126	48	115	2.0	0.9	292
14-Sep-23	106	41	105	4.3	1.2	258
15-Sep-23	105	37	111	4.9	1.3	260
16-Sep-23	120	34	111	4.7	1.0	271
17-Sep-23	107	34	110	2.8	0.2	254
18-Sep-23	73	35	117	2.6	0.7	228
19-Sep-23	90	37	118	2.5	0.8	248
20-Sep-23	114	47	114	1.6	0.7	277
21-Sep-23	128	49	93	1.4	0.7	272
22-Sep-23	131	47	98	4.0	0.9	280
23-Sep-23	130	47	114	4.3	1.0	296
24-Sep-23	125	48	102	1.7	1.2	278
25-Sep-23	132	50	104	2.6	1.2	290
26-Sep-23	123	53	108	2.5	1.1	288
27-Sep-23	118	54	103	2.5	1.2	279
28-Sep-23	129	54	109	2.3	1.3	296
29-Sep-23	132	55	107	2.5	1.2	298
30-Sep-23	128	59	103	5.2	1.1	296
कुल TOTAL	3733	1458	3094	86	31	8402
उच्चतम MAXIMUM	144	60	118	6	1	322
निम्नतम MINIMUM	73	34	55	1	0	228
औसत AVERAGE	124	49	103	3	1	280
संचयी 2023-24 Cumulative 2023-24	23272	9644	20978	585	173.3	54652
अब तक का उच्चतम All Time Max.	147	71	143	6.7	2.4	351
दिनांक Date	10.05.23	12.05.23	21.03.23	23.03.23	22.06.22	10.05.23

***Source:** RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

9. राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति
DAILY MAXIMUM DEMAND MET AT NATIONAL LEVEL

माह: सितंबर 2023 MONTH:- SEP 2023

सभी आंकड़े मेगावाट में All figures in MW

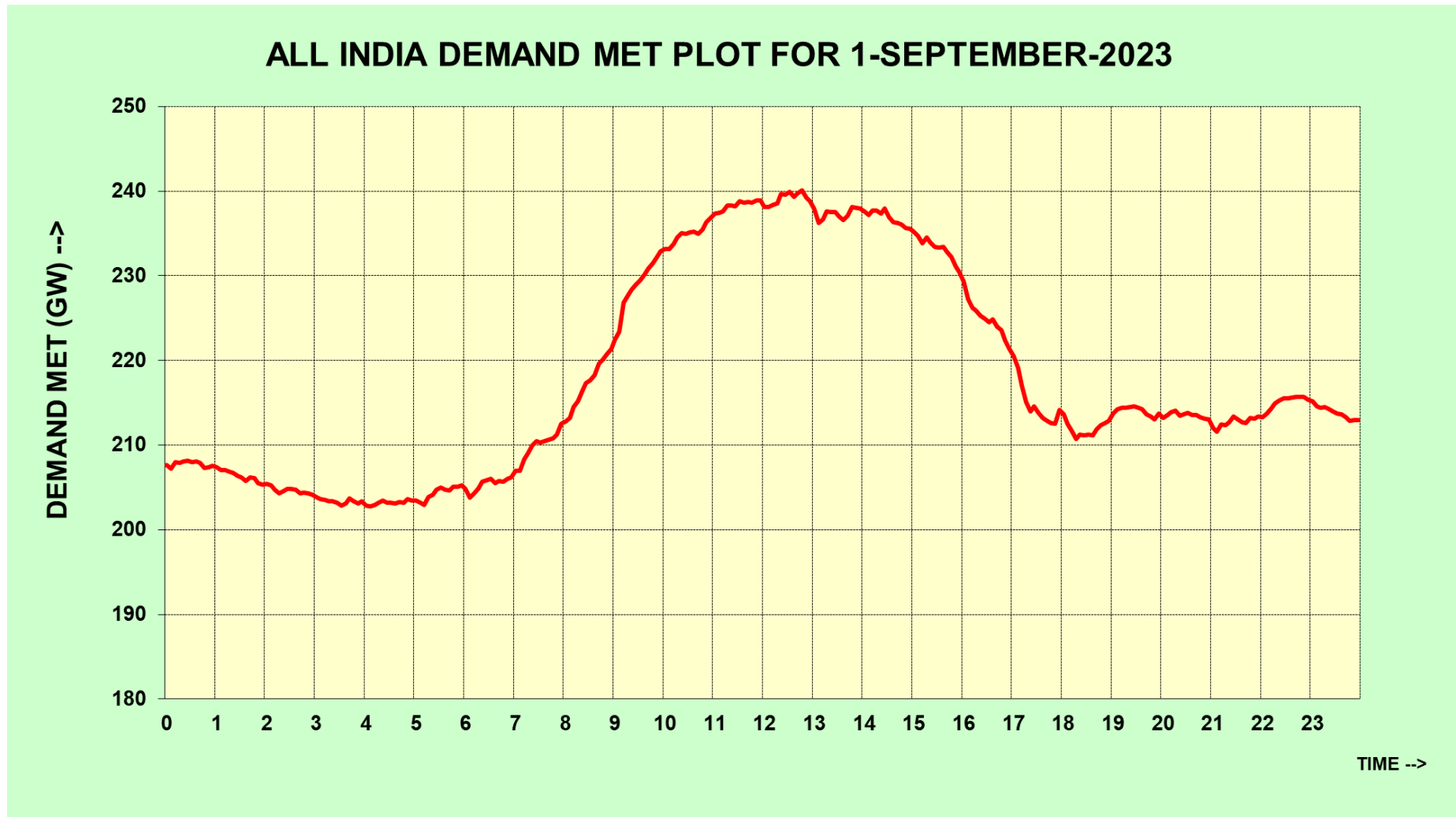
दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL	विभिन्नता फैक्टर Diversity Factor*
01-Sep-23	77621	74035	60813	27431	3251	239978	1.013
02-Sep-23	79177	73903	58271	27397	3145	238620	1.014
03-Sep-23	79130	71716	46962	26217	3480	223121	1.020
04-Sep-23	81220	72487	48844	26171	3712	226227	1.027
05-Sep-23	80005	71098	47319	26398	3258	222695	1.024
06-Sep-23	78332	67542	48486	26407	3259	219272	1.022
07-Sep-23	80494	61407	48930	25848	3124	213344	1.030
08-Sep-23	75699	60285	50684	25517	3415	209280	1.030
09-Sep-23	71897	60629	51504	26842	3552	205024	1.046
10-Sep-23	65989	59275	48816	27371	3362	196402	1.043
11-Sep-23	70281	63560	53143	28936	3490	210647	1.042
12-Sep-23	72619	63933	54066	28421	3533	212715	1.046
13-Sep-23	73456	64453	56232	26953	3439	219070	1.025
14-Sep-23	75179	65595	56667	26444	3515	221334	1.027
15-Sep-23	72742	63931	58307	27327	3670	210789	1.072
16-Sep-23	67761	60910	56285	27383	3553	207986	1.038
17-Sep-23	61073	55975	53240	27733	3351	188165	1.070
18-Sep-23	65961	58592	55168	27544	3097	195709	1.075
19-Sep-23	65120	57527	57100	25735	3087	198939	1.048
20-Sep-23	69214	60459	58665	25246	3097	206441	1.050
21-Sep-23	70847	60412	56869	24766	3048	207060	1.043
22-Sep-23	71545	59310	54545	24290	3177	202024	1.054
23-Sep-23	68440	59173	53994	23979	3182	198935	1.049
24-Sep-23	63076	56746	48799	23448	3049	187208	1.042
25-Sep-23	67719	60524	51600	24290	3385	200912	1.033
26-Sep-23	69905	61504	52780	26701	3448	206715	1.037
27-Sep-23	68723	61178	54315	27063	3422	204900	1.048
28-Sep-23	68190	57637	56290	27660	3629	205481	1.039
29-Sep-23	68478	60732	53633	26996	3486	206542	1.033
30-Sep-23	67260	62055	52320	24902	3206	201554	1.041
उच्चतम MAXIMUM	81220	74035	60813	28936	3712	239978	1.075
निम्नतम MINIMUM	61073	55975	46962	23448	3048	187208	1.013
औसत AVERAGE	71572	62886	53488	26381	3347	209570	1.039
अब तक का उच्चतम All Time Max.	81220	74035	64314	29641	3712	239978	
दिनांक Date	04.09.23	01.09.23	15.03.23	28.07.23	04.09.23	01.09.23	

* Diversity factor = (Sum of regional max demands) / All India max demand

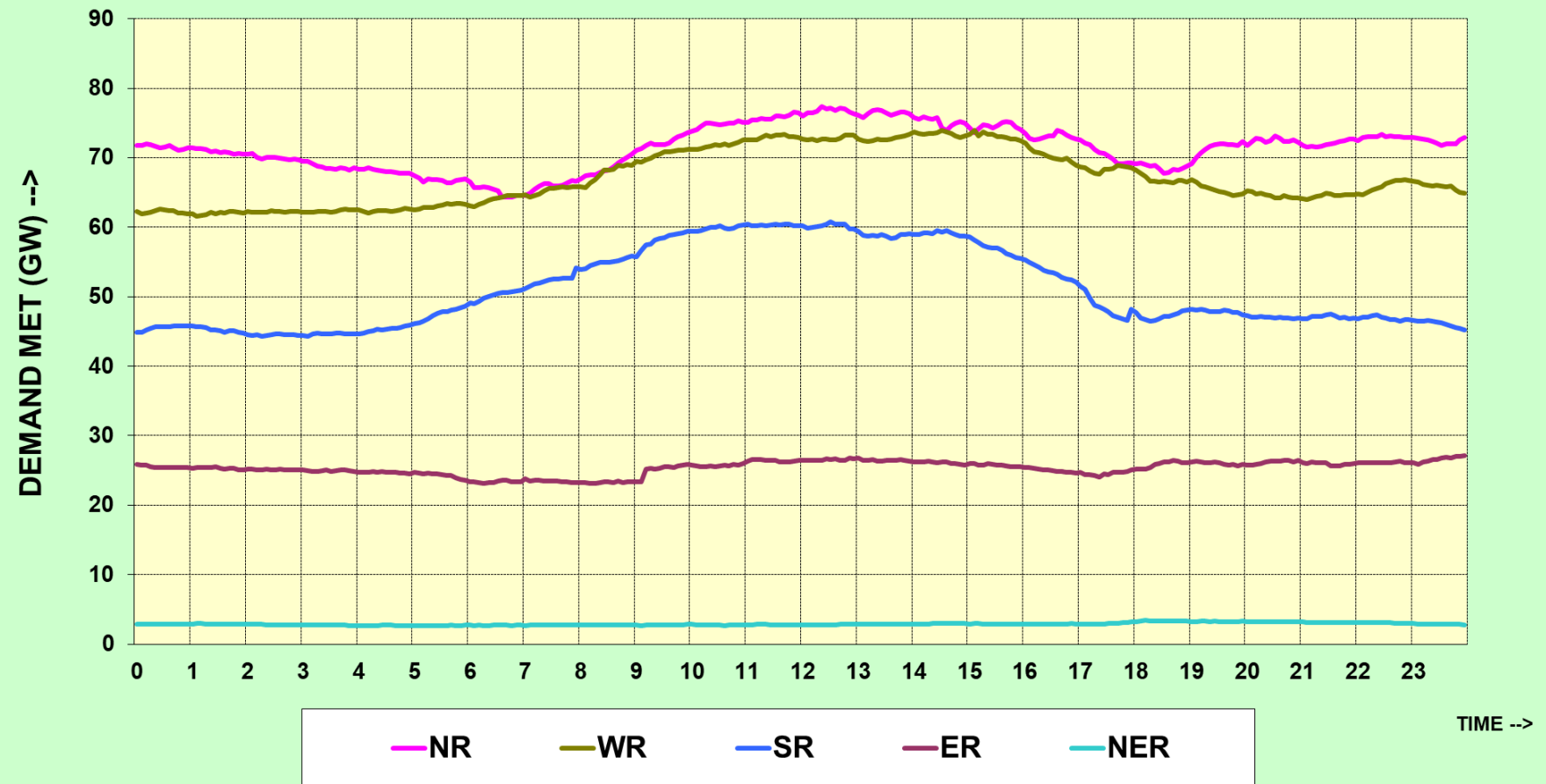
नोट : यह आंकड़े एनएलडीसी स्काडा प्रणाली में दर्ज दैनिक अधिकतम मांगपूर्ति दर्शाते हैं ।

Note: The above figures denote daily maximum demand met recorded in NLDC SCADA.

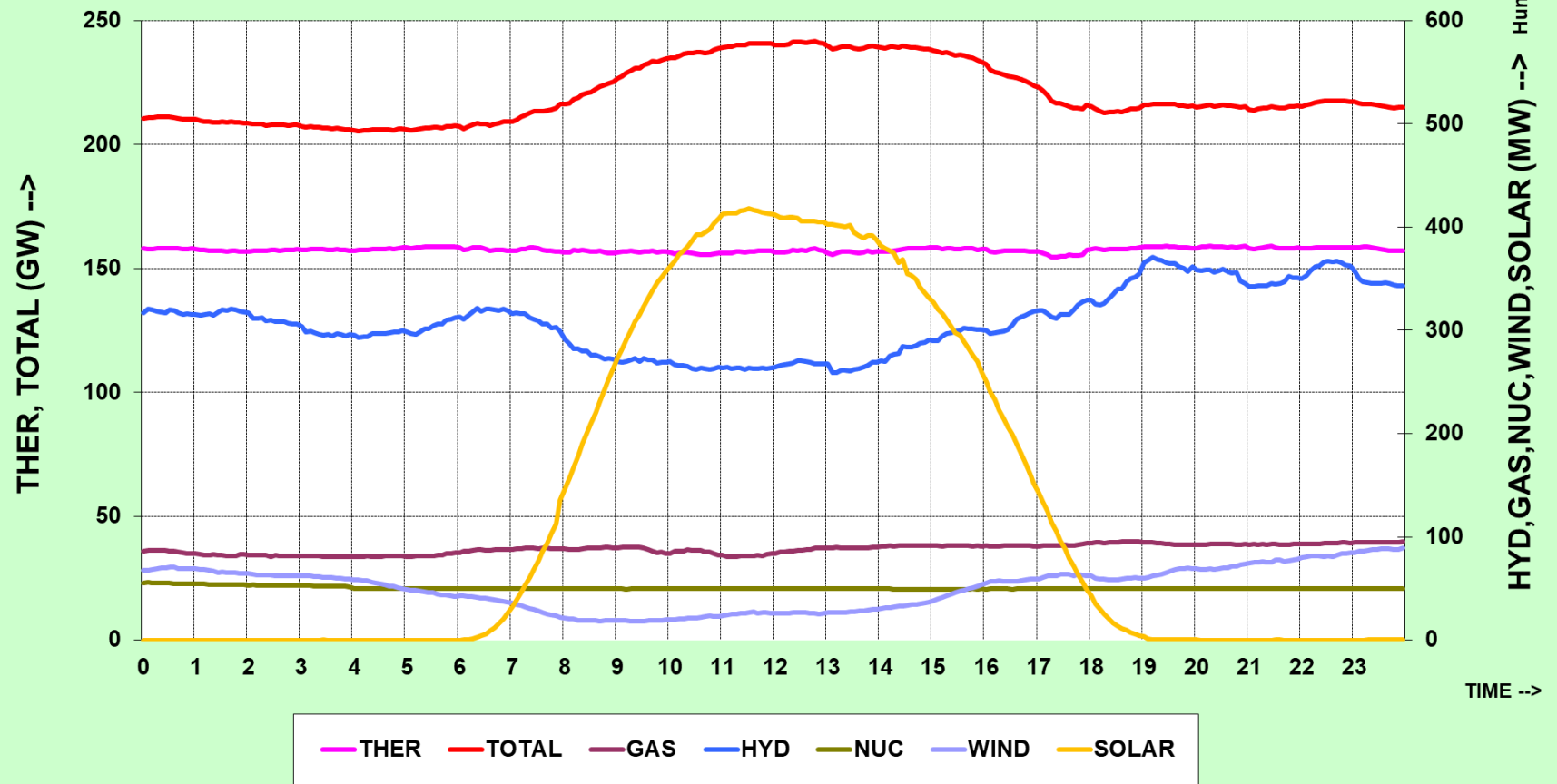
10. DEMAND AND GENERATION PLOTS FOR 01 SEPTEMBER (MAXIMUM DEMAND MET)



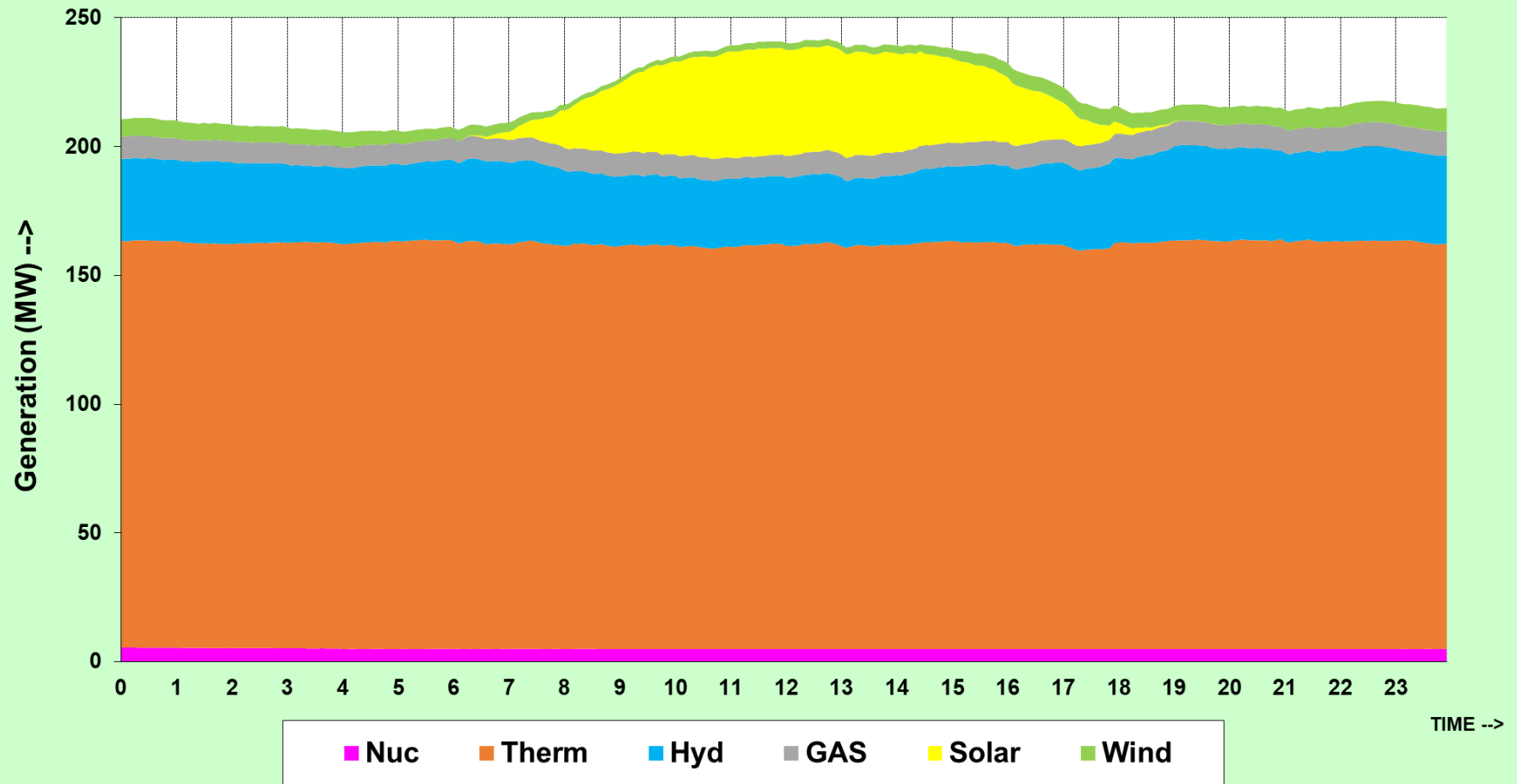
REGIONAL DEMAND MET PLOT FOR 1-SEPTEMBER-2023



ALL INDIA GENERATION PLOT FOR 1-SEPTEMBER-2023



All India Fuel-wise Generation Patterns for 1-SEPTEMBER-2023



31 अगस्त 2023 (अधिकतम माँग) की आल इंडिया माँग आपूर्ति
ALL INDIA DEMAND MET FOR 31 AUG 2023 (MAXIMUM DEMAND*)

Time (hrs)	All India Demand (MW)		Time (hrs)	All India Demand (MW)
00:05	206100		12:05	236156
00:20	206006		12:20	235887
00:35	205735		12:35	235084
00:50	205673		12:50	235537
01:05	204657		13:05	233642
01:20	204391		13:20	234533
01:35	203884		13:35	234417
01:50	203675		13:50	234481
02:05	202662		14:05	233661
02:20	202279		14:20	233173
02:35	201941		14:35	233413
02:50	201746		14:50	233117
03:05	200635		15:05	232132
03:20	200301		15:20	230980
03:35	200265		15:35	229099
03:50	200331		15:50	227830
04:05	200121		16:05	225209
04:20	200157		16:20	223135
04:35	200391		16:35	218912
04:50	201386		16:50	218574
05:05	200415		17:05	214781
05:20	201322		17:20	213317
05:35	202243		17:35	210798
05:50	201602		17:50	209292
06:05	199990		18:05	207113
06:20	201809		18:20	207854
06:35	202415		18:35	208467
06:50	202858		18:50	209864
07:05	202401		19:05	211861
07:20	204812		19:20	212636
07:35	205670		19:35	212472
07:50	206199		19:50	212104
08:05	207713		20:05	210813
08:20	211779		20:20	211041
08:35	214635		20:35	211330
08:50	216911		20:50	211697
09:05	219866		21:05	211066
09:20	224506		21:20	211529
09:35	227936		21:35	211415
09:50	229168		21:50	211937
10:05	230984		22:05	211633
10:20	232467		22:20	213063
10:35	232984		22:35	212673
10:50	233956		22:50	212097
11:05	234429		23:05	210410
11:20	234230		23:20	209951
11:35	234760		23:35	210030
11:50	235347		23:50	209647
Maximum Demand of 236598 MW met@12:11 hrs (from 1 min. interval SCADA DATA)				

* 15 minute interval SCADA DATA of instantaneous demand

11. ACTUAL POWER SUPPLY POSITION - SEPTEMBER 2023

REGION	STATE	Requirement/Availability in MU/DAY			Requirement/Availability in MU				Peak Demand/Peak Met in MW			
		Requirement	Energy met	Surplus(+)/ Deficit(-)	Requirement	Energy met	Surplus(+)/ Deficit(-)	% Shortage	Requirement	Peak Demand Met	Surplus(+)/ Deficit(-)	% Shortage
NR	Chandigarh	6	6	0.0	194	194	0	0.0	388	388	0	0.0
	Delhi	124	124	0.0	3711	3711	0	0.0	6993	6993	0	0.0
	Haryana	228	226	-1.9	6845	6789	-56	-0.8	12753	12753	0	0.0
	Himachal Pradesh	35	35	-0.1	1040	1038	-2	-0.2	1758	1758	0	0.0
	J&K(UT) and Ladakh(UT)	51	49	-1.5	1521	1476	-45	-2.9	2805	2618	-187	-6.7
	Punjab	271	271	0.0	8135	8135	0	0.0	15068	15068	0	0.0
	Rajasthan	311	309	-1.8	9339	9285	-54	-0.6	17956	17840	-116	-0.6
	Uttar Pradesh	487	487	-0.4	14615	14603	-12	-0.1	27389	27389	0	0.0
	Uttarakhand	47	47	-0.3	1408	1397	-10	-0.7	2368	2293	-75	-3.2
	Railway	3	3	0.0	104	104	0	0.0	197	197	0	0.0
	NFL	0	0	0.0	2	2	0	0.0	9	9	0	0.0
	TOTAL	1564	1558	-6.0	46914	46734	-180	-0.4	80680	80500	-180	-0.2
WR	Gujarat	423	422	-0.8	12677	12652	-24	-0.2	24829	24544	-285	-1.1
	Madhya Pradesh	253	251	-1.2	7575	7541	-35	-0.5	14873	14852	-21	-0.1
	Chhattisgarh	106	106	-0.4	3184	3173	-11	-0.3	5836	5832	-4	-0.1
	Maharashtra	526	525	-0.7	15777	15756	-22	-0.1	27678	27678	0	0.0
	Goa	13	13	0.0	403	403	0	0.0	704	704	0	0.0
	DNHDDPDCL	29	29	-0.1	878	876	-2	-0.3	1306	1306	0	0.0
	AMNSIL	18	18	0.0	548	548	0	0.0	895	895	0	0.0
	BALCO*	12	12	0.0	373	373	0	0.0	524	524	0	0.0
	TOTAL	1381	1377	-3.1	41415	41321	-94	-0.2	75027	73948	-1080	-1.4
SR	Andhra Pradesh	218	218	0.0	6550	6550	0	0.0	12343	12327	-16	-0.1
	Telangana	231	231	0.4	6915	6927	12	0.2	15389	15370	-19	-0.1
	Karnataka	239	239	0.4	7168	7180	12	0.2	14698	14698	0	0.0
	Kerala	78	79	0.1	2352	2355	4	0.2	4113	4113	0	0.0
	Tamil Nadu	339	340	0.5	10172	10187	15	0.1	17074	17035	-39	-0.2
	Puducherry	10	10	0.0	293	294	0	0.1	463	462	-1	-0.2
	TOTAL	1115	1116	1.4	33450	33493	43	0.1	60839	60839	0	0.0
ER	Bihar	139	136	-2.3	4164	4093	-70	-1.7	7387	7387	0	0.0
	Jharkhand	37	36	-1.3	1103	1065	-38	-3.5	1818	1818	0	0.0
	DVC	76	76	0.0	2288	2288	0	0.0	3636	3636	0	0.0
	Odisha	119	119	0.0	3556	3556	0	0.0	6070	6070	0	0.0
	West Bengal	200	200	0.0	5985	5985	0	0.0	10667	10667	0	0.0
	Sikkim	1	1	0.0	36	36	0	0.0	98	98	0	0.0
	Railways_ER ISTS**	0	0	0.0	5	5	0	0.0	28	28	0	0.0
	TOTAL	571	568	-3.6	17137	17028	-108	-0.6	28920	28920	0	0.0
NER	Arunachal Pradesh	3	3	0.0	90	90	0	0.0	163	163	0	0.0
	Assam	46	44	-1.2	1368	1332	-35	-2.6	2413	2413	0	0.0
	Manipur	3	3	0.0	80	80	0	0.0	209	209	0	0.0
	Meghalaya	5	5	-0.3	165	157	-8	-4.6	331	331	0	0.0
	Mizoram	2	2	0.0	51	51	0	0.0	129	129	0	0.0
	Nagaland	3	3	0.0	86	86	0	0.0	172	172	0	0.0
	Tripura	6	6	0.0	174	174	0	0.0	344	344	0	0.0
	TOTAL	67	66	-1.4	2013	1970	-43	-2.1	3678	3678	0	0.0
ALL INDIA		4698	4685	-12.7	140929	140546	-382	-0.3				

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

NOTE: BASED ON THE DATA FURNISHED BY ENTITIES ON DAILY BASIS

12. ENERGY COMPARISON OF SEPTEMBER 2023 vs SEPTEMBER 2022

REGION	STATE	Energy Requirement (MU)					Energy Met (MU)				
		Sep-22	Sep-23	Difference	% Change	Average MU/day for Sep-23	Sep-22	Sep-23	Difference	% Change	Average MU/day for Sep-23
NR	Chandigarh	176	194	19	11	6	176	194	19	11	6
	Delhi	3440	3711	271	8	124	3440	3711	271	8	124
	Haryana	6163	6845	682	11	228	6145	6789	643	10	226
	Himachal Pradesh	988	1040	53	5	35	985	1038	53	5	35
	J&K(UT) and Ladakh(UT)	1502	1521	18	1	51	1487	1476	-12	-1	49
	Punjab	7706	8135	429	6	271	7706	8135	429	6	271
	Rajasthan	8743	9339	596	7	311	8709	9285	576	7	309
	Uttar Pradesh	13250	14615	1365	10	487	13128	14603	1475	11	487
	Uttarakhand	1313	1408	95	7	47	1302	1397	95	7	47
	Railway/NFL	82	106	24	29	4	82	106	24	29	4
	TOTAL	43362	46914	3552	8	1564	43160	46734	3574	8	1558
WR	Gujarat	11406	12677	1271	11	423	11406	12652	1247	11	422
	Madhya Pradesh	6558	7575	1017	16	253	6558	7541	982	15	251
	Chhattisgarh	3191	3184	-7	0	106	3190	3173	-17	-1	106
	Maharashtra	13801	15777	1976	14	526	13801	15756	1955	14	525
	Goa	360	403	43	12	13	359	403	43	12	13
	DNHDDPDCL	827	878	51	6	29	827	876	49	6	29
	AMNSIL	430	548	118	27	18	430	548	118	27	18
	BALCO*		373	-	-	12		373	-	-	12
	TOTAL	36573	41415	4842	13	1381	36571	41321	4750	13	1377
SR	Andhra Pradesh	6003	6550	547	9	218	5994	6550	556	9	218
	Telangana	6173	6915	742	12	231	6162	6927	765	12	231
	Karnataka	5418	7168	1751	32	239	5407	7180	1773	33	239
	Kerala	2215	2352	137	6	78	2211	2355	145	7	79
	Tamil Nadu	9596	10172	576	6	339	9581	10187	606	6	340
	Puducherry	273	293	20	7	10	273	294	21	8	10
	TOTAL	29678	33450	3773	13	1115	29628	33493	3865	13	1116
ER	Bihar	3722	4164	441	12	139	3667	4093	427	12	136
	Jharkhand	1017	1103	86	8	37	976	1065	89	9	36
	DVC	2209	2288	80	4	76	2209	2288	80	4	76
	Odisha	4001	3556	-445	-11	119	4001	3556	-445	-11	119
	West Bengal	5471	5985	514	9	200	5467	5985	518	9	200
	Sikkim	45	36	-9	-20	1	45	36	-9	-20	1
	Railways_ER ISTS**		5	-	-	0		5	-	-	0
	TOTAL	16464	17137	673	4	571	16364	17028	664	4	568
NER	Arunachal Pradesh	77	90	13	17	3	77	90	13	17	3
	Assam	1219	1368	149	12	46	1219	1332	114	9	44
	Manipur	82	80	-2	-2	3	81	80	-1	-2	3
	Meghalaya	180	165	-15	-8	5	180	157	-23	-13	5
	Mizoram	48	51	3	6	2	48	51	3	6	2
	Nagaland	83	86	3	4	3	83	86	3	4	3
	Tripura	155	174	19	12	6	155	174	19	12	6
	TOTAL	1843	2013	170	9	67	1842	1970	128	7	66
ALL INDIA		127920	140929	13009	10	4698	127565	140546	12981	10	4685

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

13. DEMAND COMPARISON OF SEPTEMBER 2023 vs SEPTEMBER 2022

REGION	STATE	Peak Requirement (MW)				Peak Met (MW)			
		Sep-22	Sep-23	Difference	% Change	Sep-22	Sep-23	Difference	% Change
NR	Chandigarh	374	388	14	3.7	374	388	14	3.7
	Delhi	6687	6993	306	4.6	6687	6993	306	4.6
	Haryana	12236	12753	517	4.2	12236	12753	517	4.2
	Himachal Pradesh	1748	1758	10	0.6	1748	1758	10	0.6
	J&K(UT) and Ladakh(UT)	3137	2805	-332	-10.6	2967	2618	-349	-11.8
	Punjab	14187	15068	881	6.2	14187	15068	881	6.2
	Rajasthan	15843	17956	2113	13.3	15843	17840	1997	12.6
	Uttar Pradesh	26002	27389	1387	5.3	26002	27389	1387	5.3
	Uttarakhand	2313	2368	55	2.4	2313	2293	-20	-0.9
	Railway/NFL	166	206	40	23.8	166	206	40	23.8
WR	Gujarat	20341	24829	4488	22	20341	24544	4203	20.7
	Madhya Pradesh	10973	14873	3900	36	10973	14852	3879	35.4
	Chhattisgarh	4917	5836	919	19	4915	5832	917	18.7
	Maharashtra	22574	27678	5105	23	22550	27678	5128	22.7
	Goa	647	704	57	9	647	704	57	8.8
	DNHDDPDCL	1233	1306	73	5.9	1233	1306	73	5.9
	AMNSIL	818	895	76	9.3	818	895	76	9.3
	BALCO*		524	-	-		524	-	-
SR	Andhra Pradesh	10920	12343	1423	13.0	10920	12327	1407	12.9
	Telangana	13209	15389	2180	16.5	13191	15370	2179	16.5
	Karnataka	11822	14698	2876	24.3	11818	14698	2880	24.4
	Kerala	3902	4113	211	5.4	3902	4113	211	5.4
	Tamil Nadu	16390	17074	684	4.2	16390	17035	645	3.9
	Puducherry	444	463	19	4.3	444	462	18	4.1
ER	Bihar	7354	7387	33	0.4	6550	7387	837	12.8
	Jharkhand	1774	1818	44	2.5	1630	1818	188	11.5
	DVC	3387	3636	249	7.4	3387	3636	249	7.4
	Odisha	6439	6070	-369	-5.7	6439	6070	-369	-5.7
	West Bengal	9748	10667	919	9.4	9724	10667	943	9.7
	Sikkim	117	98	-19	-16.2	117	98	-19	-16.2
	Railways_ER ISTS**		28	28	-		28	28	-
NER	Arunachal Pradesh	144	163	20	13.6	144	163	20	13.6
	Assam	2342	2413	71	3.0	2308	2413	105	4.6
	Manipur	203	209	6	2.7	203	209	6	2.7
	Meghalaya	354	331	-23	-6.5	354	331	-23	-6.5
	Mizoram	127	129	2	1.3	127	129	2	1.3
	Nagaland	161	172	11	6.9	161	172	11	6.9
	Tripura	333	344	11	3.3	333	344	11	3.3

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

14. SCHEDULE AND DRAWAL OF CONSTITUENTS - SEPTEMBER 2023

REGION	STATE	SCHEDULE (MU)	ACTUAL DRAWAL (MU)	Over drawal(+)/ Under Drawal(-) (MU)	% OD/UD	SCHEDULE (MU/DAY)	ACTUAL DRAWAL (MU/DAY)	Over drawal(+)/ Under Drawal(-) (MU/DAY)
NR	Chandigarh	196.81	197.51	0.69	0.35	6.56	6.58	0.02
	Delhi	3265.64	3209.65	-55.99	-1.71	108.85	106.99	-1.87
	Haryana	5047.09	5028.50	-18.59	-0.37	168.24	167.62	-0.62
	Himachal Pradesh	237.85	245.13	7.28	3.06	7.93	8.17	0.24
	J&K(UT) and Ladakh(UT)	748.75	794.68	45.93	6.13	24.96	26.49	1.53
	Punjab	4828.84	4742.16	-86.68	-1.80	160.96	158.07	-2.89
	Rajasthan	3271.36	3223.69	-47.67	-1.46	109.05	107.46	-1.59
	Uttar Pradesh	6429.75	6392.77	-36.98	-0.58	214.32	213.09	-1.23
	Uttarakhand	713.86	724.88	11.02	1.54	23.80	24.16	0.37
	Railway	101.35	103.83	2.48	2.45	3.38	3.46	0.08
	NFL	2.26	2.29	0.02	1.00	0.08	0.08	0.00
	Total	24843.56	24665.08	-178.48	-0.72	828.12	822.17	-5.95
WR	Gujarat	5327.09	5286.15	-40.94	-0.77	177.57	176.20	-1.36
	Madhya Pradesh	3924.67	3849.18	-75.49	-1.92	130.82	128.31	-2.52
	Chhattisgarh	1495.83	1455.85	-39.98	-2.67	49.86	48.53	-1.33
	Maharashtra	5932.43	5833.28	-99.15	-1.67	197.75	194.44	-3.30
	Goa	391.90	391.96	0.06	0.02	13.06	13.07	0.00
	DNHDDPDCL	879.25	876.13	-3.12	-0.36	29.31	29.20	-0.10
	AMNSIL	225.83	226.47	0.64	0.28	7.53	7.55	0.02
	BALCO*	373.89	372.59	-1.30	-0.35	12.46	12.42	-0.04
	Total	18550.89	18291.61	-259.28	-1.40	618.36	609.72	-8.64
SR	Andhra Pradesh	2536.50	2475.50	-61.00	-2.40	84.55	82.52	-2.03
	Telangana	3326.88	3324.20	-2.68	-0.08	110.90	110.81	-0.09
	Karnataka	2292.50	2267.01	-25.49	-1.11	76.42	75.57	-0.85
	Kerala	1906.66	1936.27	29.61	1.55	63.56	64.54	0.99
	Tamil Nadu	4329.52	4228.41	-101.11	-2.34	144.32	140.95	-3.37
	Puducherry	279.14	272.38	-6.76	-2.42	9.30	9.08	-0.23
	Goa (SR)	64.90	68.06	3.16	4.87	2.16	2.27	0.11
	Total	14736.10	14571.83	-164.27	-1.11	491.20	485.73	-5.48
ER	Bihar	3988.25	3997.66	9.40	0.24	132.94	133.26	0.31
	Jharkhand	867.75	878.29	10.54	1.21	28.93	29.28	0.35
	DVC	-1153.36	-1146.27	7.09	-0.61	-38.45	-38.21	0.24
	Odisha	1394.95	1350.68	-44.28	-3.17	46.50	45.02	-1.48
	West Bengal	2319.66	2246.06	-73.60	-3.17	77.32	74.87	-2.45
	Sikkim	38.56	35.57	-3.00	-7.77	1.29	1.19	-0.10
	Railways_ER ISTS*	4.93	5.03	0.10	2.01	0.16	0.17	0.00
	Total	7460.76	7367.01	-93.74	-1.26	248.69	245.57	-3.12
NER	Arunachal Pradesh	86.75	89.20	2.45	2.82	2.89	2.97	0.08
	Assam	1064.58	1124.19	59.61	5.60	35.49	37.47	1.99
	Manipur	83.73	79.66	-4.07	-4.86	2.79	2.66	-0.14
	Meghalaya	32.06	27.09	-4.97	-15.49	1.07	0.90	-0.17
	Mizoram	41.74	36.66	-5.07	-12.16	1.39	1.22	-0.17
	Nagaland	79.41	74.03	-5.38	-6.78	2.65	2.47	-0.18
	Tripura	173.86	177.03	3.17	1.82	5.80	5.90	0.11
	Total	1562.12	1607.86	45.74	2.93	52.07	53.60	1.52
All India Total		67153.43	66503.40	-650.04	-0.97	2238.45	2216.78	-21.67

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

15. INTER REGIONAL EXCHANGES 2023-24

(All figures in MU)

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Fin. Year 2023-24
Name of Line							
Import of NR from WR (WR-NR)							
WR - NR 765 kV Gwalior - Agra 2xS/C	635.95	996.71	863.68	972.77	772.12	902.82	5144.05
WR-NR 765 kV Gwalior - Phagi 2xS/C	553.05	453.93	342.56	671.61	397.70	687.31	3106.16
WR-NR 765 kV Jabalpur-Orai D/C	430.29	804.53	778.33	964.77	658.94	991.15	4628.01
WR-NR 765 kV Satna-Orai	505.37	598.85	574.83	615.77	543.29	593.66	3431.77
WR-NR 765 kV Gwalior-Orai	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WR - NR HVDC Champa - Kurukshetra	804.26	1126.68	2069.11	2235.18	1955.13	2230.56	10420.92
WR - NR HVDC Vindhyachal	0.49	3.23	3.13	94.78	21.55	7.26	130.44
WR - NR 400 kV Zerda Kankroli	0.00	0.00	1.48	15.36	13.10	8.58	38.52
WR - NR 400 kV Zerda Bhinmal	0.00	0.00	0.00	12.54	13.56	23.81	49.91
WR - NR 400 kV RAPP C - Shujalpur D/C	2.28	31.04	16.05	164.49	80.51	147.53	441.90
WR - NR 220 kV Bhanpura - Ranpur	9.67	65.53	68.13	70.02	74.29	61.03	348.67
WR - NR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	0.00	38.00	38.00
WR - NR 220 kV Auraiya - Malanpur/Mehgaon	0.00	0.15	0.00	0.00	0.00	0.00	0.15
WR - NR 400 kV Vindhyachal-Rihand D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WR - NR HVDC Mundra - Mohindergarh	132.55	340.75	234.27	124.89	667.23	827.57	2327.26
WR-NR 765 kV Banaskata-Chittorgarh D/C	0.00	3.26	16.02	169.31	260.02	66.60	515.21
WR-NR 765 kV Varanasi-Vindhyachal	1030.05	1663.34	1854.30	1854.38	1551.38	1697.01	9650.46
Total WR - NR	4103.96	6088.00	6821.89	7965.87	7008.82	8282.89	40271.43
Export of NR to WR (NR-WR)							
NR - WR 765 kV Gwalior - Agra 2xS/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Gwalior - Phagi 2xS/C	0.00	5.41	0.00	0.00	0.00	0.00	5.41
NR - WR 765 kV Jabalpur-Orai D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Satna-Orai	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Gwalior-Orai	440.87	385.11	301.32	367.88	368.83	385.80	2249.81
NR - WR HVDC Champa- Kurukshetra	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR HVDC Vindhyachal	289.83	273.88	267.04	184.43	225.73	180.86	1421.77
NR - WR 400 kV Zerda Kankroli	185.75	94.43	108.34	24.78	31.17	41.36	485.83
NR - WR 400 kV Zerda Bhinmal	282.42	181.64	200.56	73.76	74.62	64.65	877.65
NR - WR 400 kV RAPP C - Shujalpur D/C	85.74	45.64	45.64	0.00	3.36	0.00	180.38
NR - WR 220 kV Bhanpura - Ranpur	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 220 kV Auraiya - Malanpur/Mehgaon	56.56	28.58	50.82	47.16	97.78	68.03	348.93
NR-WR 400 kV Rihand-Vindhyachal D/C	644.30	672.49	642.32	664.33	653.25	658.85	3935.54
NR - WR HVDC Mundra - Mohindergarh	243.86	54.36	36.25	94.16	0.00	0.00	428.63
NR - WR 765 kV Banaskata-Chittorgarh D/C	976.83	428.19	421.52	94.53	51.63	180.84	2153.54
NR-WR 765 kV Varanasi-Vindhyachal	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total NR - WR	3206.16	2169.73	2073.81	1551.03	1506.37	1580.39	12087.49

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Fin. Year 2023-24
Name of Line							
Import of NR from ER (ER-NR)							
ER - NR 765 kV Sasaram - Fatehpur	166.02	114.36	82.57	74.16	131.29	131.20	699.60
ER - NR 765 kV Gaya - Varanasi 2*S/C	224.56	107.46	28.46	47.20	75.64	115.88	599.20
ER - NR 765 kV Gaya - Balia	246.38	335.46	321.28	300.74	224.31	332.43	1760.60
ER - NR HVDC Alipurduar - Agra	0.00	0.00	375.05	784.99	444.13	750.81	2354.98
ER-NR 400 kV Patna - Balia D/C	219.16	246.32	242.35	242.23	300.66	262.01	1512.73
ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	127.03	223.63	277.35	376.44	415.21	330.20	1749.86
ER - NR 400 kV Biharshariff - Balia D/C	22.11	72.32	102.97	92.77	101.30	90.47	481.94
ER - NR 400 kV Motihari - Gorakhpur D/C	156.88	188.70	180.09	228.59	251.16	190.46	1195.88
ER - NR 400 kV Biharshariff - Varanasi D/C	67.56	47.21	34.66	24.65	44.14	47.88	266.10
ER - NR 400 kV Sasaram -Varanasi	74.99	38.68	46.09	55.64	28.71	57.56	301.67
ER - NR 400 kV Sasaram -Allahabad	51.91	33.39	21.29	28.77	17.25	52.61	205.22
ER - NR 220 kV Sahupuri-Karamnasa	78.64	82.22	82.32	43.45	51.19	29.26	367.08
ER - NR 132 kV Sahupuri - Karamnasa	1.92	0.00	0.96	1.12	0.96	2.02	6.98
ER - NR 132 kV 132 kV Rihand - Nagar Untari	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ER - NR 132 kV Rihand - Garhwa	0.46	0.00	0.00	0.00	0.00	0.00	0.46
ER-NR 400 kV Naubatpur - Balia D/C	211.13	243.59	239.56	236.53	300.12	264.57	1495.50
Total ER-NR	1648.75	1733.34	2035.00	2537.28	2386.07	2657.36	12997.80
Import of NR from NER (NER-NR)							
NER-NR HVDC Biswanath Chariali - Agra	0.00	0.00	153.62	339.28	465.76	447.38	1406.04
Total NER - NR	0.00	0.00	153.62	339.28	465.76	447.38	1406.04
Export of NR to ER (NR-ER)							
NR - ER 765 kV Sasaram - Fatehpur	0.00	2.65	7.74	3.45	0.00	4.62	18.46
NR - ER 765 kV Gaya - Varanasi 2*S/C	0.30	38.74	102.98	45.94	11.70	32.51	232.17
NR - ER 765 kV Gaya - Balia	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER HVDC Alipurduar - Agra	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Patna - Balia 2xD/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	3.47	0.00	0.00	0.00	0.00	0.00	3.47
NR - ER 400 kV Biharshariff - Balia D/C	17.15	4.18	13.06	0.00	0.00	3.12	37.51
ER - NR 400 kV Motihari - Gorakhpur D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Biharshariff - Varanasi D/C	2.90	7.31	20.53	10.39	0.22	12.39	53.74
NR - ER 400 kV Sasaram -Varanasi	0.00	0.00	0.00	0.00	1.67	0.00	1.67
NR - ER 400 kV Sasaram -Allahabad	2.25	0.00	0.00	0.82	2.28	0.00	5.35
NR - ER 220 kV Sahupuri-Karamnasa	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 132 kV Sahupuri - Karamnasa	0.01	0.96	0.48	0.96	0.00	0.58	2.99
NR - ER 132 kV 132 kV Rihand - Nagar Untari	1.74	2.20	1.08	3.37	2.39	3.46	14.24
ER - NR 132 kV Rihand - Garhwa	10.23	8.48	19.67	18.35	14.38	11.19	82.30
NR-ER 400 kV Balia- Naubatpur D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total NR - ER	38.05	64.52	165.54	83.28	32.64	67.87	451.90
Export of NR to NER (NER-NR)							
NR-NER HVDC Biswanath Chariali - Agra	261.93	291.71	104.47	0.00	0.00	0.00	658.11
Total NR - NER	261.93	291.71	104.47	0.00	0.00	0.00	658.11

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Fin. Year 2023-24
Name of Line							
Export of WR to ER (WR-ER)							
WR - ER 765 kV Ranchi - Dharamjaygarh 2xS/C	139.81	504.96	673.15	734.00	331.39	537.29	2920.60
WR - ER 765 kV Dharamjaygarh - Jharsuguda 2xD/C	652.31	469.82	490.99	593.61	768.89	177.22	3152.84
WR - ER 765 kV Durg - Jharsuguda D/C	0.00	0.64	1.75	26.78	7.42	1.32	37.91
WR - ER 400 kV Sipat - Ranchi D/C	15.97	93.84	118.29	133.75	52.57	87.51	501.93
WR - ER 400 kV Raigarh - Jharsuguda 2xD/C	0.44	6.23	8.79	6.26	28.69	11.41	61.83
WR - ER 220 kV Korba - Budhipadar D/C	36.08	80.04	73.19	22.96	28.75	29.18	270.21
WR - ER 220 kV Raigarh - Budhipadar	0.15	0.57	1.17	0.00	0.35	0.34	2.58
Total WR - ER	844.75	1156.11	1367.33	1517.34	1218.07	844.29	6947.89
Import of WR from ER (ER - WR)							
ER - WR 765 kV Ranchi - Dharamjaygarh 2xS/C	144.10	11.95	15.57	10.81	83.69	14.84	280.97
ER - WR 765 kV Dharamjaygarh - Jharsuguda 2XD/C	25.46	58.29	33.08	15.65	2.33	214.83	349.65
ER - WR 765 kV Durg - Jharsuguda D/C	394.08	204.42	148.02	71.84	73.64	163.98	1055.97
ER - WR 400 kV Sipat - Ranchi D/C	55.56	5.71	9.07	7.22	29.65	8.91	116.10
ER - WR 400 kV Raigarh - Jharsuguda 2xD/C	248.78	148.94	125.69	125.00	127.87	110.94	887.22
ER - WR 220 kV Korba - Budhipadar D/C	1.45	0.00	0.14	6.23	1.99	1.46	11.27
ER - WR 220 kV Raigarh - Budhipadar	55.98	38.46	41.41	66.29	74.76	49.37	326.26
Total ER - WR	925.41	467.76	372.98	303.04	393.92	564.32	3027.43
Export of ER to NER (ER - NER)							
ER - NER 400 kV Binaguri - Bongaigaon D/C	6.99	16.15	105.81	142.19	71.19	215.10	557.43
ER - NER 400 kV Alipurduar - Bongaigaon D/C	4.62	34.31	135.94	83.03	84.82	161.59	504.31
ER - NER 220 kV Birpara - Salakati D/C	0.54	3.12	31.19	37.25	25.19	47.94	145.22
Total ER - NER	12.16	53.58	272.93	262.48	181.20	424.63	1206.97
Import of ER from NER (NER - ER)							
NER - ER 400 kV Binaguri - Bongaigaon D/C	52.83	40.19	37.75	2.10	10.62	0.07	143.56
NER - ER 400 kV Alipurduar - Bongaigaon D/C	182.37	135.22	72.81	20.59	29.76	3.08	443.82
NER - ER 220 kV Birpara - Salakati D/C	34.59	31.32	10.78	0.62	2.83	0.14	80.27
Total NER - ER	269.78	206.73	121.33	23.31	43.22	3.29	667.65
Export of ER to SR (ER - SR)							
ER - SR HVDC Gazuwaka	375.93	395.23	208.61	178.44	321.36	201.99	1681.56
ER - SR HVDC Talchar - Kolar	1233.20	1204.28	984.75	977.61	1284.58	1076.42	6760.83
ER - SR 765 kV Angul- Srikakulam D/C	1683.42	1561.33	1412.42	1184.37	1380.27	1160.32	8382.14
Total ER - SR	3292.55	3160.85	2605.77	2340.42	2986.21	2438.74	16824.53
Import of ER from SR (SR - ER)							
SR - ER HVDC Gazuwaka	0.00	0.00	55.34	114.11	32.48	0.00	201.93
SR - ER HVDC Talchar - Kolar	0.00	0.00	0.00	0.00	0.00	0.00	0.01
SR - ER 765 kV Angul- Srikakulam D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total SR - ER	0.00	0.00	55.34	114.11	32.48	0.00	201.94

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Fin. Year 2023-24
Name of Line							
Export of WR to SR (WR-SR)							
WR - SR HVDC Bhadrawati	544.41	411.52	171.68	139.64	464.81	222.80	1954.87
WR - SR 765 kV Sholapur - Raichur 2xS/C	417.69	222.76	146.07	83.93	451.70	125.10	1447.26
WR - SR 765 kV Wardha - Nizamabad D/C	1323.51	873.44	1018.25	852.05	1313.80	864.24	6245.29
WR - SR 400KV Kolhapur-Kudgi D/C	0.00	0.00	0.00	0.00	0.06	0.00	0.06
WR-SR HVDC Raigarh-Pugalur	3004.35	1791.86	1182.91	885.62	2135.69	1450.16	10450.58
WR-SR 220kV Xeldem - Ambewadi S/C	77.16	70.18	68.17	61.74	68.61	67.55	413.40
WR-SR 220kV Ponda - Ambewadi S/C	0.29	1.54	0.28	0.16	5.23	0.04	7.53
Total WR - SR	5367.41	3371.30	2587.37	2023.14	4439.89	2729.89	20518.99
Import of WR from SR (SR - WR)							
SR - WR HVDC Bhadrawati	4.11	25.09	178.09	308.60	10.36	114.86	641.13
SR - WR 765 kV Raichur - Sholapur 2xS/C	29.13	205.27	344.82	478.51	54.56	420.76	1533.04
SR - WR 765 kV Wardha - Nizamabad D/C	0.00	3.90	0.32	18.33	0.00	19.87	42.42
SR - WR 400KV Kolhapur-Kudgi D/C	623.59	719.30	784.98	796.02	623.43	850.54	4397.87
SR-WR HVDC Pugalur-Raigarh	0.00	3.73	14.87	16.65	0.00	0.09	35.33
WR-SR 220kV Ambewadi - Xeldem S/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WR-SR 220kV Ambewadi-Ponda S/C	0.00	0.00	0.01	0.00	0.01	0.01	0.04
Total SR - WR	656.84	957.30	1323.09	1618.10	688.36	1406.13	6649.82
TOTAL ALL INDIA	20628	19721	20060	20679	21383	21447	123918

*In case of mutiple ckt / DC pole year corres. to the commissioning of final element

Date	15.1 Import-Export of NR with WR during September 2023																	
	Import of NR from WR (WR-NR)																	
	WR - NR 765 kV Gwalior - Agra 2x5/C	WR-NR 765 kV Gwalior-Phagi 2x5/C	WR-NR 765 kV Jabalpur- Orai D/C	WR-NR 765 kV Satna- Orai	WR-NR 765 kV Gwalior-Orai	WR - NR HVDC Champa- Kurukshetra	WR - NR HVDC Vindhyachal	WR - NR 400 kV Zerda- Kankroli	WR - NR 400 kV Zerda- Bhinmal	WR-NR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	WR - NR 220 kV Auraiya - Malanpur / Mehgaon	WR-NR 400 KV Rihand Vindhyachal D/C	WR - NR HVDC Mundra - M'garh	WR -NR 765 kV Banaskata - Chittorgarh D/c	WR-NR 765 kV Varanasi Vindhyachal	Total WR - NR
1-Sep-23	18.05	21.44	20.49	15.84	0.00	74.24	0.00	0.00	0.00	1.27	2.19		0.00	0.00	33.47	0.00	50.00	236.99
2-Sep-23	22.95	22.13	27.30	17.81	0.00	82.97	0.00	0.00	0.00	3.23	2.64		0.00	0.00	22.05	0.00	56.14	257.22
3-Sep-23	28.88	24.41	38.22	19.15	0.00	91.05	0.00	0.00	0.00	5.28	2.66		0.00	0.00	15.16	0.00	65.42	290.23
4-Sep-23	38.86	33.02	48.57	22.49	0.00	78.69	0.72	0.00	0.00	8.73	2.65		0.00	0.00	36.25	0.00	62.16	332.14
5-Sep-23	38.87	36.96	47.70	22.82	0.00	90.89	0.69	0.00	0.00	9.40	2.53		0.00	0.00	36.55	0.00	62.58	348.99
6-Sep-23	41.60	38.30	50.12	23.72	0.00	85.92	0.00	0.00	0.00	8.43	2.25		0.00	0.00	36.26	0.00	60.48	347.08
7-Sep-23	44.84	33.47	49.17	22.95	0.00	87.91	0.00	2.83	6.23	9.50	2.00	0.26	0.00	0.00	36.26	17.62	65.66	378.70
8-Sep-23	41.64	31.58	46.01	23.73	0.00	90.86	0.06	3.84	9.41	9.24	2.31	1.48	0.00	0.00	25.59	23.12	62.90	371.77
9-Sep-23	32.31	29.11	35.86	22.10	0.00	93.93	0.03	1.41	5.07	6.73	2.30	1.63	0.00	0.00	32.54	9.67	63.56	336.25
10-Sep-23	27.88	21.20	28.09	16.97	0.00	81.00	1.58	0.50	3.10	5.06	2.23	1.58	0.00	0.00	28.02	7.68	66.00	290.89
11-Sep-23	28.32	23.60	28.19	20.02	0.00	86.01	0.00	0.00	0.00	5.12	2.06	1.48	0.00	0.00	36.25	0.00	60.77	291.82
12-Sep-23	29.28	24.42	31.59	21.16	0.00	86.31	0.48	0.00	0.00	5.08	1.41	1.70	0.00	0.00	36.25	0.00	59.19	296.87
13-Sep-23	27.79	28.13	34.58	21.74	0.00	86.08	1.46	0.00	0.00	6.25	1.24	1.49	0.00	0.00	36.25	0.00	63.25	308.26
14-Sep-23	31.04	35.99	40.81	22.45	0.00	81.50	2.24	0.00	0.00	8.89	0.74	2.14	0.00	0.00	33.18	0.00	59.52	318.50
15-Sep-23	23.96	28.70	29.53	20.93	0.00	75.88	0.00	0.00	0.00	7.18	0.64	2.66	0.00	0.00	32.25	0.00	60.03	281.76
16-Sep-23	29.06	22.01	32.29	18.81	0.00	61.05	0.00	0.00	0.00	7.37	1.88	1.31	0.00	0.00	24.19	0.00	58.60	256.57
17-Sep-23	25.79	6.02	24.62	14.39	0.00	46.43	0.00	0.00	0.00	2.59	1.74	1.46	0.00	0.00	24.20	0.01	62.67	209.92
18-Sep-23	28.76	12.50	23.30	15.46	0.00	61.80	0.00	0.00	0.00	2.78	1.55	1.31	0.00	0.00	25.00	1.35	60.68	234.49
19-Sep-23	33.35	15.62	28.46	16.60	0.00	65.23	0.00	0.00	0.00	3.41	1.78	1.54	0.00	0.00	20.52	4.16	60.52	251.19
20-Sep-23	26.42	16.12	29.58	19.03	0.00	55.14	0.00	0.00	0.00	2.61	2.09	1.60	0.00	0.00	24.20	2.99	57.76	237.54
21-Sep-23	26.71	20.27	30.54	19.84	0.00	58.55	0.00	0.00	0.00	3.19	2.21	1.30	0.00	0.00	24.21	0.00	46.51	233.33
22-Sep-23	27.27	22.38	31.83	21.51	0.00	57.76	0.00	0.00	0.00	4.04	2.41	1.66	0.00	0.00	24.21	0.00	39.85	232.92
23-Sep-23	20.72	16.41	23.46	18.75	0.00	57.90	0.00	0.00	0.00	1.43	2.30	1.78	0.00	0.00	24.21	0.00	37.31	204.27
24-Sep-23	22.09	12.84	22.05	16.97	0.00	55.26	0.00	0.00	0.00	1.13	1.91	1.65	0.00	0.00	24.22	0.00	42.71	200.83
25-Sep-23	26.37	14.29	22.52	17.91	0.00	68.71	0.00	0.00	0.00	1.38	1.90	1.56	0.00	0.00	23.80	0.00	45.45	223.89
26-Sep-23	29.04	20.45	31.92	20.00	0.00	69.30	0.00	0.00	0.00	3.91	2.22	1.70	0.00	0.00	34.67	0.00	44.85	258.06
27-Sep-23	33.82	20.61	36.19	18.97	0.00	71.22	0.00	0.00	0.00	4.80	2.24	1.64	0.00	0.00	19.64	0.00	50.88	260.01
28-Sep-23	32.94	17.61	36.50	19.28	0.00	56.33	0.00	0.00	0.00	3.80	2.33	1.73	0.00	0.00	19.39	0.00	54.85	244.76
29-Sep-23	37.95	19.12	38.96	20.70	0.00	52.84	0.00	0.00	0.00	3.96	2.28	1.64	0.00	0.00	19.39	0.00	66.24	263.08
30-Sep-23	26.26	18.60	22.70	21.56	0.00	119.80	0.00	0.00	0.00	1.74	2.34	1.70	0.00	0.00	19.39	0.00	50.47	284.56
Total	902.82	687.31	991.15	593.66	0.00	2230.56	7.26	8.58	23.81	147.53	61.03	38.00	0.00	0.00	827.57	66.60	1697.01	8282.89
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																		

Date	15.1 Import-Export of NR with WR during September 2023																	
	Export of NR to WR (NR- WR)																	
	NR - WR 765 kV Gwalior - Agra 2x5/C	NR-WR 765 kV Gwalior-Phagi 2x5/C	NR-WR 765 kV Jabalpur- Orai D/C	NR-WR 765 kV Satna- Orai	NR-WR 765 kV Gwalior-Orai	NR - WR HVDC Champa- Kurukshetra	NR - WR HVDC Vindhyachal	NR - WR 400 kV Zerda- Kankroli	NR - WR 400 kV Zerda- Bhimnal	NR-WR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	NR - WR 220 kV Auraiya - Malanpur / Mehgaon	NR-WR 400 kV Rihand- Vindhyachal D/C	NR - WR HVDC Mundra - M'garh	NR - WR 765 kV Banaskata - Chittorgarh D/C	NR-WR 765 kV Varanasi- Vindhyachal	Total NR - WR
1-Sep-23	0.00	0.00	0.00	0.00	16.85	0.00	4.85	2.86	4.19	0.00	0.00		6.04	22.48	0.00	15.73	0.00	73.00
2-Sep-23	0.00	0.00	0.00	0.00	14.95	0.00	6.07	2.19	4.76	0.00	0.00		4.98	22.14	0.00	10.21	0.00	65.30
3-Sep-23	0.00	0.00	0.00	0.00	14.40	0.00	4.84	1.68	4.07	0.00	0.00		3.00	22.14	0.00	6.63	0.00	56.76
4-Sep-23	0.00	0.00	0.00	0.00	15.18	0.00	2.85	0.44	1.02	0.00	0.00		2.54	21.78	0.00	0.88	0.00	44.69
5-Sep-23	0.00	0.00	0.00	0.00	16.18	0.00	3.92	1.30	0.14	0.00	0.00		2.35	22.13	0.00	8.32	0.00	54.34
6-Sep-23	0.00	0.00	0.00	0.00	16.12	0.00	1.82	1.44	0.21	0.00	0.00		1.89	22.04	0.00	7.18	0.00	50.70
7-Sep-23	0.00	0.00	0.00	0.00	13.70	0.00	2.43	0.00	0.00	0.00	0.00	0.00	0.90	22.22	0.00	0.00	0.00	39.25
8-Sep-23	0.00	0.00	0.00	0.00	12.77	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.52	22.09	0.00	0.00	0.00	35.38
9-Sep-23	0.00	0.00	0.00	0.00	12.64	0.00	5.95	0.00	0.00	0.00	0.00	0.00	1.25	22.11	0.00	0.00	0.00	41.95
10-Sep-23	0.00	0.00	0.00	0.00	13.89	0.00	4.01	0.00	0.00	0.00	0.00	0.00	1.60	22.11	0.00	0.00	0.00	41.61
11-Sep-23	0.00	0.00	0.00	0.00	13.84	0.00	12.13	1.28	0.77	0.00	0.00	0.00	1.61	22.11	0.00	0.59	0.00	52.33
12-Sep-23	0.00	0.00	0.00	0.00	12.51	0.00	4.28	2.10	2.31	0.00	0.00	0.00	2.20	22.24	0.00	5.50	0.00	51.14
13-Sep-23	0.00	0.00	0.00	0.00	13.45	0.00	2.54	2.88	3.96	0.00	0.00	0.00	2.18	22.28	0.00	10.36	0.00	57.65
14-Sep-23	0.00	0.00	0.00	0.00	15.48	0.00	5.06	2.62	3.15	0.00	0.00	0.00	1.79	22.37	0.00	12.11	0.00	62.58
15-Sep-23	0.00	0.00	0.00	0.00	14.15	0.00	2.00	3.78	3.96	0.00	0.00	0.00	2.27	22.13	0.00	18.37	0.00	66.66
16-Sep-23	0.00	0.00	0.00	0.00	11.13	0.00	7.16	3.11	3.50	0.00	0.00	0.00	1.30	22.42	0.00	12.45	0.00	61.07
17-Sep-23	0.00	0.00	0.00	0.00	7.70	0.00	11.29	1.37	3.46	0.00	0.00	0.00	1.16	19.07	0.00	0.00	0.00	44.05
18-Sep-23	0.00	0.00	0.00	0.00	12.22	0.00	5.77	1.05	3.15	0.00	0.00	0.00	1.74	22.10	0.00	0.00	0.00	46.03
19-Sep-23	0.00	0.00	0.00	0.00	12.96	0.00	6.06	0.57	1.07	0.00	0.00	0.00	1.87	22.10	0.00	0.00	0.00	44.63
20-Sep-23	0.00	0.00	0.00	0.00	10.32	0.00	6.09	0.15	1.22	0.00	0.00	0.00	1.94	21.98	0.00	0.00	0.00	41.70</

Date	15.2 Import-Export of NR with ER & NER during September 2023																		
	Import of NR from ER (ER-NR)																	Import of NR from NER (NER-NR)	
	ER - NR 765 kV Sasaram - Fatehpur	ER - NR 765 kV Gaya - Varanasi 2*/C	ER - NR 765 kV Gaya Balia	ER - NR HVDC Alipurduar - Agra	ER-NR 400 kV Patna - Balia D/C	ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Balia D/C	ER - NR 400 kV Motihari - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Varanasi D/C	ER-NR 400 kV Sasaram - Varanasi	ER-NR 400 kV Sasaram - Allahabad	ER - NR 220 kV Sahupuri-Karamnasa	ER - NR 132 kV Sahupuri - Karamnasa	ER - NR 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	ER-NR 400 kV Naubatpur - Balia D/C	Total ER -NR	NER-NR HVDC Biswanath Chariali - Agra	Total NER-NR
1-Sep-23	5.12	1.39	8.78	17.95	8.78	10.61	1.34	6.80	0.47	2.14	0.17	1.87		0.00		9.03	74.45	12.34	12.34
2-Sep-23	5.15	1.54	9.55	18.67	8.63	12.76	3.96	6.96	1.05	1.88	0.37	2.01	0.00	0.00		8.55	81.08	12.83	12.83
3-Sep-23	3.77	1.26	13.21	19.38	8.36	10.88	3.45	5.36	1.19	1.68	0.65	1.87		0.00	0.00	8.27	79.33	13.24	13.24
4-Sep-23	8.37	8.98	14.19	19.37	10.90	14.57	5.50	8.37	3.27	0.48	1.86	0.58			0.00	11.23	107.67	13.25	13.25
5-Sep-23	9.20	10.98	15.20	19.22	12.26	15.94	6.99	8.78	5.61	0.41	1.91				0.00	12.72	119.22	11.96	11.96
6-Sep-23	8.94	10.71	15.15	19.49	12.49	15.60	5.44	9.21	5.22	0.44	1.71				0.00	13.03	117.43	13.86	13.86
7-Sep-23	7.52	6.19	15.18	17.66	11.03	15.47	4.70	8.06	2.60	0.74	1.63		0.00		0.00	11.37	102.15	14.45	14.45
8-Sep-23	6.79	7.35	13.72	37.48	11.30	13.30	4.84	8.11	2.02	0.63	1.87				0.00	11.58	118.99	14.45	14.45
9-Sep-23	2.70	2.50	9.29	45.42	8.90	8.51	2.15	5.77	0.85	3.40	3.45			0.00	0.00	9.39	102.33	14.30	14.30
10-Sep-23	0.00	0.00	10.01	35.80	7.76	6.19	0.95	4.11	0.00	4.20	2.90		0.48		0.00	7.63	80.03	16.33	16.33
11-Sep-23	1.05	0.00	10.25	26.82	8.87	7.03	0.47	5.17	0.00	3.77	3.16		0.00		0.00	8.71	75.30	16.83	16.83
12-Sep-23	2.89	0.37	10.45	24.74	7.66	9.44	2.10	6.03	0.00	3.82	3.27	0.00	0.00		0.00	9.48	80.25	17.17	17.17
13-Sep-23	4.56	1.86	12.51	24.59	8.39	9.89	3.63	5.43	0.50	2.65	2.62			0.00		8.23	84.86	16.93	16.93
14-Sep-23	5.98	3.78	13.84	23.19	8.14	9.53	3.64	4.67	1.26	1.82	1.63			0.00		8.06	85.54	16.81	16.81
15-Sep-23	4.29	1.47	11.92	25.30	8.50	8.21	2.02	4.76	0.26	1.01	2.51	0.52		0.00		8.39	79.16	16.97	16.97
16-Sep-23	2.41	0.00	11.82	24.14	8.53	6.19	0.26	4.89	0.00	2.21	1.30	1.19		0.00	0.00	8.39	71.33	16.89	16.89
17-Sep-23	0.00	0.00	8.66	24.00	5.97	2.33	0.00	2.55	0.00	2.54	0.85	0.90			0.00	5.56	53.36	16.07	16.07
18-Sep-23	0.00	0.00	9.47	23.15	6.05	2.80	0.00	2.70	0.00	2.47	1.01		1.06		0.00	5.64	54.35	16.20	16.20
19-Sep-23	1.32	0.00	9.50	25.15	5.66	6.80	0.00	4.12	0.00	2.21	1.23	1.23			0.00	5.24	62.46	17.06	17.06
20-Sep-23	1.97	0.00	9.65	24.60	5.91	8.42	0.46	4.94	0.00	2.27	1.23	1.10	0.48		0.00	5.48	66.51	16.86	16.86
21-Sep-23	6.70	5.86	10.14	24.54	9.92	15.48	5.31	7.84	2.51	1.83	2.07	1.91			0.00	9.98	104.09	17.01	17.01
22-Sep-23	9.08	12.66	11.07	24.58	12.70	18.43	6.74	10.55	5.58	1.42	2.09	2.21			0.00	13.24	130.35	16.78	16.78
23-Sep-23	5.55	9.40	6.60	23.60	12.45	18.44	5.91	10.77	2.15	1.65	1.86	1.66	0.00		0.00	12.87	112.91	16.90	16.90
24-Sep-23	3.84	5.41	7.22	24.04	10.45	16.73	5.00	9.33	2.85	1.48	1.99	1.69	0.00		0.00	10.20	100.23	17.15	17.15
25-Sep-23	3.40	3.42	8.25	24.50	7.19	14.24	3.69	7.73	1.93	1.78	1.80	1.30			0.00	8.93	88.16	13.67	13.67
26-Sep-23	5.61	7.47	10.72	23.99	9.59	13.59	4.00	7.51	2.66	1.59	1.83	1.91	0.00		0.00	9.35	99.82	11.58	11.58
27-Sep-23	4.33	4.64	10.75	22.94	8.21	11.72	3.37	6.37	1.62	1.74	1.69	1.89		0.00	0.00	8.08	87.35	12.21	12.21
28-Sep-23	4.43	2.79	11.77	26.80	5.98	8.99	1.90	4.50	1.30	1.87	1.55	2.08			0.00	5.53	79.49	13.13	13.13
29-Sep-23	1.94	1.16	12.33	29.13	5.35	8.94	0.53	4.21	0.99	2.02	1.40	1.72			0.00	4.80	74.52	12.11	12.11
30-Sep-23	4.29	4.69	11.23	30.57	6.08	9.17	2.12	4.86	1.99	1.41	1.00	1.62			0.00	5.61	84.64	12.04	12.04
Total	131.20	115.88	332.43	750.81	262.01	330.20	90.47	190.46	47.88	57.56	52.61	29.26	2.02	0.00	0.00	264.57	2657.36	447.38	447.38
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																			

[illegible]

Date	15.3 Import and Export of ER with WR & NER during September 2023																							
	Export of WR to ER (WR-ER)								Import of WR from ER (ER - WR)								Export of ER to NER (ER-NER)				Import of ER from NER (NER-ER)			
	WR - ER 765 kV Dharamjaygarh - Ranchi 2x5/C	WR - ER 765 kV Dharamjaygarh - Jharsuguda Q/C	WR - ER 765 kV Durg - Jharsuguda D/C	WR - ER 400 kV Sipat - Ranchi D/C	WR - ER 400 kV Raigarh - Jharsuguda- 2xD/C	WR - ER 220 kV Korba - Budhipadar D/C	WR - ER 220 kV Raigarh - Budhipadar	Total WR - ER	ER - WR 765 kV Dharamjaygarh - Ranchi 2x5/C	ER-WR 765 kV Dharamjaygarh - Jharsuguda D/C	ER-WR 765 kV Durg - Jharsuguda D/C	ER - WR 400 kV Sipat - Ranchi D/C	ER - WR 400 kV Raigarh - Jharsuguda- 2xD/C	ER - WR 220 kV Korba - Budhipada r D/C	ER - WR 220 kV Raigarh - Budhipadar	Total ER - WR	ER - NER 400 kV Binaguri - Bongaigaon D/C	ER - NER 400 kV Alipurduar - Bongaigaon D/C	ER - NER 220 kV Birpara - Salakati D/C	Total ER - NER	NER - ER 400 kV Binaguri - Bongaigaon D/C	NER - ER 400 kV Alipurduar - Bongaigaon 2xD/C	NER - ER 220 kV Birpara - Salakati D/C	Total NER - ER
1-Sep-23	12.8	21.5	0.2	0.7	0.0	0.0	0.0	35.15	0.0	0.0	1.5	0.2	7.2	0.4	3.2	12.50	1.22	0.35	0.37	1.94	0.05	1.11	0.03	1.19
2-Sep-23	16.9	22.1	0.2	1.7	0.0	0.0	0.0	41.01	0.0	0.0	1.2	0.1	6.0	0.2	3.2	10.73	2.43	1.87	0.46	4.76	0.02	0.53	0.11	0.65
3-Sep-23	24.5	5.1	0.0	3.1	0.0	0.0	0.0	32.71	0.0	4.1	2.4	0.0	8.0	0.5	2.7	17.63	3.60	2.97	1.05	7.63	0.00	0.01	0.00	0.01
4-Sep-23	19.6	1.1	0.0	2.6	0.0	0.2	0.0	23.38	0.0	11.0	3.6	0.3	6.9	0.1	1.9	23.74	5.33	5.32	1.46	12.10	0.00	0.01	0.00	0.01
5-Sep-23	18.0	1.5	0.0	2.0	0.0	0.3	0.0	21.85	0.0	17.5	4.9	0.2	6.3	0.0	2.0	31.02	4.21	3.56	1.10	8.87	0.00	0.00	0.00	0.00
6-Sep-23	20.2	0.1	0.0	2.5	0.0	0.3	0.0	23.11	0.0	23.6	6.5	0.2	5.2	0.0	2.0	37.56	6.28	6.19	1.63	14.11	0.00	0.00	0.00	0.00
7-Sep-23	26.1	0.5	0.0	4.4	0.3	0.8	0.0	32.07	0.0	17.4	3.9	0.0	2.7	0.0	1.4	25.49	5.60	4.90	1.37	11.87	0.00	0.05	0.00	0.05
8-Sep-23	21.9	0.8	0.0	3.5	0.3	1.0	0.0	27.41	0.0	21.6	5.2	0.1	2.9	0.0	1.1	30.96	7.64	2.84	1.32	11.80	0.00	0.22	0.00	0.22
9-Sep-23	19.9	1.8	0.0	3.4	0.4	1.2	0.1	26.78	0.0	10.3	4.1	0.2	2.5	0.0	0.8	17.79	10.46	3.88	1.76	16.11	0.00	0.39	0.00	0.39
10-Sep-23	25.1	5.1	0.4	5.0	0.8	1.2	0.1	37.65	0.0	4.5	2.7	0.1	1.7	0.0	0.8	9.78	9.71	5.30	1.91	16.92	0.00	0.01	0.00	0.01
11-Sep-23	22.9	6.0	0.0	3.4	0.1	0.4	0.0	32.64	0.0	3.9	4.1	0.3	4.0	0.0	1.6	13.90	8.08	6.59	1.88	16.55	0.00	0.00	0.00	0.00
12-Sep-23	20.2	4.0	0.0	2.5	0.0	0.2	0.0	26.90	0.3	7.5	6.4	0.5	5.6	0.1	1.9	22.36	8.02	7.24	1.97	17.23	0.00	0.00	0.00	0.00
13-Sep-23	18.9	2.2	0.0	2.3	0.0	0.8	0.0	24.24	0.1	11.7	7.3	0.4	4.7	0.0	1.3	25.60	7.17	5.81	1.64	14.63	0.00	0.00	0.00	0.00
14-Sep-23	16.7	0.9	0.0	2.4	0.1	1.1	0.0	21.13	0.1	18.3	7.8	0.4	2.6	0.0	0.9	30.04	8.21	6.82	1.81	16.83	0.00	0.00	0.00	0.00
15-Sep-23	16.1	2.5	0.0	2.4	0.0	1.4	0.0	22.44	0.4	8.4	7.1	0.4	3.3	0.0	0.9	20.42	8.65	7.40	1.95	18.01	0.00	0.00	0.00	0.00
16-Sep-23	21.8	1.3	0.0	3.5	0.0	1.4	0.0	27.95	0.1	5.3	6.1	0.2	4.3	0.0	1.4	17.45	10.11	9.42	2.33	21.86	0.00	0.00	0.00	0.00
17-Sep-23	28.2	12.1	0.5	6.1	2.5	2.1	0.1	51.57	0.0	1.4	3.3	0.1	1.9	0.0	1.0	7.68	10.42	9.75	2.37	22.54	0.00	0.00	0.00	0.00
18-Sep-23	25.1	11.0	0.0	5.4	1.1	1.9	0.0	44.48	0.0	2.3	3.9	0.0	1.6	0.0	1.3	9.07	8.55	7.05	1.84	17.44	0.00	0.00	0.00	0.00
19-Sep-23	22.7	11.1	0.0	4.3	0.1	1.4	0.0	39.48	0.1	0.3	4.8	0.1	3.5	0.0	1.4	10.18	8.36	6.10	1.63	16.09	0.00	0.00	0.00	0.00
20-Sep-23	15.7	6.5	0.0	3.3	0.2	1.3	0.0	27.00	0.2	2.4	8.9	0.1	4.1	0.0	1.7	17.37	7.98	5.66	1.48	15.12	0.00	0.00	0.00	0.00
21-Sep-23	10.0	6.1	0.0	1.7	0.1	1.2	0.0	19.03	1.4	0.9	13.9	0.6	4.5	0.0	1.5	22.76	5.34	3.26	1.12	9.71	0.00	0.00	0.00	0.00
22-Sep-23	4.6	5.7	0.0	0.7	0.1	1.1	0.0	12.26	3.1	2.1	14.2	0.9	4.4	0.0	2.0	26.79	5.01	3.26	1.17	9.44	0.00	0.70	0.00	0.71
23-Sep-23	3.2	5.8	0.0	0.9	0.5	1.3	0.0	11.67	3.3	2.9	8.0	1.0	3.0	0.0	1.8	20.02	6.08	4.57	1.61	12.26	0.00	0.00	0.00	0.00
24-Sep-23	8.4	7.1	0.1	1.9	0.4	1.3	0.0	19.34	1.8	2.0	4.9	0.5	1.9	0.0	1.7	12.79	6.11	5.18	1.71	13.00	0.00	0.00	0.00	0.00
25-Sep-23	11.3	5.5	0.0	2.7	0.8	1.4	0.0	21.70	1.0	2.9	4.8	0.3	1.9	0.0	1.3	12.22	6.32	5.02	1.68	13.01	0.00	0.00	0.00	0.00
26-Sep-23	11.5	5.6	0.0	1.7	0.4	1.0	0.0	20.14	1.4	2.9	6.0	0.7	3.7	0.0	2.0	16.78	6.99	4.87	1.75	13.61	0.00	0.04	0.00	0.04
27-Sep-23	17.1	9.3	0.0	2.9	0.6	1.0	0.0	30.86	0.6	4.4	4.1	0.4	2.7	0.0	1.9	14.15	7.66	5.52	1.75	14.94	0.00	0.00	0.00	0.00
28-Sep-23	21.6	5.9	0.0	3.9	0.5	1.3	0.0	33.19	0.0	6.5	3.3	0.1	1.2	0.0	1.3	12.43	8.91	6.93	1.89	17.73	0.00	0.00	0.00	0.00
29-Sep-23	24.5	7.3	0.0	4.4	0.6	1.3	0.0	38.08	0.0	4.7	3.4	0.1	1.3	0.0	1.9	11.36	10.17	7.00	1.95	19.12	0.00	0.00	0.00	0.00
30-Sep-23	11.8	2.0	0.0	2.4	1.7	1.2	0.0	19.07	0.9	14.1	5.6	0.5	1.2	0.0	1.5	23.77	10.47	6.97	1.97	19.41	0.00	0.00	0.00	0.00
Total	537.29	177.22	1.32	87.51	11.41	29.18	0.34	844.29	14.84	214.83	163.98	8.91	110.94	1.46	49.37	564.32	215.10	161.59	47.94	424.63	0.07	3.08	0.14	3.29

Date	15.4 Import and Export of SR with ER & WR during September 2023																							
	Export of ER to SR (ER - SR)				Import of ER from SR (SR - ER)				Export of WR to SR (WR-SR)							Import of WR from SR (SR - WR)								
	ER - SR HVDC Gazuwaka	ER - SR HVDC Talchar -Kolar	ER-SR 765 kV Angul- Srikakulam D/C	Total ER - SR	SR - ER HVDC Gazuwaka	SR - ER HVDC Talchar -	SR-ER 765 kV Angul- Srikakulam D/C	Total SR - ER	WR - SR HVDC Bhadravati	WR - SR 765 kV Sholapur- Raichur 2x5/C	WR-SR 765 kV Wardha - Nizamabad D/C	WR-SR 400KV Kolhapur-Kudgi D/C	WR-SR HVDC Raigarh- Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	Total WR - SR	SR - WR HVDC Bhadravati	SR - WR 765 kV Raichur - Sholapur 2x5/C	SR-WR 765 kV Wardha - Nizamabad D/C	SR-WR 400KV Kolhapur- Kudgi D/C	SR-WR HVDC Raigarh- Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	Total SR - WR
1-Sep-23	8.75	44.06	39.53	92.34	0.00	0.00	0.00	0.00	20.86	8.97	35.15	0.00	47.87	2.46	0.00	115.31	0.00	2.76	0.00	27.67	0.00	0.00	0.00	30.43
2-Sep-23	8.79	36.74	41.62	87.15	0.00	0.00	0.00	0.00	10.47	6.75	36.58	0.00	34.68	2.43	0.00	90.92	0.00	3.60	0.00	24.01	0.00	0.00	0.00	27.61
3-Sep-23	8.79	34.28	28.80	71.87	0.00	0.00	0.00	0.00	7.27	1.53	14.21	0.00	15.34	1.23	0.00	39.59	0.00	24.68	1.25	31.43	0.00	0.00	0.00	57.37
4-Sep-23	5.15	25.40	28.21	58.75	0.00	0.00	0.00	0.00	3.48	0.00	8.85	0.00	22.03	2.29	0.00	36.65	3.68	34.76	4.00	34.83	0.00	0.00	0.00	77.27
5-Sep-23	3.59	27.27	24.19	55.04	0.00	0.00	0.00	0.00	0.00	0.00	4.82	0.00	19.16	2.36	0.00	26.34	8.38	37.07	6.90	35.23	0.00	0.00	0.00	87.58
6-Sep-23	7.54	26.13	26.44	60.10	0.00	0.00	0.00	0.00	0.00	0.00	7.61	0.00	19.68	2.37	0.00	29.66	21.62	39.01	4.74	38.26	0.09	0.00	0.00	103.72
7-Sep-23	7.53	30.09	25.96	63.59	0.00	0.00	0.00	0.00	0.00	0.16	9.76	0.00	18.30	2.32	0.00	30.53	14.60	31.50	0.46	34.92	0.00	0.00	0.00	81.49
8-Sep-23	7.48	26.09	29.86	63.43	0.00	0.00	0.00	0.00	0.00	1.11	14.34	0.00	34.38	2.41	0.00	52.24	11.58	28.38	0.57	33.43	0.00	0.00	0.00	73.96
9-Sep-23	7.47	30.09	29.65	67.21	0.00	0.00	0.00	0.00	0.00	0.60	17.18	0.00	24.78	2.33	0.00	44.89	7.16	24.09	0.00	34.55	0.00	0.00	0.00	65.80
10-Sep-23	5.61	30.09	28.94	64.63	0.00	0.00	0.00	0.00	0.00	0.90	18.56	0.00	24.18	2.30	0.00	45.95	7.99	22.91	0.00	33.02	0.00	0.00	0.00	63.92
11-Sep-23	2.69	30.08	41.43	74.19	0.00	0.00	0.00	0.00	0.00	4.16	30.54	0.00	24.89	2.26	0.00	61.85	7.16	9.81	0.00	27.42	0.00	0.00	0.00	44.39
12-Sep-23	2.71	31.28	44.19	78.17	0.00	0.00	0.00	0.00	0.00	3.22	34.94	0.00	33.07	2.36	0.00	73.60	7.17	8.30	0.00	31.13	0.00	0.00	0.00	46.59
13-Sep-23	2.74	30.09	46.29	79.11	0.00	0.00	0.00	0.00	0.00	3.64	38.67	0.00	32.14	2.28	0.00	76.73	7.17	6.17	0.00	30.45	0.00	0.00	0.00	43.80
14-Sep-23	2.72	33.76	38.05	74.52	0.00	0.00	0.00	0.00	0.00	2.09	28.65	0.00	49.33	2.34	0.00	82.41	7.21	18.14	0.00	35.83	0.00	0.00	0.00	61.18
15-Sep-23	2.67	43.70	44.57	90.94	0.00	0.00	0.00	0.00	0.00	2.52	35.83	0.00	61.97	2.40	0.00	102.72	7.27	8.83	0.00	30.45	0.00	0.00	0.00	46.56
16-Sep-23	2.67	34.13	46.05	82.85	0.00	0.00	0.00	0.00	9.16	4.55	38.68	0.00	66.64	2.38	0.00	121.43	1.66	10.80	0.00	28.13	0.00	0.00	0.00	40.59
17-Sep-23	2.67	33.94	41.19	77.80	0.00	0.00	0.00	0.00	11.19	7.62	37.98	0.00	70.77	2.19	0.00	129.75	0.34	10.48	0.00	24.91	0.00	0.00	0.00	35.73
18-Sep-23	10.62	37.49	42.30	90.41	0.00	0.00	0.00	0.00	10.33	9.01	41.17	0.00	74.03	2.18	0.00	136.73	1.87	9.05	0.00	24.34	0.00	0.00	0.00	35.26
19-Sep-23	12.50	36.45	46.15	95.09	0.00	0.00	0.00	0.00	14.84	10.04	47.17	0.00	60.88	2.16	0.00	135.11	0.00	2.15	0.00	20.31	0.00	0.00	0.00	22.46
20-Sep-23	12.55	40.92	47.05	100.53	0.00	0.00	0.00	0.00	15.91	8.46	50.20	0.00	94.98	2.11	0.00	171.66	0.00	4.52	0.00	22.94	0.00	0.00	0.00	27.46
21-Sep-23	9.51	45.58	50.95	106.04	0.00	0.00	0.00	0.00	10.24	10.35	49.56	0.00	85.58	2.17	0.00	157.90	0.00	1.20	0.00	20.72	0.00	0.00	0.00	21.92
22-Sep-23	7.50	46.72	52.75	106.96	0.00	0.00	0.00	0.00	14.19	13.28	46.92	0.00	94.46	2.19	0.00	171.04	0.00	0.59	0.00	22.63	0.00	0.00	0.00	23.22
23-Sep-23	7.49	44.47	45.06	97.03	0.00	0.00	0.00	0.00	18.80	3.72	30.42	0.00	97.84	2.22	0.00	153.00	0.00	6.13	0.00	24.04	0.00	0.00	0.00	30.17
24-Sep-23	7.49	39.72	38.95	86.16	0.00	0.00	0.00	0.00	12.07	1.16	24.65	0.00	65.07	2.18	0.00	105.12	0.00	13.92	0.00	25.68	0.00	0.00	0.00	39.60
25-Sep-23	7.50	42.54	40.90	90.94	0.00	0.00	0.00	0.00	12.07	2.18	28.57	0.00	68.82	2.14	0.00	113.79	0.00	10.03	0.00	25.23	0.00	0.00	0.00	35.26
26-Sep-23	7.50	37.94	44.54	89.97	0.00	0.00	0.00	0.00	12.60	2.67	32.14	0.00	40.42	2.29	0.00	90.12	0.00	7.54	0.00	24.82	0.00	0.00	0.00	32.36
27-Sep-23	7.45	39.70	36.73	83.88	0.00	0.00	0.00	0.00	11.88	3.25	26.59	0.00	52.13	2.30	0.00	96.14	0.00	12.46	0.25	27.09	0.00	0.00	0.00	39.80
28-Sep-23	7.38	37.08	36.69	81.15	0.00	0.00	0.00	0.00	7.31	5.78	28.53	0.00	50.90	2.36	0.00	94.88	0.00	11.11	1.69	25.26	0.00	0.00	0.00	38.06
29-Sep-23	7.38	37.33	39.40	84.11	0.00	0.00	0.00	0.00	12.81	3.73	27.53	0.00	33.00	2.33	0.00	79.40	0.00	8.30	0.00	26.13	0.00	0.00	0.00	34.43
30-Sep-23	7.59	43.28	33.89	84.77	0.00	0.00	0.00	0.00	7.31	3.65	18.43	0.00	32.83	2.21	0.01	64.45	0.00	12.46	0.00	25.66	0.00	0.00	0.00	38.13
TOTAL	201.99	1076.42	1160.32	2438.74	0.00	0.00	0.00	0.00	222.80	125.10	864.24	0.00	1450.16	67.55	0.04	2729.89	114.86	420.76	19.87	850.54	0.09	0.00	0.01	1406.13

16. भूटान , नेपाल, बांग्लादेश एवं म्यान्मार के साथ अंतरराष्ट्रीय विद्युत विनिमय

अप्रैल 2023 से मार्च 2024 April 2023 to March 2024

अंतरराष्ट्रीय विद्युत विनिमय [भारत 'से निर्यात'/'को आयात']
Transnational Exchange ('Export from'/'Import to' India)

माह MONTH	भूटान BHUTAN		नेपाल NEPAL		बांग्लादेश BANGLADESH		म्यान्मार MYANMAR	
	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)
अप्रैल 2023 APR'23	19.40	41.40	357.99	0.00	687.30	0.00	0.73	0.00
मई 2023 MAY'23	2.27	117.99	271.80	0.34	775.10	0.00	0.61	0.00
जून 2023 JUN'23	0.00	626.15	80.00	88.86	743.92	0.00	0.77	0.00
जुलाई 2023 JUL'23	0.00	1291.66	0.00	223.72	780.77	0.00	1.02	0.00
अगस्त 2023 AUG'23	0.00	1244.52	0.00	281.61	767.79	0.00	0.94	0.00
सितंबर 2023 SEP'23	0.00	924.25	0.00	323.59	-754.88	0.00	0.77	0.00
कुल Total	21.67	4245.97	709.78	918.12	3000.00	0.00	4.85	0.00

* Based on daily operational data

Date	16.1 Import from neighbouring countries during September 2023													
	(All figures in MU)													
	Import from Bhutan								Import from Bangladesh		Import from Nepal			Import from Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwar D/C*	400 kV Jigmeling-Alipurdwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Sep-23	18.11	5.27	1.32	0.00	3.83	11.54	0.40	0.55	0.00	0.00	0.60	9.97	0.00	0.00
2-Sep-23	18.14	5.28	1.55	0.09	3.97	11.94	0.00	0.73	0.00	0.00	0.55	10.12	0.00	0.00
3-Sep-23	18.07	5.24	1.62	0.00	3.77	11.34	0.00	0.36	0.00	0.00	0.00	8.94	0.00	0.00
4-Sep-23	18.44	4.62	1.90	0.09	3.64	10.95	0.00	0.00	0.00	0.00	0.00	9.14	0.00	0.00
5-Sep-23	16.48	4.69	1.90	0.10	3.30	9.91	0.00	0.08	0.00	0.00	0.12	9.06	0.00	0.00
6-Sep-23	14.88	4.34	1.67	0.00	2.84	9.32	0.00	0.24	0.00	0.00	0.00	9.18	0.00	0.00
7-Sep-23	14.01	3.95	1.69	0.00	2.78	8.41	0.00	0.69	0.00	0.00	0.01	9.00	0.00	0.00
8-Sep-23	13.68	3.85	1.76	0.00	3.47	10.47	0.00	0.69	0.00	0.00	0.59	9.05	0.00	0.00
9-Sep-23	13.41	3.65	1.48	0.00	2.82	8.48	0.00	0.68	0.00	0.00	0.55	9.99	0.00	0.00
10-Sep-23	14.35	3.98	1.54	0.00	2.71	8.17	0.00	0.82	0.00	0.00	0.34	9.52	0.00	0.00
11-Sep-23	13.57	3.77	1.52	0.00	2.98	8.98	0.27	0.66	0.00	0.00	1.07	9.36	0.00	0.00
12-Sep-23	15.54	2.57	1.53	0.00	3.33	10.03	1.14	0.09	0.00	0.00	1.16	8.92	0.00	0.00
13-Sep-23	13.60	3.77	1.70	0.00	2.31	9.42	1.18	0.17	0.00	0.00	0.84	8.79	0.00	0.00
14-Sep-23	12.09	3.25	1.37	0.00	2.51	7.56	1.12	0.11	0.00	0.00	0.77	8.68	0.00	0.00
15-Sep-23	12.05	3.27	1.63	0.00	2.37	7.12	1.25	0.11	0.00	0.00	0.37	8.92	0.00	0.00
16-Sep-23	10.20	2.64	0.99	0.00	2.16	6.51	1.29	0.26	0.00	0.00	0.96	9.86	0.00	0.00
17-Sep-23	10.40	2.71	1.07	0.00	2.02	6.06	1.31	0.36	0.00	0.00	0.84	8.81	0.00	0.00
18-Sep-23	9.61	2.50	0.94	0.00	2.40	6.17	1.31	0.15	0.00	0.00	1.34	3.97	0.00	0.00
19-Sep-23	8.55	2.89	0.70	0.00	2.65	4.00	1.13	0.14	0.00	0.00	1.22	9.92	0.00	0.00
20-Sep-23	8.02	3.25	0.57	0.00	2.69	4.08	1.22	0.22	0.00	0.00	0.08	9.74	0.00	0.00
21-Sep-23	12.25	5.37	1.63	0.00	4.28	6.46	1.19	0.42	0.00	0.00	0.00	11.28	0.00	0.00
22-Sep-23	15.87	5.74	1.63	0.00	4.13	6.25	1.25	0.44	0.00	0.00	0.62	11.87	0.00	0.00
23-Sep-23	18.00	5.22	1.54	0.00	4.53	6.83	1.31	0.51	0.00	0.00	1.32	14.10	0.00	0.00
24-Sep-23	18.09	5.28	1.50	0.00	5.04	7.60	1.31	0.58	0.00	0.00	1.26	13.46	0.00	0.00
25-Sep-23	18.12	5.26	1.52	0.00	4.62	6.95	1.31	0.53	0.00	0.00	1.30	13.02	0.00	0.00
26-Sep-23	17.76	5.10	1.42	0.00	3.45	5.19	1.31	0.49	0.00	0.00	1.22	11.49	0.00	0.00
27-Sep-23	14.78	3.33	1.39	0.00	3.21	4.82	1.07	0.42	0.00	0.00	0.92	11.52	0.00	0.00
28-Sep-23	12.67	3.44	1.30	0.00	2.80	4.21	1.29	0.35	0.00	0.00	0.66	11.82	0.00	0.00
29-Sep-23	11.87	3.14	1.50	0.00	2.62	3.96	1.20	0.34	0.00	0.00	0.69	11.67	0.00	0.00
30-Sep-23	11.69	3.09	1.13	0.00	2.51	3.80	1.03	0.35	0.00	0.00	0.62	12.64	0.00	0.00
Total	424.31	120.47	43.02	0.28	95.75	226.52	23.89	11.54	0.00	0.00	20.02	303.80	0.00	0.00
Based on SEM/Energy meter data for links where available (*Mangdechu generation receipt at APD through a bypassed arrangement at 400kV Punatsanchu station)														

Date	16.2 Export to neighbouring countries during September 2023 (All figures in MU)													
	Export to Bhutan								Export to Bangladesh		Export to Nepal			Export to Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurdwār D/C*	400 kV Jigmeling-Alipurdwār D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Sep-23	0.00	0.00	0.00	0.20	0.00	0.00	0.00	0.00	21.89	3.44	0.00	0.00	0.00	0.00
2-Sep-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.81	3.71	0.00	0.00	0.00	0.00
3-Sep-23	0.00	0.00	0.00	0.08	0.00	0.00	0.00	0.00	21.91	2.94	0.04	0.00	0.00	0.01
4-Sep-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.16	21.78	2.77	0.25	0.00	0.00	0.03
5-Sep-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.79	2.77	0.00	0.00	0.00	0.02
6-Sep-23	0.00	0.00	0.00	0.41	0.00	0.00	0.00	0.00	21.90	3.28	0.03	0.00	0.00	0.01
7-Sep-23	0.00	0.00	0.00	0.20	0.00	0.00	0.00	0.00	21.79	3.34	0.00	0.00	0.00	0.03
8-Sep-23	0.00	0.00	0.00	0.02	0.00	0.00	0.00	0.00	21.84	3.26	0.00	0.00	0.00	0.02
9-Sep-23	0.00	0.00	0.00	0.11	0.00	0.00	0.00	0.00	22.01	3.57	0.00	0.00	0.00	0.03
10-Sep-23	0.00	0.00	0.00	0.12	0.00	0.00	0.00	0.00	22.38	3.50	0.00	0.00	0.00	0.03
11-Sep-23	0.00	0.00	0.00	0.08	0.00	0.00	0.00	0.00	21.83	3.54	0.00	0.00	0.00	0.03
12-Sep-23	0.00	0.00	0.00	0.36	0.00	0.00	0.00	0.00	21.82	3.54	0.00	0.00	0.00	0.03
13-Sep-23	0.00	0.00	0.00	0.11	0.00	0.00	0.00	0.00	21.83	3.76	0.00	0.00	0.00	0.03
14-Sep-23	0.00	0.00	0.00	0.28	0.00	0.00	0.00	0.00	21.79	3.80	0.00	0.00	0.08	0.03
15-Sep-23	0.00	0.00	0.00	0.23	0.00	0.00	0.00	0.00	21.74	3.54	0.00	0.00	0.00	0.03
16-Sep-23	0.00	0.00	0.00	0.38	0.00	0.00	0.00	0.00	22.00	3.54	0.00	0.00	0.00	0.03
17-Sep-23	0.00	0.00	0.00	0.25	0.00	0.00	0.00	0.00	22.63	3.56	0.00	0.00	0.00	0.03
18-Sep-23	0.00	0.00	0.00	0.49	0.00	0.00	0.00	0.00	21.83	3.67	0.00	0.00	0.00	0.03
19-Sep-23	0.00	0.00	0.00	0.60	0.00	0.00	0.00	0.00	21.45	3.46	0.00	0.00	0.00	0.03
20-Sep-23	0.00	0.00	0.00	0.51	0.00	0.00	0.00	0.00	19.55	3.52	0.00	0.00	0.00	0.03
21-Sep-23	0.00	0.00	0.00	0.21	0.00	0.00	0.00	0.00	21.39	3.41	0.19	0.00	0.00	0.03
22-Sep-23	0.00	0.00	0.00	0.19	0.00	0.00	0.00	0.00	21.51	3.12	0.00	0.00	0.00	0.03
23-Sep-23	0.00	0.00	0.00	0.35	0.00	0.00	0.00	0.00	21.67	3.24	0.00	0.00	0.00	0.03
24-Sep-23	0.00	0.00	0.00	0.37	0.00	0.00	0.00	0.00	21.18	3.15	0.00	0.00	0.00	0.03
25-Sep-23	0.00	0.00	0.00	0.38	0.00	0.00	0.00	0.00	21.85	3.24	0.00	0.00	0.00	0.02
26-Sep-23	0.00	0.00	0.00	0.35	0.00	0.00	0.00	0.00	21.75	3.54	0.00	0.00	0.00	0.03
27-Sep-23	0.00	0.00	0.00	0.50	0.00	0.00	0.00	0.00	21.20	3.50	0.00	0.00	0.00	0.03
28-Sep-23	0.00	0.00	0.00	0.45	0.00	0.00	0.00	0.00	21.19	3.46	0.00	0.00	0.00	0.04
29-Sep-23	0.00	0.00	0.00	0.22	0.00	0.00	0.00	0.00	21.59	3.20	0.00	0.00	0.00	0.03
30-Sep-23	0.00	0.00	0.00	0.45	0.00	0.00	0.00	0.00	21.34	3.36	0.00	0.00	0.00	0.03
Total	0.00	0.00	0.00	7.90	0.00	0.00	0.00	0.16	650.24	101.73	0.51	0.00	0.08	0.77
Based on SEM/Energy meter data for links where available (*Mangdechu generation receipt at APD through a bypassed arrangement at 400kV Punatsanchu station)														

17. एसटीओए (द्विपक्षीय एवं सामूहिक) एवं डी एस एम बिलिंग का ब्योरा – सितंबर 2023

STOA (BILATERAL AND COLLECTIVE) & DSM BILLING DETAILS - SEPTEMBER 2023

द्विपक्षीय एसटीओए BILATERAL SHORT TERM OPEN ACCESS

सितंबर 2023	SEP 2023	Apr'23- SEP'23		
नोडल क्षे.भा.प्रे.के. का नाम Name of Nodal RLDC	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved(MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved (MU)
ऊ. क्षे. NR	1036	6399	6381	28814
प.क्षे. WR	1253	1889	5967	12744
द.क्षे. SR	1103	3005	6038	16591
पू.क्षे. ER	901	1022	4681	5645
पूर्वोत्तर क्षे. NER	188	87	843	440
कुल TOTAL	4481	12402	23910	64234

एसटीओए SHORT TERM OPEN ACCESS

	सामूहिक एसटीओए Collective STOA	द्विपक्षीय एसटीओए Bilateral STOA		
माह MONTH	क्रेताओं / विक्रेताओं की सं. NO. of Buyers/Sellers	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)
अप्रैल 2023 Apr'23	5580	6941	3322	9714
मई 2023 May'23	5798	6626	3469	8418
जून 2023 Jun'23	5783	6966	3925	10583
जुलाई 2023 Jul'23	6428	6719	3872	10780
अगस्त 2023 Aug'23	5954	6805	4841	12338
सितंबर 2023 Sep'23	6440	6506	4481	12402
कुल TOTAL	35983	40563	23910	64234

मासिक डी एस एम बिलिंग का ब्योरा* 2021-22
MONTHLY DSM BILLING DETAILS* 2021-22

अनंतिम आँकड़े
Provisional data
subject to change
करोड़ रु. में (RS. IN CRORES)

क्षेत्र REGION →	उत्तरी क्षेत्र NORTH	पश्चिमी क्षेत्र WEST	दक्षिणी क्षेत्र SOUTH	पूर्वी क्षेत्र EAST	पूर्वोत्तर क्षेत्र NORTH EAST
सप्ताह WEEK ↓					
28.08.23 to 03.09.23	286.03	437.85	490.01	389.67	---
04.09.23 to 10.09.23	178.21	300.89	346.54	257.26	---
11.09.23 to 17.09.23	139.78	233.17	337.06	260.06	---
18.09.23 to 24.09.23	97.57	318.24	382.05	282.50	---
25.09.23 to 01.10.23	142.46	234.00	330.38	---	---

* Amount shown is Payable to DSM pool ^Provisional Data

18. पावर मार्केट की सूचना (स्रोत : आई ई एक्स. एच पी एक्स.आई एच.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्स्चेंज के माध्यम से विनिमय – माह: सितंबर 2023 EXCHANGES THROUGH POWER EXCHANGES - SEPTEMBER 2023						
क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU			
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)	
			विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase
1	पंजाब Punjab	उत्तरी क्षेत्र NR	26.12	221.35	117.46	302.01
2	हरियाणा Haryana		120.50	175.63	26.69	151.09
3	राजस्थान Rajasthan		90.41	92.33	186.31	118.62
4	दिल्ली Delhi		224.19	10.30	142.69	43.80
5	उत्तर प्रदेश Uttar Pradesh		198.87	275.26	56.65	227.81
6	उत्तराखंड Uttarakhand		0.31	128.69	11.56	85.57
7	हिमाचल प्रदेश Himachal Pradesh		57.25	1.15	73.85	25.95
8	जम्मू और कश्मीर J & K		38.42	2.46	72.69	29.41
9	केन्द्र शासित चंडीगढ़ UT Chandigarh		22.71	5.31	1.87	0.57
10	ए डी हाइड्रो AD Hydro		23.50	0.00	5.36	0.00
11	कारगम वॉंगतू Karcham Wangtoo		32.50	0.00	5.48	0.00
12	श्री सीमेंट Shree Cement		20.28	0.00	0.74	0.00
13	नाथपा झाकरी Nathpa Jhakri		0.00	0.00	0.00	0.00
14	लॉको बुधिल Lanco Budhil		3.61	0.00	0.62	0.00
15	सिंगरौली Singrauli		0.22	0.00	5.16	0.00
16	दादरी-I Dadri TPS		2.99	0.00	3.05	0.00
17	दादरी-II Dadri stg-II		2.94	0.00	2.47	0.00
18	ऊंचाहार-I Unchahar-I TPS		1.18	0.00	0.65	0.00
19	ऊंचाहार-II Unchahar-II TPS		0.30	0.00	1.08	0.00
20	ऊंचाहार-III Unchahar-III TPS		0.61	0.00	0.47	0.00
21	रिहंद-I Rihand STPS-I		0.00	0.00	3.22	0.00
22	रिहंद-II Rihand STPS-II		0.00	0.00	2.25	0.00
23	रिहंद-III Rihand STPS-III		0.00	0.00	3.61	0.00
24	एपीसीपीएल झुझर APCL Jhajjar		0.00	0.00	4.02	0.00
25	सैज Sainj HEP		21.53	0.00	2.60	0.40
26	दादरी गैस Dadri GPP		0.00	0.00	0.00	0.00
27	औरंगा गैस NTPC Auranga		0.00	0.00	0.00	0.00
28	अंता गैस NTPC Anta		0.00	0.00	0.00	0.00
29	रामपुर एर्बईपी RAMPUR HEP		0.00	0.00	0.00	0.00
30	ऊंचाहार-IV UNCHAHAR-IV		2.36	0.00	1.79	0.00
31	एनटीपीसी कोलहाम NTPC KOLDHAM		0.00	0.00	0.00	0.00
32	कोटेश्वर Koteswar		0.00	0.00	0.00	0.00
33	टांडा स्टेज-२ Tando Stg-2		0.00	0.00	2.48	0.00
34	अदानी सौर एनर्जि टू लिमिटेड Adani Solar Energy Jodhpur Two Limited		10.18	0.00	0.00	0.00
35	सिंगोली भटवारी हेप Singoli Bhatwari HEP		5.68	0.00	1.60	0.69
36	रेलवे Railways		0.00	27.13	0.00	8.82
37	टिहरी Tehri		0.00	0.00	0.00	0.00
38	Renew Solar Energy (Uttarakhand Three) Private Limited (RSE/TP1)		0.00	0.00	0.00	0.00
39	सोरंग एच ई सी. Sorang HEP, (Himachal Sorang Power Private Ltd)		0.00	0.00	3.56	0.00
40	अदानी सौर ऊर्जा जैसलमेर वन लिमिटेड (सौर) Adani Hybrid Energy Jaisalmer One Limited (Solar)		0.00	0.00	0.00	0.00
41	अदानी सौर ऊर्जा जैसलमेर टू लिमिटेड (सौर) Adani Hybrid Energy Jaisalmer Two Limited (Solar)		0.00	0.00	0.00	0.00
42	अदानी सौर ऊर्जा जैसलमेर थ्री लिमिटेड (सौर) Adani Hybrid Energy Jaisalmer Three Limited (Solar)		0.00	0.00	0.00	0.00
43	अयाना रिन्यूएबल पावर वन लिमिटेड Ayana Renewable Power One Private Limited		0.00	0.00	0.00	0.00
44	Adani Hybrid Energy Jaisalmer Four Limited (Solar)अदानी सौर ऊर्जा जैसलमेर चार लिमिटेड (सौर)		0.00	0.00	0.00	0.00
45	रिनीव सौर ऊर्जा प्राइवेट लिमिटेड Renew Solar Urja private limited		0.00	0.00	0.00	0.00
46	NEA-NR एन एन ई ए एन और		22.18	0.00	0.00	0.00
47	रेनूव सूर्या रवि प्राइवेट लिमिटेड (सौर) Renew Surya Ravi Private Limited (Solar)		15.41	0.00	0.64	0.00
48	अजुंदा पावर प्राइवेट लिमिटेड (सौर) Azure Power Maple Pvt Ltd (Solar)		0.00	0.00	0.00	0.00
49	अदानी ग्रीन हाइब्रिड सौर जैसलमेर Adani Green Hybrid Solar Jaisalmer		0.00	0.00	0.00	0.00
50	अदानी ग्रीन हाइब्रिड पवन जैसलमेरAdani Green Hybrid Wind Jaisalmer		0.00	0.00	0.00	0.00
51	अदानी ग्रीन सौर जैसलमेर Adani Green Solar Jaisalmer		0.00	0.00	0.00	0.00
52	ABC Renewable Energy (RJ-01) Private Limited एबीसी रीनोवैबल एनर्जी (आर-01) प्राइवेट लिमिटेड		0.00	0.00	0.00	0.00
53	SBSR Power Cleantech Eleven Private Limited एसबीएसआर पावर क्लीन्टेक लिमिटेड		2.79	0.00	0.00	0.00
54	टनकपुर पावर स्टेशन Tanakpur Power Station		0.00	0.00	0.00	0.00
55	उरी पावर स्टेशन Uri I Power Station		0.00	0.00	0.00	0.00
56	उरी पावर स्टेशन Uri II Power Station		0.00	0.00	0.00	0.00
57	चमेरा पावर स्टेशन Chamera I Power Station		0.00	0.00	0.00	0.00
58	चमेरा पावर स्टेशन Chamera II Power Station		0.00	0.00	0.00	0.00
59	चमेरा पावर स्टेशन Chamera III Power Station		0.00	0.00	0.00	0.00
60	धौलीगंगा पावर स्टेशन Dhauliganga Power Station		0.00	0.00	0.00	0.00
61	दुलहासठी पावर स्टेशन Dulhashti Power Station		0.00	0.00	0.00	0.00
62	किशनगंगा पावर स्टेशन Kishanganga Power Station		0.00	0.00	0.00	0.00
63	पार्वती पावर स्टेशन Parbati III Power Station		0.00	0.00	0.00	0.00
64	सेवा पावर स्टेशन Sewa II Power Station		0.00	0.00	0.00	0.00
65	सलाब पावर स्टेशन Salal Power Station		0.00	0.00	0.00	0.00
66	ADANI SOLAR ENERGY JAISALMER ONE PRIVATE LIMITED AHS अदानी सौर ऊर्जा जैसलमेर वन प्राइवेट लिमिटेड ए एच अय		0.00	0.00	0.00	0.00
67	पार्वती H-ई प्रोजेक्ट Parbati H HE Project		0.00	0.00	0.00	0.00
68	एनटीपीसी अंता जीपीएस NTPC ANTAS GPS RLNG		0.00	0.00	0.00	0.00
69	एनटीपीसी औरंगा जीपीएस NTPC AURANGA GPS RLNG		2.18	0.00	0.00	2.18
70	एनटीपीसी दादरी जीपीएस NTPC DADRI GPS RLNG		2.78	0.00	0.00	2.78
71	अदानी सौर ऊर्जा जैसलमेर चार लिमिटेड ADANI HYBRID ENERGY JAISALMER FOUR LIMITED AHW		0.00	0.00	0.00	0.00
72	NTPC ANTAS GPS spot RF एनटीपीसी आन्टा जीपीएस स्पॉट आरएफ		1.66	0.00	0.00	1.66
73	GMR Bajaji Hosi Hydropower Private Limited जीएमआर बाजोली हाइड्रो पावर प्राइवेट लिमिटेड		0.00	0.00	1.00	1.00

18. पावर मार्केट की सूचना (स्रोत : आई ई एक्स. एच पी एक्स.आई एच.) POWER MARKET INFORMATION (Source IEX & PXIL) पावर एक्चेंज के माध्यम से विनिमय – माह: सितंबर 2023 EXCHANGES THROUGH POWER EXCHANGES - SEPTEMBER 2023								
क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU					
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)		Total (I-DAM+RTM)	
			विक्रय Sale	अप्रच Purchase	विक्रय Sale	अप्रच Purchase	विक्रय Sale	अप्रच Purchase
74	मध्य प्रदेश Madhya Pradesh	पश्चिमी क्षेत्र WR	152.24	47.22	404.66	51.65	556.91	98.87
75	महाराष्ट्र Maharashtra		51.12	442.54	41.85	278.75	92.97	721.29
76	गुजरात Gujarat		64.85	470.04	18.11	87.01	82.96	557.05
77	छत्तीसगढ़ Chhattisgarh		295.36	3.74	203.43	2.51	498.79	6.26
78	गोवा डब्लू और Goa WR		5.33	6.67	4.24	2.13	9.57	8.80
79	जिंदल पावर JINDAL POWER		75.16	0.00	1.41	2.07	76.57	2.07
80	लॉको Lanco		0.00	0.00	0.00	0.00	0.00	0.00
81	ए सी सी आई एल ACIL		9.81	0.00	5.30	0.07	15.10	0.07
82	जिंदल स्टील JINDAL STEEL		71.34	0.00	0.82	0.00	72.16	0.00
83	बालको 2 BALCO 2		3.19	0.00	5.78	0.00	8.96	0.00
84	जीएमआर वरौरा GMR Warora		15.67	0.00	1.37	0.00	17.04	0.00
85	एस्सार पावर एम पी Essar Power MP Ltd		10.77	0.00	10.06	0.28	20.82	0.28
86	के एस के महानदी KSK Mahanadi		0.00	0.00	0.00	0.00	0.00	0.00
87	जिंदल पावर लिमिटेड Jindal Power Ltd. Stg-II		0.04	0.00	0.88	0.00	0.92	0.00
88	रायगढ़ एनर्जी लिमिटेड Raigarh Energy Ltd.		21.32	0.00	9.15	0.00	30.47	0.00
89	डी बी पावर DB Power		199.83	0.00	0.64	0.00	200.47	0.00
90	धारीवाल Dhariwal		0.00	0.00	0.00	0.00	0.00	0.00
91	जेपी निगरी JAYPEE NIGRIE		151.45	0.00	0.04	3.56	151.49	3.56
92	डीजेन मेगा पावर DGEN Mega power		0.97	0.00	3.08	0.00	4.05	0.00
93	एम बी पावर MB Power		139.13	0.00	10.65	0.00	149.78	0.00
94	अदानी पावर स्टेज 3 Adani Power STG3		0.00	0.00	0.00	0.00	0.00	0.00
95	टोरेंट पावर Torrent Power		0.25	0.00	0.00	0.00	0.25	0.00
96	मारुति क्लीन कोल पावर Maruti Clean Coal Power		10.15	0.00	1.57	0.00	11.72	0.00
97	झाबुआ पावर Jhabua Power		10.13	0.00	10.04	0.00	20.18	0.00
98	विंध्याचल स्टेज-1 Vindhyachal stg-1		0.00	0.00	2.25	0.00	2.25	0.00
99	विंध्याचल स्टेज-2 Vindhyachal stg-2		0.00	0.00	1.60	0.00	1.60	0.00
100	विंध्याचल स्टेज-3 Vindhyachal stg-3		0.00	0.00	1.64	0.00	1.64	0.00
101	विंध्याचल स्टेज-4 Vindhyachal stg-4		0.00	0.00	1.10	0.00	1.10	0.00
102	विंध्याचल स्टेज-5 Vindhyachal stg-5		0.00	0.00	1.12	0.00	1.12	0.00
103	मौदा स्टेज-1 Mouda Stg-1		0.00	0.00	2.74	0.00	2.74	0.00
104	सिपाट स्टेज-1 Sipat Stg-1		0.00	0.00	0.95	0.00	0.95	0.00
105	सिपाट स्टेज-2 Sipat Stg-2		1.92	0.00	1.22	0.00	3.13	0.00
106	सासन Sasan UMPP		0.00	0.00	24.86	0.00	24.86	0.00
107	भिलाई एनएससीएस HSPCL Bhilai		0.00	0.00	0.00	0.00	0.00	0.00
108	कोरबा स्टेज-3 Korba Stg 3		0.00	0.00	0.23	0.00	0.23	0.00
109	टीआरएन एनर्जी TRN Energy		0.00	0.00	7.45	0.00	7.45	0.00
110	गंधार Gandhar		0.00	0.00	0.00	0.80	0.00	0.80
111	कोरबा Korba		0.27	0.00	1.49	0.00	1.76	0.00
112	आर के एम पावर RKM Power		7.11	0.00	21.33	0.00	28.43	0.00
113	कावस Kawas		0.00	0.00	0.00	0.48	0.00	0.48
114	मौदा स्टेज-2 Mouda Stg-2		0.00	0.00	3.97	0.00	3.97	0.00
115	एस के एस पावर SKS Power		36.35	0.00	2.62	0.00	38.98	0.00
116	एन टी पि सी शोलापुर NTPC Sholapur		0.84	0.00	3.96	0.00	4.81	0.00
117	एन टी पि सी गदखरा NTPC Gadakhara		0.03	0.00	7.19	0.00	7.22	0.00
118	एन टी पि सी लारा NTPC Lara		0.00	0.00	1.46	0.00	1.46	0.00
119	रायपुर एनर्जी Raipur Energen		0.00	0.00	0.00	0.00	0.00	0.00
120	आर्सेलर मितल ArcelorMittal Nippon steel (ESIL)		0.00	34.26	0.00	16.71	0.00	50.97
121	खरगीन Khargone STPP		1.71	0.00	5.26	0.00	6.97	0.00
122	कोस्टल गुजरात पावर लिमिटेड Coastal Gujarat Power Limited		0.00	0.00	0.00	0.00	0.00	0.00
123	कंटीन्यूम पावर ट्रेडिंग Continuum Power Trading (TN) Private Limited		0.00	0.00	0.00	0.00	0.00	0.00
124	ओस्ट्रो एनर्जी प्राइवेट लिमिटेड Ostro Energy Private Limited (OEPL)		0.00	0.00	0.00	0.00	0.00	0.00
125	दमन डी/दादरा और नगर हवेलीDaman & Diu/Dadar and Nagar Haveli		0.00	10.97	3.80	21.73	3.80	32.70
126	अदानी विंड एनर्जी Adani Wind Energy Kutchh Fire Limited		0.00	0.00	0.00	0.00	0.00	0.00
127	आर जी पि पि एल RGPPL		0.03	0.00	0.00	1.04	0.03	1.04
128	Kakrapar Atomic Power Station काकरपर अटॉमिक पावर स्टेशन		0.00	0.00	0.00	0.00	0.00	0.00
129	BALCO बालको		0.00	5.90	0.00	24.41	0.00	30.31
130	अदानी विंड एनर्जी कच कोर लिमिटेड नखलरान ADANI WIND ENERGY KUTCHH FOUR Ltd. Nakhlarana		46.43	0.00	0.00	0.00	46.43	0.00
131	एनटीपीसी झनोर गंधार जीपीएस NTPC Jhanor Gandhar GPS RLNG		0.00	0.00	0.00	0.00	0.00	0.00
132	एनटीपीसी कावस गपद NTPC Kawas GPS RLNG		0.00	0.00	0.00	0.00	0.00	0.00
133	रातनागरी गैस और पावर प्राइवेट लिमिटेड Ratnagiri Gas and Power Private Limited-LTRLNG		0.00	0.00	0.00	0.00	0.00	0.00
134	विंड फाइव एनर्जी लिमिटेड WIND FIVE RENERGY LIMITED		0.88	0.00	0.00	0.00	0.88	0.00
135	एनटीपीसी झनोर गंधार जीपीएस कॉम गैस NTPC Jhanor Gandhar GPS_COMGAS		0.00	0.00	0.00	0.00	0.00	0.00
136	धारीवाल एस टी यू DHARIWAL STU OTH		2.70	0.00	0.00	0.00	2.70	0.00
137	धारीवाल आइ एस टी DHARIWAL ISTS		0.76	0.00	0.00	0.00	0.76	0.00
138	इनोक्स ग्रीन एनर्जी सर्विसेस लिमिटेड INOX GREEN ENERGY SERVICES LIMITED		2.01	0.00	0.00	0.00	2.01	0.00
137	अदानी पावर लिमिटेड रायपुर टीपीपी Adani Power Limited Raipur TPP		49.45	0.00	3.85	0.00	53.29	0.00
138	टाटा पावर कंपनी लिमिटेड (MTPS) The Tata Power Co Ltd (MTPS)		1.02	0.00	3.33	0.00	4.35	0.00
139	NTPC Ltd Jhanor Gandhar Gas Power Project ComRLNG		1.81	0.00	0.00	0.00	1.81	0.00
140	NTPC KAWAS GPS COMGAS		1.71	0.00	0.00	0.00	1.71	0.00
141	मसाया सोलर एनर्जी प्राइवेट लिमिटेड Masaya Solar Energy Private Limited		5.35	0.00	0.87	0.00	6.21	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एस. एच.पी.एस.आई.एल.)						
POWER MARKET INFORMATION (Source IEX & PXL)						
पावर एक्चेंज के माध्यम से विनिमय - माह: सितंबर 2023						
EXCHANGES THROUGH POWER EXCHANGES - SEPTEMBER 2023						
क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU			
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)	
			विक्रय Sale	अप्रय Purchase	विक्रय Sale	अप्रय Purchase
142	आंध्र प्रदेश Andhra Pradesh	दक्षिणी क्षेत्र SR	54.15	253.38	203.84	59.69
143	कर्नाटक Karnataka		79.57	101.87	30.10	135.46
144	केरल Kerala		3.35	72.61	3.67	220.46
145	तमिलनाडु Tamilnadu		129.85	114.04	161.74	52.51
146	पांडिचेरी Pondicherry		28.77	0.00	13.30	0.20
147	गोवा एस और Goa SR		0.00	3.62	0.00	1.40
148	तेलंगाना TELANGANA		70.06	552.14	117.13	468.04
149	कोस्टल एनर्जेन Costal Energen		0.86	0.00	0.00	0.86
150	आई एल और एफ एस IL&FS		69.65	0.00	4.37	0.00
151	सिंहद्रि स्टेज -2 Simhadri Stg-2		0.00	0.00	2.25	0.00
152	एम्पलसी इंडिया स्टेशन 1 विस्तार NLC India Station I Expansion		0.00	0.00	1.13	0.00
153	एन एल सी स्टेज-1 NLC TPS-I Stg-1		0.00	0.00	0.00	0.00
154	एन एल सी स्टेज-2 NLC TPS-II Stg-2		0.00	0.00	0.00	0.00
155	रामगुंडम स्टेज-1 Ramagundam Stg-1		0.00	0.00	2.70	0.00
156	तलचर स्टेज -2 Talcher Stg-2		0.00	0.00	1.40	0.00
157	रामगुंडम स्टेज-3 Ramagundam Stg-3		0.00	0.00	0.62	0.00
158	सेम्कोर्प गायात्री Sembcorp Gayatri		31.67	0.00	6.22	0.00
159	वल्लूर Vallur		0.00	0.00	0.00	0.00
160	कुड्गी NTPC Kudgi		3.47	0.00	5.70	0.00
161	सिंहद्रि स्टेज -1 Simhadri Stg-1		0.00	0.00	2.28	0.00
162	एन एल सी टी पी एस -11 एक्सपन NLC TPS-II Expn		0.00	0.00	0.00	0.00
163	एनटीपीएल NTPL		0.00	0.00	2.81	0.00
164	न्यूवेल न्यू थर्मल पावर प्रोजेक्ट Neyveli New Thermal Power Project		0.00	0.00	3.14	0.00
165	रामगुंडम फ्लोटिंग सोलर पी वी स्टेशन Ramagundam Floating Solar PV Station		6.78	0.00	0.00	0.00
166	सिंहद्रि फ्लोटिंग सोलर Simhadri Floating solar		1.47	0.00	0.00	0.00
167	सिंहद्रि एफएसपी 15 एससी प्रोजेक्ट Simhadri FSP 15 MW Project		2.12	0.00	0.04	0.00
168	सेम्कोर्प एनर्जी इंडिया लिमिटेड Sembcorp energy india limited		8.04	0.00	1.82	0.00
169	जे एन डब्ल्यू रिनैव एनर्जी टू लिमिटेड JSW RENEW ENERGY TWO LIMITED		0.00	0.00	0.25	0.00
170	ओस्ट्रो कन्नड़ पावर प्राइवेट लिमिटेड Ostro Kannada Power Private Limited		5.94	0.00	2.51	0.00
171	सिम्हापुरी एनर्जी लिमिटेड Simhapuri Energy Limited		4.17	0.00	6.04	3.02
172	परमपूज्य सोलर एनर्जी प्राइवेट लिमिटेड PARAMPUJYA SOLAR ENERGY PRIVATE LIMITED		0.00	0.00	0.00	0.00
173	पश्चिम बंगाल West Bengal	पूर्वी क्षेत्र EH	20.12	336.88	99.80	138.34
174	ओडिशा Odisha		42.23	63.16	152.43	81.36
175	बिहार Bihar		112.47	41.78	304.93	39.02
176	झारखंड Jharkhand		20.59	10.47	73.29	5.25
177	सिक्किम Sikkim		11.97	0.00	9.71	0.19
178	दामोदर घाटी परियोजना DVC		11.99	32.47	10.43	79.17
179	मैथन पावर लिमिटेड Maithon P. Ltd		0.00	0.00	0.00	0.00
180	आधुनिक पा. लिमिटेड Adhunik P.Ltd		0.05	0.00	4.31	0.00
181	चुराचन Churachen HEP		7.18	0.00	0.13	0.00
182	जीएमआर कमलंगा GMR Kamalanga		0.94	0.00	0.03	0.00
183	खिदल आई पी टी एल JPTL		130.95	0.00	1.76	0.00
184	जोरथंग Jorethang		0.00	0.00	1.44	0.00
185	फरक्का-III Farakka-III		0.01	0.00	0.96	0.00
186	बाढ़-II Barh Stage-II		1.13	0.00	3.84	0.00
187	कहालगॉन-II Kahalgaon Stage-II		0.00	0.00	4.09	0.00
188	फरक्का Farakka		0.53	0.00	2.68	0.00
189	कहालगॉन Kahalgaon		0.53	0.00	2.84	0.00
190	तालचर Talcher		0.00	0.00	0.74	0.00
191	तीस्ता-III Teesta-III		0.00	0.00	0.00	0.00
192	दिखु Dichu HEP		0.00	0.00	4.66	0.00
193	ताशिदिंग Tashiding HEP		0.00	0.00	1.11	0.00
194	केबीपूएनएल KBUNL		0.00	0.00	0.58	0.00
195	दर्रापल्ली Darlipalli		0.00	0.00	1.37	0.00
196	नबीनगर NPGC Nabhinagar		0.00	0.00	13.22	0.00
197	नेपाल विद्युत प्राधिकरण Nepal Electricity Authority		240.22	0.00	0.00	0.00
198	Barh STPS Stage-I बाढ़ एसटीपीएस स्टेज-1		1.27	0.00	4.33	0.00
199	NTPC North Karanpura STPS नॉर्थ करनपुरा		0.00	0.00	0.00	0.00
200	Druk Green Power Corporation Limited-Bhutan ड्रुक ग्रीन पावर कॉर्पोरेशन लिमिटेड-भूटान		0.00	0.00	0.00	0.00
201	NTPC BARH Unit 2 एनटीपीसी बाढ़ 1		0.00	0.00	0.00	0.00
202	सिक्किम ऊर्जा लिमिटेड Sikkim Urja Limited		20.56	0.00	24.55	0.00

18. पावर मार्केट की सूचना (स्रोत : आई ई एक्स. एच पी एक्स.आई एच.)								
POWER MARKET INFORMATION (Source IEX & PXIL)								
पावर एक्चेंज के माध्यम से विनिमय – माह:- सितंबर 2023								
EXCHANGES THROUGH POWER EXCHANGES - SEPTEMBER 2023								
क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU					
			Integrated Day Ahead Market (I-DAM)		Real Time Market (RTM)		Total (I-DAM+RTM)	
			विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase	विक्रय Sale	क्रय Purchase
203	अरुणाचल प्रदेश Arunachal Pradesh	पूर्वोत्तर क्षेत्र NER	16.96	0.00	2.63	2.92	19.59	2.92
204	असम Assam		1.04	20.41	6.26	131.03	7.29	151.44
205	मणिपुर Manipur		1.31	0.00	18.00	1.84	19.30	1.84
206	मेघालय Meghalaya		7.23	13.58	0.87	10.40	8.10	23.97
207	मिज़ोरम Mizoram		17.67	0.00	0.00	0.00	17.67	0.00
208	नागालैंड Nagaland		4.76	0.00	2.24	0.00	7.00	0.00
209	त्रिपुरा Tripura		0.00	4.84	12.16	5.58	12.16	10.42
210	अगरतला जीटीपीपी Agartala GTPP_AGO		0.00	0.00	0.03	0.00	0.03	0.00
211	असम जीबीपीपी Assam GBPP_KTD		0.00	0.00	0.10	0.00	0.10	0.00
212	पलाताना Palatana		0.20	0.00	3.17	0.00	3.37	0.00
213	Doyang HEP द्रोंयांग हेप		0.00	0.00	0.00	0.00	0.00	0.00
214	बोंगाईर्गाँव Bongaigaon TPP		5.78	0.00	1.90	0.00	7.68	0.00
215	कामेंग Kameng HEP		2.46	0.00	1.56	0.00	4.02	0.00
216	रंगान्दी हाइड्रो पावर स्टेशन Ranganadi Hydro Power Station		0.00	0.00	0.00	0.00	0.00	0.00
217	लोकटक पावर स्टेशन LOKTAK POWER STATION		0.00	0.00	0.00	0.00	0.00	0.00
			3582	3582	2924	2924	6506	6506

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM

MONTH : SEPTEMBER 2023

RE Source & Unit wise break up (01.09.2023-30.09.2023)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	0	0	3	2
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	27	9	0	0
5	Small Hydro	0	0	6	1
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	27	9	9	3

RECs Issued (01.09.2023-30.09.2023)

Sr.No.	Non Solar	Solar	Total
1	293749	78844	372593

Redemption of REC (01.09.2023-30.09.2023)

Sr.No.	Non Solar	Solar	Total
1	1049079	30738	1079817

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM
MONTH : SEPTEMBER 2023

RE Source & Unit wise break up (Apr'23-Sep'23)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	339	11	437	13
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	131	41	65	3
5	Small Hydro	42	7	42	7
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	5	1	5	1
9	Bio-fuel cogeneration	8	2	23	5
	Total	525	62	572	29

RECs Issued (Apr'23-Sep'23)

Sr.No.	Non Solar	Solar	Total
1	2045469	629492	2674961

Redemption of REC (Apr'23-Sep'23)

Sr.No.	Non Solar	Solar	Total
1	3022695	433378	3456073

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM

MONTH : SEPTEMBER 2023

RE Source & Unit wise break up Since Inception to Sep'23

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	3055	517	2893	516
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	1400	533	1250	468
5	Small Hydro	686	40	698	42
6	Others	4	2	3	1
7	Geothermal	0	0	0	0
8	Biomass	402	37	378	35
9	Bio-fuel cogeneration	817	90	383	55
	Total	6364	1219	5605	1117

RECs Issued since Inception to Sep'23

Sr.No.	Non Solar	Solar	Total
1	89925609	13040230	102965839

Redemption of REC since Inception to Sep'23

Sr.No.	Non Solar	Solar	Total
1	70550880	12061766	82612646

REC Closing balance as on 30.09.2023

Sr.No.	Non Solar	Solar	Total
1	19374729	978464	20353193

20.Details of Grid Events during the Month of September 2023



Northern Region

Sl No.	Category of Grid Event (GI 1or 2/ GD-1 to GD-3)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Uttar Pradesh	03-Sep-2023 00:05	03-Sep-2023 00:25	00:20	0	750	0.000	1.021	56899	73429	i) During antecedent condition, 220kV Saranath(UPI)-Gaipur(UPI) Ckt was opened from Saranath(UPI) end due to overloading of ICTs at Saranath(UPI). ii) As reported, at 00:05 hrs, 400/220kV 500MVA ICT-1 at Rasra(UPI) tripped due to overloading. iii) Since 220kV Saranath(UPI)-Gaipur(UPI) Ckt was already open, with the tripping of 400/220kV 500MVA ICT-1 at Rasra(UPI), major load of Rasra and Gaipur complex was fed by Gorakhpur(UPI). iv) Due to this, SPS operated due to overloading of 400/220kV ICTs (1*500MVA, 1*315MVA and 1*240MVA) at 400/220kV Gorakhpur-1(UPI) and 132kV Gorakhpur-new(UPI)-Maharajganj(Ct) Ckt. 132kV Gorakhpur-new(UPI)-Kasaya Ckt, 220kV Gorakhpur-1(UPI)-Deoria(UPI) Ckt and 220kV Gorakhpur-1(UPI)-Gorakhpur-2(UPI) Ckt-2 tripped. (220kV Gorakhpur-1(UPI)-Gorakhpur-2(UPI) Ckt-1 should also have tripped due to SPS operation at Gorakhpur-1(UPI), but it didn't trip. Exact reason need to be analysed) v) As per SCADA SOE, 220kV Rasra(UPI)-Deoria(UPI) Ckt-1 also tripped during the same time. (Exact reason yet to be shared) vi) As per PMU at Lucknow(PG), no fault is observed in the system. vii) As per SCADA, change in demand of approx. 735MW is observed in UP control area. But, as reported by SLDC-UP, load loss of approx. 750MW occurred in UP control area. viii) As reported, at 00:07 hrs, 220kV Rasra_220(UPI)-Deoria(UPI) Ckt tripped due to broken jumper. ix) As per SCADA SOE, 220kV Rasra(UPI)-Rasra_220(UPI) Ckt also tripped during the same time. x) Complete blackout occurred at 220/132kV Rasra_220(UPI) S/s.	1) 400/220kV 500MVA ICT-1 at Rasra(UPI) 2) 132kV Gorakhpur-new(UPI)-Maharajganj(FCI) Ckt 3) 132kV Gorakhpur-new(UPI)-Kasaya Ckt 4) 220kV Gorakhpur-1(UPI)-Deoria(UPI) Ckt 5) 132kV Gorakhpur-new(UPI)-Mohaddipur(UPI) Ckt 6) 220kV Gorakhpur-1(UPI)-Gorakhpur-2(UPI) Ckt-2 7) 220kV Rasra(UPI)-Deoria(UPI) Ckt
2	GI-2	Uttar Pradesh	03-Sep-2023 00:12	03-Sep-2023 00:32	00:20	0	380	0.000	0.527	56588	72118	i) At 00:07 hrs, 220kV Saranath(UPI)-Gaipur(UPI) Ckt was again charged from Saranath(UPI) end due to tripping of 400/220kV 500MVA ICT-1 at Rasra(UPI). (Previously 220kV Saranath(UPI)-Gaipur(UPI) Ckt was opened from Saranath(UPI) end due to overloading of ICTs at Saranath(UPI) at 00:05 hrs) ii) Since 220kV Rasra(UPI)-Deoria(UPI) Ckt and 400/220kV 500MVA ICT-1 at Rasra(UPI) already tripped, load of Rasra & Gaipur complex was fed by Saranath(UPI). iii) Due to this, at 00:12 hrs, 400/220kV 315MVA ICT-4 at Saranath(UPI) tripped due to over-loading. (ICT-4 tripped on O/C protection operation before SPS operation. O/C setting of ICT and settings of SPS need to be reviewed). iv) At the same time, SPS also operated due to overloading of 400/220kV ICTs (3*315MVA and 1*500MVA; among these ICT-4 tripped already) at 400/220/132kV Saranath(UPI) and 132 kV Saranath(UPI)-Saidpur(UPI) Ckt, 220kV Saranath(UPI)-Aamgaon(UPI) Ckt, 220kV Saranath(UPI)-Harhua(UPI) Ckt and 220kV Saranath(UPI)-Ghaipur(UPI) Ckt tripped. v) As per PMU at Lucknow(PG), no fault is observed in the system. vi) As per SCADA, change in demand of approx. 470MW is observed in UP control area. But, as reported by SLDC-UP, load loss of approx. 380MW occurred in UP control area.	1) 400/220kV 315MVA ICT-4 at Saranath(UPI) 2) 132 kV Saranath(UPI)-Saidpur(UPI) Ckt 3) 220kV Saranath(UPI)-Aamgaon(UPI) Ckt 4) 220kV Saranath(UPI)-Harhua(UPI) Ckt 5) 220kV Saranath(UPI)-Ghaipur(UPI) Ckt
3	GD-1	Rajasthan	04-Sep-2023 12:37	04-Sep-2023 13:13	00:36	0	585	0.000	0.732	67572	79890	i) 220/132kV Debari(RS) has connectivity from 220kV Amber(RS) and 400/220kV Chittorgarh(RS). ii) During antecedent condition, 187MW was coming to 220/132kV Amber(RS) through 220kV Amber(RS)-Debari(RS) ckt and rest of the power was flowing through 220/132kV 500MVA ICT-1 & 2 at Amber(RS) feeding 132kV feeders. At 400/220kV Chittorgarh(RS) 400/220kV 315MVA ICT-1&2 were carrying approx. 368MW each. iii) As reported, at 12:37hrs, 220 kV Amber(RS)-Kankroli(PG) (RS) Ckt tripped due to O/C E/F relay operation at Amber(RS) end. (Exact reason and location of fault yet to be shared) iv) As per PMU at Kankroli(PG), B-N phase to earth fault is observed with delayed fault clearance time of 1300ms. v) Due to loss of supply, complete blackout occurred at 220/132kV Amber(RS). vi) With this, only source left for Debari was 400/220kV Chittorgarh. Loading of 400/220kV 315 MVA ICT-1 & 2 at Chittorgarh(RS) increased further and then both the ICTs tripped due to overloading. (Exact value of increase in loading is yet to be received from Rajasthan). vii) As per SCADA, change in demand of approx. 585MW is observed in Rajasthan control area.	1) 220 kV Amber(RS)-Kankroli(PG) (RS) Ckt 2) 400/220 kV 315 MVA ICT-1 at Chittorgarh(RS) 3) 400/220 kV 315 MVA ICT-2 at Chittorgarh(RS)
4	GI-1	Uttar Pradesh	05-Sep-2023 09:56	07-Sep-2023 10:51	00:55	187	0	0.293	0.000	63839	76167	i) During antecedent condition, 210 MW Unchahar III TPS - UNIT 1 (carrying ~187MW), 210 MW Unchahar II TPS - UNIT 1 (was going under shutdown) and 220kV feeders to Kanpur (ckt-1 & II) were connected to 220kV Bus-3. Rest of the elements were connected to other 220kV buses. ii) As reported, at 09:56hrs, while taking shutdown of 210 MW Unchahar II TPS - UNIT 1 on boiler tube leakage, when generator CB was opened, Y-ph pole of the breaker didn't open. iii) Due to this, pole discrepancy relay operated but breaker didn't open and remained stuck in closed position. iv) Further, manual tripping command was initiated to trip the breaker but as Y-ph pole of the breaker was in stuck condition, LBB of the generator breaker operated. v) Due to LBB operation, all the elements connected at 220kV Bus-3 i.e., 210 MW Unchahar II TPS - UNIT 1 and 220kV feeders to Kanpur (ckt-1 & II) tripped. vi) As per PMU at Kanpur(PG), no fault in system is observed. vii) As per SCADA, generation loss of approx. 187MW at Unchahar TPS occurred.	1) 220 kV Kanpur(PG)-Unchahar(NT) (PG) Ckt-1 2) 220 kV Kanpur(PG)-Unchahar(NT) (PG) Ckt-3 3) 210 MW Unchahar II TPS - UNIT 1 4) 210 MW Unchahar II TPS - UNIT 1
5	GI-2	Uttar Pradesh	05-Sep-2023 12:50	05-Sep-2023 14:30	01:40	0	0	0.000	0.000	67091	78696	i) During antecedent condition, 765 kV Obra_C_TPS-Unnao (UPI) ckt and 765 kV Anpara_CILAN-Unnao (UPI) (UPI) Ckt were carrying approx. 846MW & 913MW respectively. ii) As reported, at 12:50 hrs, 765 kV Obra_C_TPS-Unnao (UPI) ckt tripped on B-N phase to earth fault, fault was in 2-1(148km) from Unnao end. At the same time, 765 kV Anpara_CILAN-Unnao (UPI) (UPI) Ckt also tripped on Tee differential protection operation at Unnao end. iii) As per PMU at Unnao(PG), B-N phase to earth fault with no A/R operation is observed. iv) As per DR of 765 kV Obra_C_TPS-Unnao (UPI) ckt at Unnao end, B-N fault in 2-1 with no A/R operation at Unnao end and unsuccessful A/R operation at Obra_C end is observed. v) With the tripping of aforementioned 765kV lines, MW loading of 765kV Anpara_C-Obra_B ckt increased to ~947MW. vi) As a remedial actions, units of Anpara-A,B,C&D thermal plants were back down to technical minimum. Line loading in the complex came within safe limit by 13:20hrs. vii) As reported and as per SCADA data, generation backdown of approx. 1240MW at Anpara generation complex occurred in UP control area within 45minutes.	1) 765 kV Anpara_CILAN-Unnao (UPI) (UPI) Ckt-1 2) 765 kV Obra_C_TPS-Unnao (UPI) ckt
6	GD-1	HP	06-Sep-2023 06:44	06-Sep-2023 06:59	00:15	40	150	0.079	0.219	50371	68491	i) As reported, at 06:44 hrs, 220 kV Kunihar-Baddi ckt-1&2 tripped on over current earth fault protection operation. (Exact reason and location of fault yet to be shared) ii) At the same time, 220 kV Jeori-Kunihar(HP) Ckt, 220 kV Kunihar-Pinjore(HP) Ckt also tripped on overvoltage. iii) With the tripping of aforementioned elements load of 220/132kV Kunihar(HP) got affected. iv) As per PMU, Y-B fault converted into three phase fault with delayed clearance in 880ms is observed. v) As per SCADA, change in demand of approx. 150MW in HP control area and HP hydro generation loss of approx. 40MW (Bhabha HEP) is observed. vi) Further at 07:22 hrs, while restoration again multiple elements tripping occurred at Kunihar S/s and 220/132kV Kunihar S/s became dead. vii) As per PMU & DR, no fault was in system at 07:22hrs. viii) As per SCADA, change in demand of approx. 250MW in HP control area and is observed at 07:22hrs.	1) 220 kV Wangtoo-Bhabha-Kunihar(HP) ckt (Tconnection) 2) 220 kV Jeori-Kunihar(HP) Ckt 3) 220 kV Baddi-Kunihar(HP) Ckt-1 4) 220 kV Baddi-Kunihar(HP) Ckt-2 5) 220/132kV ICT-1 at Kunihar(HP) 6) 220/132kV ICT-2 at Kunihar(HP) 7) 220/132kV ICT-3 at Kunihar(HP)
7	GD-1	Uttar Pradesh	06-Sep-2023 01:48	06-Sep-2023 02:43	00:55	0	130	0.000	0.174	52124	74681	i) 220/132kV Nanauta(UPI) S/s has double main bus scheme at both 220 & 132 kV level. There are three (03) 220kV lines connected at 220kV Nanauta(UPI) i.e. 220kV lines to Shaharanpur(PG), Badhalkalan(UPI) & Shamli(UPI). ii) During antecedent condition, 220kV lines to Shaharanpur(PG), Badhalkalan(UPI) & Shamli(UPI) were carrying 100MW (towards Nanauta), 43MW (towards Nanauta) and 20MW (towards Shamli) respectively. iii) As reported, at 01:48 hrs, Y-N phase to earth fault occurred on 220 kV Nanauta-Badhalkalan (UPI) Ckt, fault was at ~22.4km from Badhalkalan end. On this fault, distance protection at Nanauta end operated however, line CB of Badhalkalan line at Nanauta end didn't open and remained stuck. iv) Due to this, LBB protection of line CB at Nanauta end of Badhalkalan ckt operated. All three (03) 220kV lines connected at 220kV Nanauta(UPI) tripped and 220/132kV Nanauta(UPI) became dead. v) As per PMU at Shaharanpur(PG), Y-N phase to earth fault with fault with delayed clearance in 320ms is observed. vi) As per SCADA, change in demand of approx. 115MW is observed in UP control area. As reported by SLDC-UP, load loss of approx. 130MW occurred in UP control area.	1) 220 kV Nanauta-Badhalkalan (UPI) Ckt 2) 220 kV Nanauta(UPI)-Shaharanpur(PG) (UPI) Ckt 3) 220 kV Nanauta-Shamli (UPI) Ckt 4) 220/132kV 200MVA ICT-1 at Nanauta(UPI) 5) 220/132kV 200MVA ICT-1 at Nanauta(UPI)
8	GI-2	Uttar Pradesh	07-Sep-2023 12:37	07-Sep-2023 13:35	00:58	282	0	0.453	0.000	62257	79493	i) 400/220kV Rosa(UPI) S/s has double main transfer bus scheme as 400 & 220kV level. ii) During antecedent condition, 300 MW Rosa TPS - UNIT-2 (connected at 220kV side) was carrying approx. 282MW and 400/220kV 200MVA ICT-1&2 were carrying approx. 65MW each. iii) As reported, at 12:37 hrs, R-Y fault occurred on 220kV Rosa-Badaun(UPI) Ckt-2. Rosa end distance protection sensed fault in 2-2 (93km). Line CB at Rosa end didn't open and LBB also failed to operate. iv) Further, adjacent 220kV feeders at Rosa and tripped from remote end and 400/220kV 200MVA ICTs at Rosa tripped on O/C E/F protection operation. v) At the same time, 300 MW Rosa TPS - UNIT 2 tripped on standby earth fault protection operation and 220kV Shahjahanpur(PG)-Shahjahanpur(UPI) Ckt also tripped from Shahjahanpur(PG) end only. (as per SOE). vi) As per PMU at Bareilly(PG), R-Y phase to phase fault with delayed clearance in 1280ms is observed. vii) Generation loss of approx. 282MW occurred due to tripping of 300 MW Rosa TPS - UNIT 2.	1) 400/220kV 200MVA ICT-1 at Rosa(UPI) 2) 400/220kV 200MVA ICT-2 at Rosa(UPI) 3) 300 MW Rosa TPS - UNIT 2 4) 220kV Shahjahanpur(UPI)-Rosa TPS(UPI) Ckt-1 5) 220kV Shahjahanpur(UPI)-Rosa TPS(UPI) Ckt-2 6) 220kV Rosa-Badaun(UPI) Ckt-1 7) 220kV Rosa-Badaun(UPI) Ckt-2 8) 220kV Rosa-Dohma(UPI) Ckt-1 9) 220kV Shahjahanpur(PG)-Shahjahanpur(UPI) Ckt

20.Details of Grid Events during the Month of September 2023



Northern Region

Sl No.	Category of Grid Event (GI for 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
9	GI-1	Punjab	08-Sep-2023 10:21	08-Sep-23 11:09	00:48	280	70	0.484	0.096	57809	72607	i) During antecedent condition, 210 MW Guru Gobind Singh TPS (Ropar) - UNIT 4 (carrying ~162MW) & UNIT 5 (carrying ~112MW) and 220kV feeders to Kharar & Mohali were connected to 220kV Bus-1 section-III. Rest of the elements were connected to Bus-1 section-I&II and Bus-2. ii) On 08th sept, 2023, Unit-6 at GGSSTP was in light up condition after attending boiler tube leakage. At 10:21hrs, generator breaker was closed to synchronize the unit however, R-ph pole of the breaker didn't close. iii) Due to this, pole discrepancy relay operated but this time B-ph pole of the breaker didn't not open and remained stuck in closed position. iv) Further, manual tripping command was initiated to trip the breaker but as B-ph pole of the breaker was in stuck condition, I&II of the generator breaker operated. v) Due to I&II operation, all the elements connected at 220kV Bus-1 section-III i.e., 210 MW Guru Gobind Singh TPS (Ropar) - UNIT 4 & 5 and 220kV feeders to Kharar & Mohali tripped. vi) As per PMU at Jalandhar(PG), no fault in system is observed. vii) As per SCADA, generation loss of approx. 280MW at Ropar GGSSTP and load loss of approx. 70MW in Punjab control area.	1) 210 MW Guru Gobind Singh TPS (Ropar) - UNIT 4 2) 210 MW Guru Gobind Singh TPS (Ropar) - UNIT 5 3) 220kV GGSSTP-Kharar ckt 4) 220kV GGSSTP-Mohali ckt
10	GI-2	Uttar Pradesh	09-Sep-2023 15:24	09-Sep-2023 19:30	04:06	500	0	0.938	0.000	53328	66254	i) During antecedent condition, 490MW Unit-1&2 at Dadri Thermal Stage-2(NTPC) were generating approx. 256MW & 254MW respectively. Supply to FGD (Flue gas desulphurization) and PA fan was coming from common 11kV bus. ii) As reported at 15:24hrs, fault occurred in 11kV feeder to FGD (Flue gas desulphurization). This feeder tripped with the delay of approx. 350msec on O/C E/F protection operation. iii) As per communication with Dadri TPS, during the fault, voltage of 11kV bus dropped to ~20kV and as soon fault cleared, motor of PA fan drew current in the range of ~6.5kA. Further, PA fan motor tripped on O/C which further led to the tripping of rail and then units tripped on flame failure. iv) As per PMU at Maharanibagh(PG), no fault in system is observed. v) As per SCADA, change in generation of approx. 500MW at Dadri Thermal Stage-2(NTPC) is observed.	1) 490 MW Dadri Thermal stage-2 - UNIT 1 2) 490 MW Dadri Thermal stage-2 - UNIT 2
11	GD-1	Delhi	10-Sep-2023 17:08	10-Sep-2023 17:13	00:05	0	125	0.000	0.221	46035	56475	i) 220kV Badapur(DTL) has double main single breaker bus scheme. It has source from 220 kV Tuglakabad-Badapur (DTL) Ckt-1&2 and 220 kV Ballabhgarh(BB)-Badapur (DTL) Ckt-1&2. ii) During antecedent condition, part load of 220kV Okhla and 220kV Sarita Vihar was fed from 220kV Badapur(DTL) via 220kV Badapur-Okhla ckt-1&2 and 220kV Badapur-Sarita Vihar ckt-1&2 respectively. iii) As reported, at 17:08hrs, B-ph conductor near wavetrap of 220 kV Ballabhgarh(BB)-Badapur (DTL) Ckt-2 at Badapur end damaged and created R-N fault. iv) As per DR of Ballabhgarh(BB) end, R-N fault followed by Y-N & B-N fault with total fault clearance time of ~800msec is observed. Fault distance recorded at Ballabhgarh end was ~24.7km. v) As reported, on this fault, 220 kV Ballabhgarh(BB)-Badapur (DTL) Ckt-1&2 tripped from Ballabhgarh(BB) end only, 220 kV Tuglakabad-Badapur (DTL) Ckt-1&2 tripped from Badapur end in 2-4 and 220kV Badapur-Okhla ckt-1&2 tripped on O/C E/F protection operation. vi) As per PMU at Ballabhgarh(PG), R-N fault which later converted into three phase fault with fault clearance time of ~1240msec is observed. vii) With the tripping of aforementioned lines, 220kV Badapur S/A became dead and load feeding from Badapur to 220kV Okhla and 220kV Sarita Vihar affected. viii) As reported by SLDC Delhi, load loss of approx. 125MW occurred in Delhi control area. ix) At 17:13 hrs, 220kV bus coupler breaker at Sarita Vihar S/A and Okhla S/A was closed and load was normalized.	1) 220 kV Tuglakabad-Badapur (DTL) Ckt-1 2) 220 kV Tuglakabad-Badapur (DTL) Ckt-2 3) 220 kV Ballabhgarh(BB)-Badapur (DTL) Ckt-1 4) 220 kV Ballabhgarh(BB)-Badapur (DTL) Ckt-2 5) 220kV Badapur-Okhla ckt-1 6) 220kV Badapur-Okhla ckt-2
12	GD-1	Rajasthan	10-Sep-2023 14:28	10-Sep-2023 17:45	03:17	220	0	0.419	0.000	52504	62000	i) During antecedent condition, total MW generation of MSUPL RE station was approx. 220MW and it was evacuated through 220 kV Bhadla2(PG)-MSUPL (MSUPL) Ckt. ii) As reported, at 14:28hrs, 220 kV Bhadla2(PG)-MSUPL (MSUPL) Ckt tripped on B-N phase to earth fault. iii) As per PMU at Bhadla2(PG), B-N fault with unsuccessful A/R operation is observed. iv) As per SCADA, generation loss of approx. 220MW at MSUPL RE station is observed.	1) 220 KV Seora_SL_BHD2_PG (Mega_SuryaUrja)-Bhadla_2_(PG) (Mega_SuryaUrja) Ckt-1
13	GD-1	Uttar Pradesh	11-Sep-2023 05:36	11-Sep-2023 06:36	01:00	0	465	0.000	0.817	41261	56825	i) 220/132kV Azamgarh2(UP) S/s has double main transfer bus scheme at both 220 & 132 kV level. ii) As reported, at 05:36 hrs, 220kV Bus-bar protection operated at Azamgarh2(UP) which resulted into tripping of all the elements connected to 220kV Bus-1 & 2 at Azamgarh2(UP). (Exact reason of busbar protection operation yet to be shared) iii) Due to loss of supply at 132kV level of Azamgarh2(UP), both 132kV Bus-1 & 2 at Azamgarh2(UP) also became dead which resulted into total blackout of 220/132kV Azamgarh2(UP) S/s. iv) As per PMU at Varanasi(PG), Y-N phase to earth fault with fault clearance time of 80ms is observed. v) As per SCADA, change in demand of approx. 465MW is observed in UP control area.	1) 400/220 kV 500 MVA ICT-1 at Azamgarh2(UP) 2) 400/220 kV 500 MVA ICT-2 at Azamgarh2(UP) 3) 220/132 kV 160 MVA ICT-1 at Azamgarh2(UP) 4) 220/132 kV 160 MVA ICT-2 at Azamgarh2(UP) 5) 220/132 kV 160 MVA ICT-3 at Azamgarh2(UP) 6) 220kV Azamgarh2-Harauri(UP) Ckt 7) 220kV Azamgarh2-Jaunpur(UP) Ckt
14	GD-1	Uttar Pradesh	12-Sep-2023 06:43	12-Sep-2023 07:15	00:32	0	125	0.000	0.207	47396	60478	i) 220/132kV Skandira (Agra2) (UP) S/s has double main transfer bus scheme at both 220 & 132 kV level. However, during antecedent condition, all the elements were connected to 220kV Bus-1 only. 220kV Bus-2 and transfer were not in service condition. ii) As reported, at 06:43 hrs, 220kV Agra2(UP)-Agra2(UP) Ckt-2 tripped due to Y-ph and B-ph CT blast. Fault distance was 0.7 km from Skandira (Agra2) (UP). iii) At the same time, 220kV bus-bar protection operated at Skandira (Agra2) (UP) which resulted into tripping of all the elements connected to 220kV Bus-1 at Skandira (Agra2) (UP). iv) As per DR of 220kV Busbar at Skandira (Agra2) (UP), B-N phase to earth fault with fault current of ~13.87kA followed by Y-N phase to earth fault with fault current of ~13.98kA is observed. v) As per PMU at Agra(PG), B-N phase to earth fault followed by Y-N phase to earth fault with fault clearance time of 80ms is observed. vi) Due to tripping of 220kV Bus-1, supply to 132kV side of Skandira (Agra2) (UP) also lost and 132kV Bus-1 & 2 at Skandira (Agra2) (UP) also became dead which resulted into total blackout of 220/132kV Skandira (Agra2) (UP) S/s. (all 132kV feeders at Agra2(UP) opened manually) vii) As per SCADA, change in demand of approx. 125MW is observed in UP control area.	1) 220 KV Auraiya(NT)-Agra2(UP) (PG) Ckt-1 2) 220 KV Auraiya(NT)-Agra2(UP) (PG) Ckt-2 3) 220kV Agra2(UP)-Agra2(UP) Ckt-1 4) 220kV Agra2(UP)-Agra2(UP) Ckt-2 5) 220kV Agra2(UP)-Kirawali Ckt 6) 220/132kV 160MVA ICT-1 at Agra2(UP) 7) 220/132kV 160MVA ICT-2 at Agra2(UP) 8) 220/132kV 100MVA ICT-3 at Agra2(UP)
15	GI-2	Uttar Pradesh	13-Sep-2023 07:50	13-Sep-2023 08:48	00:58	308	124	0.625	0.204	49310	60833	i) During antecedent condition, 200 MW Obra TPS - UNIT 12 was running through station transformer and generating approx. 124MW. ii) As reported, at 07:50 hrs, 400/220 kV 315 MVA ICT-1 at Obra-B(UP) tripped on R-ph differential protection operation due to blast of 220kV R phase bushing of transformer. Firefighting system installed to protect ICT from fire, operated automatically and protected the complete transformer from the spreading of fire and further damage to other equipment and ICT itself. iii) At the same time, 400/220 kV 315 MVA ICT-2 and 240 MVA ICT-3 at Obra-B(UP) also tripped on over-current earth-fault protection operation. iv) During the same time, "SPS" related to overloading of remaining ICT after tripping of any ICT at Obra TPS" operated and 220kV Obra-A(UP)-Rewa Road(UP) Ckt-1 & 2 and 220kV Obra-A(UP)-Mirzapur(UP) Ckt tripped. v) As per PMU at Allahabad(PG), R-N phase to earth fault with fault clearance time of 80ms is observed. vi) As per SCADA, change in demand of approx. 345MW is observed in UP control area. vii) As reported by SLDC-UP, load loss of approx. 308MW is observed in UP control area and generation loss of approx. 124MW is observed at Obra-B(UP) .	1) 400/220 kV 315 MVA ICT-1 at Obra-B(UP) 2) 400/220 kV 315 MVA ICT-2 at Obra-B(UP) 3) 400/220 kV 240 MVA ICT-3 at Obra-B(UP) 4) 200 MW Obra TPS - UNIT 12 5) 220kV Obra-A(UP)-Rewa Road(UP) Ckt-1 6) 220kV Obra-A(UP)-Rewa Road(UP) Ckt-2 7) 220kV Obra-A(UP)-Mirzapur(UP) Ckt
16	GD-1	Uttarakhand	14-Sep-2023 17:19	14-Sep-2023 18:08	00:49	108	0	0.200	0.000	53939	67824	i) During antecedent condition, 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP were generating approx. 36MW each respectively. ii) As reported, at 12:24 hrs, 220 kV Singoli Bhatwari(Singoli(TLHP))-Srinarag(UK) (PTCU) Ckt-1 & 2 tripped on Y-B-N double phase to ground fault with fault distance of 70.81km and 71.13km from Srinarag(UK) end respectively. As per DR, fault sensed in zone-1 at both the ends for both the lines; DT received at Singoli Bhatwari end. iii) The power generated by 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP were evacuating through 220 kV Singoli Bhatwari(Singoli(TLHP))-Srinarag(UK) (PTCU) Ckt-1 & 2. Hence, due to tripping of both 220 kV Singoli Bhatwari(Singoli(TLHP))-Srinarag(UK) (PTCU) Ckt-1 & 2, 33MW Unit-1, 2 and 3 at Singoli Bhatwari HEP tripped due to loss of evacuation path and blackout occurred at 220kV Singoli Bhatwari HEP. iv) As per PMU at Musaffarnagar(UP), Y-B-N double phase to ground fault is observed with fault clearing time of 80 ms. v) As per SCADA, generation loss of approx. 108MW at Singoli Bhatwari HEP is observed.	1) 220 KV Singoli Bhatwari (Singoli(TLHP))-Srinarag(UK) (PTCU) Ckt-1 2) 220 KV Singoli Bhatwari (Singoli(TLHP))-Srinarag(UK) (PTCU) Ckt-2 3) 33MW Unit-1 at Singoli Bhatwari HEP 4) 33MW Unit-2 at Singoli Bhatwari HEP 5) 33MW Unit-3 at Singoli Bhatwari HEP

20.Details of Grid Events during the Month of September 2023



Northern Region

Sl No.	Category of Grid Event (GI for 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
17	GI-1	Jammu & Kashmir	14-Sep-2023 14:58	14-Sep-2023 16:27	01:29	0	145	0.000	0.198	60432	73270	i) 220/132kV Ziankote S/s have two bus at 220kV side i.e., main bus & reserve bus. ii) During antecedent condition, 220kV Ziankote was operating in bus split mode via. 220kV Amargah(INDGRID)–Ziankote(JK) D/C (carrying 126MW each) was feeding Ziankote load. 220kV Wagwora-Ziankote(JK) Ckt-2 (carrying 84MW) was connected at other bus and feeding Alusteng. 220kV Wagwora-Ziankote(JK) Ckt-1 was not in service. iii) As reported, at 14:58 hrs, 220 KV Amargah(INDGRID)-Ziankote(JK) (POD JK) Ckt-2 tripped from Ziankote(JK) end only on B-N phase to earth fault with fault current of 1.043kA from Ziankote(JK) end. (Exact reason of fault yet to be shared) iv) As per PMU at Amargah(PG), B-N phase to earth fault is observed with delayed fault clearing time of 1200ms. v) As per SCADA, change in demand of approx. 145MW is observed in J&K control area.	1) 220 KV Amargah(INDGRID)-Ziankote(JK) (POD JK) Ckt-2
18	GD-1	Uttarakhand	16-Sep-2023 18:45	16-Sep-2023 19:52	01:07	140	0	0.289	0.000	48425	61120	i) During antecedent condition, 70 MW Unit-2 & 4 at Dhauliganga(NH) and 220 KV Jaulvi(PG)-Dhauliganga(NH) (PG) Ckt-2 were connected to 220kV Bus-2 at Dhauliganga (NH) and only 70 MW Unit-2 & 4 at Dhauliganga(NH) was running and generating approx. 70MW each and total MW generation of 140MW was evacuating through 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-2 only. 220kV Bus-1 at Dhauliganga (NH) was not in service. ii) As reported, at 18:45hrs, bus bar protection operated at 220kV Bus-2 at Dhauliganga(NH) during synchronization of 70 MW Unit-3 at Dhauliganga(NH), as there was fault in Y-ph or CB (SO2 found in CB chamber) and 70 MW Unit-2 & 4 at Dhauliganga(NH) and 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-2 tripped. iii) As per PMU at 400kV Jaulvi(PG), Y-N phase to ground fault is observed in the system with fault clearance time of 80ms. iv) As per SCADA, generation loss of approx. 140MW is observed at Dhauliganga(NH). v) 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-2 and 70 MW Unit-4 at Dhauliganga(NH) revived at 19:52 hrs and 20:03 hrs respectively.	1) 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-2 2) 70 MW Unit-2 at Dhauliganga(NH) 3) 70 MW Unit-4 at Dhauliganga(NH)
19	GD-1	Uttarakhand	16-Sep-2023 21:00	20-Sep-2023 23:06	02:06	68	0	0.140	0.000	48641	64129	i) During antecedent condition, 70 MW Unit- 2 & 4 at Dhauliganga(NH) and 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-2 were connected to 220kV Bus-2 at Dhauliganga (NH) and only 70 MW Unit- 4 at Dhauliganga(NH) was running and generating approx. 68MW and total MW generation of 68MW was evacuating through 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-2 only. 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-1 was not in service. ii) As reported, at 21:00 hrs, bus bar protection operated at 220kV Bus-2 at Dhauliganga(NH) (Exact reason of busbar protection operation yet to be shared) and 70 MW Unit- 4 at Dhauliganga(NH) and 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-2 tripped. iii) As per PMU at 400kV Jaulvi(PG), Y-N phase to ground fault is observed in the system with fault clearance time of 120ms. iv) As per SCADA, generation loss of approx. 68MW is observed at Dhauliganga(NH). v) 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-1 revived at 23:06 hrs on 20th September, 2023.	1) 220 KV Jaulvi (PG)-Dhauliganga(NH) (PG) Ckt-2 2) 70 MW Unit-4 at Dhauliganga(NH)
20	GI-1	Jammu & Kashmir	16-Sep-2023 17:36	16-Sep-2023 22:16	04:40	0	250	0.000	0.437	47735	57150	i) 220/132kV Ziankote S/s have two bus at 220kV side i.e., main bus & reserve bus. ii) During antecedent condition, 220kV Ziankote was operating in bus split mode via. 220kV Amargah(INDGRID)–Ziankote(JK) D/C (carrying 117MW each) was feeding Ziankote load. 220kV Wagwora-Ziankote(JK) Ckt-2 (carrying 133MW) was connected at other bus and feeding Alusteng. 220kV Wagwora-Ziankote(JK) Ckt-1 was not in service. iii) As reported, at 17:36 hrs, 220 KV Amargah(INDGRID)-Ziankote(JK) (POD JK) Ckt-1 tripped on Y-B phase to phase fault with fault current of Iy=3.714kA and Ibr=3.402kA and fault distance of 17.89km from Amargah end. (Exact reason of fault yet to be shared) iv) At the same time, 220 KV Amargah(INDGRID)-Ziankote(JK) (POD JK) Ckt-2 also tripped on Y-B phase to phase fault with fault current of Iy=3.733kA and Ibr=3.401kA and fault distance of 17.91km from Amargah end. (Exact reason of fault yet to be shared) v) As per PMU at Amargah(PG), Y-B phase to phase fault is observed with delayed fault clearing time of 120ms. vi) As per SCADA, change in demand of approx. 250MW is observed in J&K control area.	1) 220 KV Amargah(INDGRID)-Ziankote(JK) (POD JK) Ckt-1 2) 220 KV Amargah(INDGRID)-Ziankote(JK) (POD JK) Ckt-2
21	GD-1	Rajasthan	17-Sep-2023 10:34	17-Sep-2023 13:32	02:58	205	0	0.448	0.000	45770	49801	i) During antecedent condition, 400/33 KV 330 MVA ICT-1 at AYANA1 SL_BKN_PG (ARBP1PL) was carrying total MW generation of Ayana Solar which was approx. 205MW. ii) As reported, at 10:34hrs, 400/33 KV 330 MVA ICT-1 at AYANA1 SL_BKN_PG (ARBP1PL) tripped on over current protection operation due to flashover on LV-02 isolator clamps and IPS tube connector. iii) As per PMU at Ayana(PG), R-Y phase to phase fault converted to 3-phase fault is observed with delayed fault clearance time of 600ms. iv) As per SCADA, generation loss of approx. 205MW is observed in Ayana Solar.	1) 400/33 KV 330 MVA ICT-1 at AYANA1 SL_BKN_PG (ARBP1PL)
22	GD-1	Rajasthan	17-Sep-2023 00:51	17-Sep-2023 02:01	01:10	110	0	0.251	0.000	43865	60246	i) 220/132kV Dausa(RS) has double main and transfer bus scheme at both 220kV and 132kV level. ii) As reported, at 00:51hrs, R and Y phase CT blast occurred at Lakot end of 220 KV Lakot(RS)-Dausa(RS) (PG) Ckt. iii) At the same time, bus bar protection operated at Dausa(RS) and all the elements connected to both 220kV Bus-1 and 2 at Dausa(RS) tripped. (Exact reason of bus bar protection operation at Dausa(RS) yet to be shared) iv) Due to loss of supply to 132kV level also 220/132kV Dausa(RS) S/s became dead. v) As per PMU at Bassi(PG), R-N phase to earth fault with delayed fault clearance time of 200ms followed by Y-N phase to earth fault with fault clearance time of 120ms are observed. vi) As per SCADA, change in demand of approx. 110MW is observed in Rajasthan control area.	1) 220 KV Lakot(RS)-Dausa(RS) (PG) Ckt 2) 220 KV Bassi(PG)-Dausa(RS) (PG) Ckt-1 3) 220 KV Bassi(PG)-Dausa(RS) (PG) Ckt-2 4) 220 KV Sawaimadhopur(RS)-Dausa(RS) (PG) Ckt
23	GD-1	Uttar Pradesh	18-Sep-2023 13:53	18-Sep-2023 14:23	00:30	0	325	0.000	0.172	49556	56847	i) During antecedent condition, 400/220 KV 315 MVA ICT-1 at Obra-B(UP), 220kV Obra-A(UP)-Rewa Road(UP) Ckt-1 & 2 and 220kV Obra-A(UP)-Mirzapur(UP) Ckt were not in service. 220kV Obra-A(UP)-Robertganj(UP) Ckt and 220kV Obra-A(UP)-Sahapur(UP) Ckt were fed radially through 400/220 KV 315 MVA ICT-2 and 240 MVA ICT-3 at Obra-B(UP). Active power loading of 400/220 KV 315 MVA ICT-2 and 240 MVA ICT-3 at Obra-B(UP) were approx. 202 MW and 157MW respectively. 200 MW Obra TPS – UNIT 12 was running through station transformer and generating approx. 120MW. ii) As reported, at 13:53 hrs, 400/220 KV 240 MVA ICT-3 at Obra-B(UP) tripped on over-current earth-fault protection operation. (Exact reason yet to be shared.) iii) Due to tripping of 400/220 KV 240 MVA ICT-3 at Obra-B(UP), 400/220 KV 315 MVA ICT-2 at Obra-B(UP) got overloaded and tripped on directional over-current protection operation. iv) As 220kV Obra-A(UP)-Robertganj(UP) Ckt and 220kV Obra-A(UP)-Sahapur(UP) Ckt were fed radially through 400/220 KV 315 MVA ICT-2 and 240 MVA ICT-3 at Obra-B(UP), 220/132kV Obra-A(UP) S/s became dead due to tripping of both the ICTs and loss of supply. v) During the same time, 200 MW Obra TPS – UNIT 12 also tripped as station supply failed due to tripping of both the ICTs. vi) As per PMU at Allahabad(PG), R-N phase to earth fault with fault clearance time of 80ms is observed. vii) As per SCADA, change in demand of approx. 325MW is observed in UP control area. viii) As per SCADA, change in generation of approx. 120MW is observed at Obra-B(UP). As reported by SLDC-UP, generation loss of approx. 110MW is observed at Obra-B(UP).	1) 400/220 KV 315 MVA ICT-2 at Obra-B(UP) 2) 400/220 KV 240 MVA ICT-3 at Obra-B(UP) 3) 200 MW Obra TPS - UNIT 12
24	GD-1	Rajasthan	21-Sep-2023 21:14	21-Sep-2023 23:32	02:18	0	190	0.000	0.315	43865	60246	i) 220/132kV Khushkhera(RS) has double main bus scheme at both 220kV and 132kV level. ii) As reported, at 00:51hrs, Y phase CT blast occurred at Khushkhera(RS) end of 220 KV Khushkhera(RS)-Neemrana(PG) Ckt. iii) As per DR at Khushkhera(RS) end of 220 KV Khushkhera(RS)-Neemrana(PG) Ckt, line tripped on Y-N phase to earth fault with fault current of ~11.94kA from Khushkhera(RS) end and fault clearing time of ~310ms, fault sensed in zone-4 at Khushkhera(RS) end. iv) At the same time, bus bar protection operated at Khushkhera(RS) and all the elements connected to both 220kV Bus-1 and 2 at Khushkhera(RS) tripped. v) Due to loss of supply to 132kV level also 220/132kV Khushkhera(RS) S/s became dead. vi) As per PMU at Bhiwadi(PG), Y-N phase to earth fault converted to R-N double phase to earth fault is observed with delayed fault clearance time of 600ms. vii) As per SCADA, change in demand of approx. 190MW is observed in Rajasthan control area.	1) 220 KV Khushkhera(RS)-Neemrana(PG) Ckt 2) 220 KV Khushkhera(RS)-Bhiwadi(PG) Ckt 3) 220 KV Khushkhera(RS)-Alwar(RS) Ckt 4) 220 KV Khushkhera(RS)-Kishanghar-Bas(RS) Ckt 5) 220/132kV 160 MVA ICT-1 at Khushkhera(RS) 6) 220/132kV 160 MVA ICT-2 at Khushkhera(RS)
25	GI-1	Delhi	22-Sep-2023 13:38	22-Sep-2023 13:44	00:06	0	125	0.000	0.188	57561	66602	i) 220kV side of 400/220kV Mandaula(PG) has double main & transfer bus scheme and 220/66kV Narela has double main bus scheme. ii) During antecedent condition, 220 KV Mandaula(PG)-Narela(DV) (DTL) Ckt-2 was already under planned shutdown, and load of 220kV Narela S/s was connected through 220 KV Mandaula(PG)-Narela(DV) (DTL) Ckt-1. iii) As reported, at 13:38 hrs, 220 KV Mandaula(PG)-Narela(DV) (DTL) Ckt-1 tripped on R-N phase to ground fault with fault current of 8.89kA and fault distance of 13.21km from Mandaula(PG) end. (Exact reason of fault yet to be shared) iv) As per PMU at Mandaula(PG), R-N phase to earth fault is observed with delayed fault clearing time of 160ms. v) As per SCADA, change in demand of approx. 125 MW is observed in Delhi control area. vi) As reported, load of 220kV Narela S/s was restored at 13:44 hrs through 220kV DSIDC Bawana-Narela ckt-1&2.	1) 220 KV Mandaula(PG)-Narela(DV) (DTL) Ckt-1

20.Details of Grid Events during the Month of September 2023



Northern Region

Sl No.	Category of Grid Event (GI for 2/ (GD-1 to GD-3))	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
						Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
26	GI-1	Haryana	23-Sep-2023 10:45	23-Sep-2023 12:11	01:26	0	0	0.000	0.000	55363	58056	i) During antecedent condition, 220 KV Panipat(THV)-Panipat(BB) (HVPNL) Ckt-1, 2, 3 & 4 were carrying approx. 28MW each. ii) As reported by BBMB, at 10:45hrs, 220 KV Panipat(THV)-Panipat(BB) (HVPNL) Ckt-4 tripped on B-N phase to earth fault with fault distance of 7.38 km from Panipat(THV) end. As per DR at Panipat(BB) end, fault sensed in zone-1 at Panipat(BB) end; fault current was ~3.92KA from Panipat(BB) and fault clearing time was ~55ms. (Exact reason of fault yet to be shared) iii) Due to delay in fault clearance from Panipat(THV) end, 220 KV Panipat(THV)-Panipat(BB) (HVPNL) Ckt-1, 2 & 3 also tripped from Panipat(BB) end only on 2-2/2-3 distance protection operation with fault distance of 56.66km, 38.41 km and 15.31km respectively from Panipat(THV) end. iv) As per DR at Panipat(BB) end, fault sensed in zone-3 for Ckt-1 & 2 and in zone-2 for Ckt-3 at Panipat(BB) end; fault current was ~3.92KA, ~3.80KA and ~3.64KA from Panipat(BB) respectively and fault clearing time was ~780ms, ~780ms and ~80ms respectively. No relay operated at Panipat(THV) end. v) As per PMU at Panipat(BB), Y-N phase to earth fault with delayed clearance of 760ms is observed. (There is phase sequence issue at BBMB stations that's why Y-N fault is recorded as B-N fault at Panipat(BB)) vi) As per SCADA, no change in demand is observed in Haryana control area.	1) 220 KV Panipat(THV)-Panipat(BB) (HVPNL) Ckt-1 2) 220 KV Panipat(THV)-Panipat(BB) (HVPNL) Ckt-2 3) 220 KV Panipat(THV)-Panipat(BB) (HVPNL) Ckt-3 4) 220 KV Panipat(THV)-Panipat(BB) (HVPNL) Ckt-4
27	GI-2	Haryana	24-Sep-2023 00:23	24-Sep-2023 02:44	02:23	0	0	0.000	0.000	46977	58783	i) During antecedent condition, 800 KV HVDC Kurukshetra(PG) Pole-1, 2, 3 & 4 were carrying 725 MW, 721 MW, 470 MW and 477 MW respectively from Champa to Kurukshetra. Total power order was approx. 2500MW. ii) As reported, at 00:23hrs, 800 KV HVDC Kurukshetra(PG) Pole-01 blocked due to false latching of External Block in Lane-1 Main-1 & Main-2. iii) At the same time, 800 KV HVDC Kurukshetra(PG) Pole-03 also blocked due to protective pole isolation failure in Pole-1 and CAT-B initiation at Champa end. iv) As reported, Pole 1 Lane 1 was put into maintenance to avoid further tripping due to external block until identification of root cause for latching of protection. v) As per PMU at Kurukshetra(PG), no fault is observed in the system, but fluctuation in voltage is observed. vi) As per SCADA, no load loss is observed in Haryana control area.	1) 800 KV HVDC Kurukshetra(PG) Pole-01 2) 800 KV HVDC Kurukshetra(PG) Pole-03
28	GD-1	Delhi	25-Sep-2023 13:27	25-Sep-2023 13:30	00:03	52	275	0.099	0.464	52696	59229	i) During antecedent condition, 220 KV Preet Vihar-Harsh Vihar (DTL) Ckt-1 & 2 were feeding the load of 220KV Dew nager, Park Street(part load), I.P. Stn & Rajhat S/s through 220KV Harsh Vihar-Preet Vihar-Patpargang-I.P. Stn-Pragati link. STG at Pragati(DTL) was also connected to this link and was generating approx. 52MW. ii) As reported, at 13:27 hrs, bus bar protection operated at 220KV Bus-1 & 2 at Preet Vihar(DTL) which resulted into tripping of all the elements connected to both the buses and 220/33kV Preet Vihar(DTL) S/s became dead. (Exact reason of bus bar protection operation yet to be shared) iii) Due to tripping of Harsh Vihar-Preet Vihar-Patpargang link the load of 220KV 220KV Dew nager, Park Street(part load), I.P. Stn & Rajhat got interrupted. iv) As reported by SLDC-Delhi, generation loss of approx. 52MW occurred due to tripping of STG at Pragati(DTL) during the same time. v) As per PMU at Maharambagh(PG), no fault is observed in the system. vi) As per SCADA, change in demand of approx. 420MW is observed in Delhi control area. vii) As reported by SLDC-Delhi, load loss of approx. 275MW is occurred in Delhi control area. viii) At 13:30 hrs, 220KV Patpargang-Gazipur Ckt was charged and supply was extended to I.P. Stn, rajhat, Pragati & Park Street S/s and load normalized.	1) 220 KV Preet Vihar-Harsh Vihar (DTL) Ckt-1 2) 220 KV Preet Vihar-Harsh Vihar (DTL) Ckt-2 3) 220 KV Preet Vihar-Patpargang (DTL) Ckt-1 4) 220 KV Preet Vihar-Patpargang (DTL) Ckt-2 5) 220/33kV 160MVA ICT-1 at Preet Vihar (DTL) 6) 220/33kV 100MVA ICT-2 at Preet Vihar (DTL) 7) STG at Pragati(DTL)
29	GD-1	Uttar Pradesh	27-Sep-2023 20:09	27-Sep-2023 20:51	00:42	0	300	0.000	0.439	52451	68272	i) 220KV Allahabad Rewaroad(UP) has double main bus scheme and both the buses are connected through bus coupler. ii) During antecedent condition, 220KV Obra-A(UP)-Mirzapur Ckt was under shutdown to avoid overloading of 400/220KV ICTs at Obra-B(UP) and 220KV Mirapur S/s was radially fed from 220KV Allahabad Rewaroad (UP). As a standby arrangement, 220KV Allahabad Rewaroad (UP)-Obra-A(UP) Ckt-1 was connected with 220KV Obra-A(UP)-Mirapur Ckt through jumpering. iii) As reported, at 19:14 hrs, 220KV Allahabad Rewaroad (UP)-Obra-A(UP) Ckt-1 (now 220KV Allahabad Rewaroad (UP)-Mirapur Ckt as a standby arrangement) tripped on R-N phase to earth fault with fault distance of 21.47km from Allahabad Rewaroad(UP); fault sensed in zone-1. iv) After clearance line was tried to be energized at 20:09 hrs. During charging, again R-N phase to earth fault occurred on the line. As reported by SLDC-UP, R-phase jumper was found broken at location no. 457. v) Due to delay in opening of CB at Allahabad Rewaroad(UP) end of 220KV Allahabad Rewaroad (UP)-Obra-A(UP) Ckt-1, LBB protection operated at 220KV Allahabad Rewaroad(UP). vi) Due to isolator contact issue bus tied condition was reflecting and elements connected to both the buses tripped resulting in complete blackout of 220/132/33kV Allahabad Rewaroad(UP) S/s. vii) Load of areas fed from 132KV Minto Park, 132KV Kareli, 132KV Sarai Aqil and 132KV Meja road was affected. 220KV Sirathu and 220KV Allahabad Cantt were getting supply from 220KV Fatehpur(UP). viii) As per PMU at Allahabad(PG), R-N phase to earth fault with delayed fault clearance time of 160ms is observed. ix) As per SCADA, change in demand of approx. 390MW is observed in UP control area. x) As reported by SLDC-UP, load loss of approx. 300MW is observed in UP control area.	1) 220KV Allahabad Rewaroad (UP)-Obra-A(UP) Ckt-1 2) 220KV Allahabad Rewaroad (UP)-Allahabad400(UP) Ckt-1 3) 220KV Allahabad Rewaroad (UP)-Allahabad400(UP) Ckt-2 4) 220KV Allahabad Rewaroad (UP)-Allahabad(PG) Ckt-1 5) 220KV Allahabad Rewaroad (UP)-Allahabad(PG) Ckt-2 6) 220KV Allahabad Rewaroad (UP)-Sirathu Ckt 7) 220KV Allahabad Rewaroad (UP)-Allahabad Cantt Ckt 8) 220KV Allahabad Rewaroad (UP)-Mirapur Ckt 9) 220/132kV 160MVA ICT-1 at Allahabad Rewaroad(UP) 10) 220/132kV 200MVA ICT-2 at Allahabad Rewaroad(UP) 11) 220/132kV 200MVA ICT-3 at Allahabad Rewaroad(UP)
30	GD-1	Uttar Pradesh	27-Sep-2023 20:25	27-Sep-2023 21:05	00:40	0	570	0.000	0.840	52341	67942	i) During antecedent condition, 220/132kV Allahabad Cantt(UP) & 220/132kV Sirathu(UP) S/s were getting supplies from 220/132kV Fatehpur(UP) in radial mode. ii) As reported, at 20:25 hrs, 220 KV Fatehpur(PG)-Fatehpur(UP) (PG) Ckt-1 tripped on B-N phase to earth fault from Fatehpur(UP) end and auto-reclosed successfully from Fatehpur(PG) end. iii) Again B-N phase to earth fault occurred in 220 KV Fatehpur(PG)-Fatehpur(UP) (PG) Ckt-1 within reclaim time and line finally tripped from Fatehpur(PG) end also. iv) With the tripping of 220 KV Fatehpur(PG)-Fatehpur(UP) (PG) Ckt-1, 220KV Fatehpur(UP)- Sirathu(UP) Ckt and 220KV Fatehpur(UP)- Allahabad Cantt(UP) Ckt started taking power from 220KV Bus-1 at Fatehpur(UP) through back feeding from 220/132kV 200MVA ICT-2 at Fatehpur(UP). v) Due to this, 220/132kV 200MVA ICT-1 at Fatehpur(UP) got overloaded and finally tripped on over-current protection operation causing supply failure of 220KV Fatehpur(UP)-Sirathu(UP) Ckt and 220KV Fatehpur(UP)- Allahabad Cantt(UP) Ckt which resulted into complete blackout of 220/132kV Allahabad Cantt(UP) & 220/132kV Sirathu(UP). vi) Load of areas fed from 132KV Khaga, Mahan, Hussaing, OPH, Kareilly, Manor, Manjapur, Barwar and Sirathu were affected. vii) As per PMU at Fatehpur(PG), two consecutive B-N phase to earth faults with fault clearance time of 80ms are observed. viii) As per SCADA, change in demand of approx. 540MW is observed in UP control area. ix) As reported by SLDC-UP, load loss of 570MW occurred in UP control area.	1) 220 KV Fatehpur(PG)-Fatehpur(UP) (PG) Ckt-1 2) 220/132kV 200MVA ICT-1 at Fatehpur(UP) 3) 220/132kV 200MVA ICT-2 at Fatehpur(UP) 4) 220KV Fatehpur(UP)- Sirathu(UP) Ckt 5) 220KV Fatehpur(UP)- Allahabad Cantt(UP) Ckt
31	GI-2	Uttar Pradesh	27-Sep-2023 12:53	27-Sep-2023 14:02	01:09	285	0	0.512	0.000	55632	65423	i) During antecedent condition, 400/220 KV 315 MVA ICT-1 at Obra-B(UP) was not in service. 400KV Obra-B(UP)-Sultanpur(UP) Ckt, 400/220 KV 240 MVA ICT-3 at Obra-B(UP), 200 MW Obra TPS - UNIT 11 and 13 were connected to 400KV Bus-1 at Obra-B(UP) and rest of the elements were connected to 400KV Bus-2 at Obra-B(UP), 200 MW Obra TPS - UNIT 11 and 13 were generating approx. 285MW in total. ii) As reported, at 12:53 hrs, 220KV Obra-A(UP)-Robertnagar(UP) Ckt tripped on Y-B phase to phase fault with fault distance of 1.4km from Obra-A(UP) end; fault sensed in zone-1. (Exact reason yet to be shared) iii) At the same, 400/220 KV 240 MVA ICT-3 at Obra-B(UP) tripped on over-current earth-fault protection operation and bus bar protection of 400KV Bus-1 also operated. (Reason of ICT tripping and bus bar protection operation yet to be shared by UP) iv) As per PMU at Allahabad(PG), Y-B phase to phase fault with fault clearance time of 80ms is observed. v) As reported by SLDC-UP, no change in demand is observed in UP control area. vi) As per SCADA and as reported by SLDC -UP, generation loss of approx. 285MW is observed at Obra-B(UP).	1) 220KV Obra-A(UP)-Robertnagar(UP) Ckt 2) 400/220 KV 240 MVA ICT-3 at Obra-B(UP) 3) 400KV Obra-B(UP)-Sultanpur(UP) Ckt 4) 200 MW Obra TPS - UNIT 11 5) 200 MW Obra TPS - UNIT 13
32	GI-2	Uttarakhand	28-Sep-2023 15:45	28-Sep-2023 16:26	00:41	0	0	0.000	0.000	56005	64265	i) 765/400KV Koteswar(PG) has double main bus scheme at 400KV level. During antecedent condition, 765/400KV 800MVA ICT-2 at Koteswar(PG), 400KV Koteswar(PG)-Tehr(TH) (PG) Ckt-1 & 3 and 400KV Koteswar(PG)- Koteswar (TH) (PG) Ckt-2 were connected to 400KV Bus-2 at Koteswar(PG) and rest of the elements were connected to 400KV Bus-1 at Koteswar(PG). 765/400KV 800MVA ICT-4 at Koteswar(PG) was under shutdown. Only 100MW Unit-4 at Koteswar HEP was running during antecedent condition. ii) As reported, at 15:45hrs, 400KV side bay (410) of 765/400KV 800MVA ICT-4 at Koteswar(PG) was closed for CSD adjustment. However, at the same time bus bar protection of 400KV Bus-2 operated. This resulted into tripping of all the elements connected to Bus- 2 and Bus-2 became dead. (Exact reason of bus bar protection operation yet to be shared) iii) As per PMU at 400KV Koteswar(PG), B-N phase to ground fault is observed in the system with fault clearance time of 80ms. iv) As per SCADA, no load loss or generation loss is observed in Uttarakhand control area.	1) 400KV Bus-2 at Koteswar (PG) 2) 765/400KV 800MVA ICT-2 at Koteswar(PG) 3) 400KV Koteswar(PG)-Tehr(TH) (PG) Ckt-1 4) 400KV Koteswar(PG)-Tehr(TH) (PG) Ckt-3 5) 400KV Koteswar(PG)- Koteswar (TH) (PG) Ckt-2

20.Details of Grid Events during the Month of September 2023



Western Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GI-2	WR	02-Sep-23 17:18	02-Sep-23 20:30	03:12	-	-	-	-	77937	68275	At 17:15 Hrs/ 02-09-2023, 400 kV Bhilai-Bus-2 tripped on LBB operation of Main breaker of 400 kV Raita-Bhilai-1 due to R-phase stuck during auto recloser. 400 kV Bhilai-Kirnapur-1 and 400 kV Bhilai-Marwa-1 tripped on Over Voltage Stage-II protection operation. 400 kV Bhilai-Korba West Extension-1 tripped due to faulty LBB cable connection in Main and Tie bays. No load loss was occurred during the above event.	Tripping of follwing elements- 1. 400 kV Raita-Bhilai-1 2. 400 kV Bhilai-Kirnapur-1 3. 400 kV Bhilai-Marwa-1 4. 400 kV Bhilai-Korba West Extension-1
2	GD-1	WR	06-Sep-23 11:33	06-Sep-23 12:52	01:19	2.5	186	0.000	0.003	71024	66711	At 11:33 hrs/ 06-09-2023, B-phase tension disc insulator string of 220 kV Rajgarh-Rajgarh(MP)-1 failed and gantry to gantry conductor of bay fell on 220 kV Rajgarh(MP)-Bus-1 creating Y-B-E phase to phase to Ground fault in Bus-1. Bus bar protection didn't operate. 220 kV Rajgarh(MP)-Khiledi-1 and 220 kV Rajgarh(MP)-Rajgarh(PG)-2 tripped from remote ends on Zone-4 protection operation. 220/132 kV Rajgarh(MP)-ICT-1,2&3 were hand tripped resulting in blackout at 220 kV Rajgarh(MP). Generation loss of 2.5 MW at Ambethi and Load loss of 186 MW occurred due to the event.	Tripping of follwing elements- 1. 220 kV Rajgarh(MP)-Rajgarh(PG)-1 2. 220 kV Rajgarh(MP)-Rajgarh(PG)-2 3. 220 kV Rajgarh(MP)-Khiledi-1 4. 220 kV Rajgarh-Bus-1 5. 220 kV Rajgarh-Bus-2
3	GI-2	WR	13-Sep-23 13:17	13-Sep-23 14:50	01:33	-	-	-	-	69776	61240	At 13:17 hrs/ 13-09-2023, Spare ICT bus isolator came in induction zone resulting in Bus fault in 400 kV Akola-Bus-2. 400 kV Akola-Wardha-1, 400 kV Akola-Aurangabad-1&2 tripped from Akola end only. 400 kV Akola-Wardha-2 tripped and 400/220 kV Akola-ICT-1&2 also tripped. No load loss occurred during the event,	Tripping of follwing elements- 1. 400 kV Akola-Bus-2 2. 400 kV Akola-Wardha-1&2 3. 400 kV Akola-Aurangabad-1&2 4. 400/220 kV Akola-ICT-1&2
4	GI-2	WR	15-Sep-23 04:49	15-Sep-23 06:43	01:54	475	-	0.007	-	68884	56060	At 04:49 hrs/ 15-09-2023,Korba Unit-7 (500 MW) tripped due fault in AVR, due to non opening of B-phase breaker, LBB operated on pole discrepancy resulting in tripping of 400 kV Korba(NTPC)-Bus-3 and connected elements tripped at NTPC Korba. Generation loss of 475 MW due to tripping of Korba Unit-7 (500 MW) occurred at Korba (NTPC) thermal power plant due to the event.	Tripping of follwing elements- 1. 400 kV Korba(NTPC)-Bus-3 2. 400 kV Korba-Birsinghpur-2 3. 400 kV Korba-Raipur-4 4. Korba Unit-7 (500 MW)
5	GD-1	WR	17-Sep-23 12:13	17-Sep-23 22:11	09:58	65	-	0.001	-	61557	53312	At 12:13 hrs/ 17-09-2023, 220 kV Bhuj-Kotda Madh-1 tripped on B-E fault due to nuisance done by unknown person at tower location-7. Generation loss of 65 MW occurred at Kotda Madh (Alfanar) wind power plant due to loss of evacuation path.	Tripping of follwing elements- 1. 220 kV Bhuj-Kotda Madh-1
6	GD-1	WR	17-Sep-23 20:55	18-Sep-23 03:54	06:59	44.43	-	0.001	-	64409	52973	At 12:13 hrs/ 17-09-2023, 220 kV Bhuj-Vadva-1 tripped on R-E fault due to suspension disc insulator failure at tower location at 208. Generation loss of 44.43 MW occurred at Vadva (GIWEL-II) wind power plant due to loss of evacuation path.	Tripping of follwing elements- 1. 220 kV Bhuj-Vadva-1
7	GI-2	WR	18-Sep-23 03:25	18-Sep-23 04:31	01:06	-	-	-	-	61095	48989	At 03:25 hrs/ 18-09-2023, during charging of 400 kV Karad-Kolhapur(MH)-2 while closing of line side isolator heavy sparking was observed in Y-phase line side isolator. It was decided to opening bus side isolator (Bus-2), while opening of Bus side isolator heavy sparking occurred at bus isolator and 400 kV Kolhapur(MH)-Bus-2 and connected elements tripped on Bus bar protection operation . No load loss occurred during the event.	Tripping of follwing elements- 1. 400 kV Kolhapur(MH)-Bus-2 2. 400 kV Kolhapur(MH)-Kolhapur(PG)-1&2 3. 400 kV Kolhapur(MH)-Solapur(PG)-2 4. 400/220 kV Kolhapur(MH)-ICT-1&2
8	GD-1	WR	19-Sep-23 22:40	20-Sep-23 23:48	01:08	136	-	0.002	-	69180	54263	At 22:40 hrs/ 19-09-2023, 220 kV Bhuj-Gadhsisa-1 tripped on R-E fault. During patrolling stay wire was found broken and Y-phase insulator was faulty at tower location 48. Generation loss of 136 MW occurred at Gadhsisa (Renew Power) wind power plant due to loss of evacuation path.	Tripping of follwing elements- 1. 220 kV Bhuj-Gadhsisa-1

20.Details of Grid Events during the Month of September 2023



Western Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
9	GD-1	WR	20-Sep-23 01:15	20-Sep-23 07:51	06:36	245	-	0.004	-	68037	52451	At 01:15 hrs/20-09-2023, 220 kV Bhuj-Naranpar-1 tripped on R-E fault due to broken R phase jumper fell over LV side of 220/33 kV Naranpar-ICT-2. Generation loss of 245 MW occurred at Naranpar (GIWEL) wind power plant due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Bhuj-Naranpar-1 2. 220/33 kV Naranpar-ICT-2
10	GI-2	WR	23-Sep-23 03:12	23-Sep-23 12:11	08:59	1734	-	0.027	-	63268	51182	At 03:12 hrs/ 23-09-2023, 400 kV Wardha-Mouda-2 tripped on R-B phase to phase fault due to lightning strike which resulted in tripping of Mouda-Unit-1 (500 MW) and Mouda-Unit-3&4 (660 MW). Generation loss of 1734 MW occurred at Mouda (NTPC) thermal power plant due to units tripping.	Tripping of following elements- 1. 400 kV Wardha-Mouda-2 2. Mouda-Unit-1 (500 MW) 3. Mouda-Unit-3&4 (660 MW)
11	GI-2	WR	24-Sep-23 18:46	24-Sep-23 20:26	01:40	-	-	-	-	65880	56441	At 18:46 hrs/ 24-09-2023, 400 kV Bhopal-Itarsi-MP-2 tripped on Y-B phase to phase fault, due to delayed opening of bay 400 kV Bhopal-MP-Bus-2 tripped on LBB protection operation. No load loss occurred due to the above event	Tripping of following elements- 1. 400 kV Bhopal-MP-Bus-2 2. 400 kV Bhopal-MP-Itarsi-MP-2 3. 400 kV Bhopal-MP-BR-2
12	GD-1	WR	25-Sep-23 05:29	25-Sep-23 15:57	10:28	37	-	0.001	-	61324	51933	At 05:29 hrs/ 25-09-2023, 220 kV Nakhatrana-Dedhiya-1 tripped on Y-E fault due to Tension insulator failure. Generation loss of 37 MW occurred at Dedhiya (AWEK4L) wind power plant due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Nakhatrana-Dedhiya-1
13	GD-1	WR	26-Sep-23 17:46	26-Sep-23 18:01	00:15	-	360	-	0.006	68055	58469	At 17:39 hrs/ 26-09-2023, 400/220 kV Magarwada(PG)-ICT-1 tripped due to mal operation of master trip relays due to DC earth fault. 400/220 kV Magarwada(PG)-ICT-2 was under planned shutdown prior to the tripping of 400/220 kV Magarwada(PG)-ICT-1 which caused loss of supply to 220 kV Magarwada(PG). Due to this, 220 kV Magarwada(DD) and 220 kV Ringanwada was fed through 220 kV Vapi(PG)-Magarwada(DD)-D/C. At 17:46 hrs, 220 kV Vapi(PG)-Magarwada(DD)-1 tripped on R-phase to earth fault due to conductor snapping between tower location 60 and 61. 220 kV Vapi(PG)-Magarwada(DD)-2 got overloaded and tripped at 17:46 hrs. Due to this supply got interrupted at 220kV Magarwada(DD) and Ringanwada, 220 kV Magarwada(PG)-Magarwada(DD)-D/C and 220 kV Magarwada(PG)-Ringanwada-D/C were hand tripped. Load loss of 360 MW occurred at Magarwada(DD) and Ringanwada due to the event.	Tripping of following elements- 1. 400/220 kV Magarwada(PG)-ICT-1 2. 220 kV Vapi(PG)-Magarwada-1&2

20.Details of Grid Events during the Month of September 2023
Southern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Name of Elements (Tripped/Manually opened)
	(GI for 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Andhra Pradesh	01-Sep-23 12:45	01-Sep-23 14:45	2hrs	0	558	0.00%	0.92%	54843	60389	Complete Outage of 220kV/110kV Pydibhimavaram SS, 220kV/132kV Garividi SS, 220kV/132kV Tekkali SS and 220kV/132kV Bobbili SS and Tripping of 220kV Bus-1 and Bus-2 of 400kV/220kV Maradam SS: As per the reports submitted, the triggering incident was Y-N fault in 220kV Bus-2 of 400kV/220kV Maradam SS. Immediately, 220kV Bus-2 BBP operated and all the elements connected to the Bus-2 tripped. Consequently, 400kV/220kV Maradam ICT-1 which was feeding 220kV Maradam Bus-1 tripped on overloading. This resulted in complete outage of 220kV/110kV Pydibhimavaram SS, 220kV/132kV Garividi SS, 220kV/132kV Tekkali SS and 220kV/132kV Bobbili SS as these SS were being radially fed from 220kV Maradam Bus-1 & Bus-2.	1. 400kV/220kV Maradam ICT-1&2 2. 220kV Maradam Garividi Line-1 3. 220kV Maradam Bobbili Line-2 4. 220kV Maradam Pydibhimavaram Line-1
2	GD-1	Tamil Nadu	05-Sep-23 02:56	05-Sep-23 04:45	1hr 49mins	114	0	0.27%	0.00%	42895	37358	Complete Outage of 400kV/110kV Thappugundu SS of TANTRANSCO: During antecedent conditions, 400kV Palavadi Rasipalayam Line-4 and 400kV Anaikadavu Thappagundu Line-2 were under outage. As per the reports submitted, the triggering incident was RYB-N fault in 230kV Anaikadavu Othakalamandapam Line-1&2 and the lines tripped. Subsequently, over voltage protection operated in 400kV Anaikadavu Thappagundu Line-1 and 400kV Anaikadavu Rasipalayam Line-2 at Anaikadavu end and in 400kV Palavadi Rasipalayam Line-2 at Rasipalayam end and the lines tripped. Tripping of 400kV Anaikadavu Thappagundu Line- 1 resulted in loss of power supply to 400kV/110kV Thappugundu SS which led to complete outage of the SS.	1. 400kV Anaikadavu Thappagundu Line-1 2. 400kV Anaikadavu Rasipalayam Line-2 3. 400kV Palavadi Rasipalayam Line-2 4. 230kV Anaikadavu Othakalamandapam Line-1&2
3	GD-1	Tamil Nadu	08-Sep-23 11:03	08-Sep-23 11:12	9mins	191	0	0.36%	0.00%	52771	50085	Complete Outage of 230kV Sprng_Pugalur Wind Generating Station: As per the reports submitted, the triggering incident was RB fault in 400kV Karaikudi Pugalur Line-2. At the same time, the 230kV Pugalur Sprng_Pugalur line tripped only at the Sprng_Pugalur end on operation of the voltage surge relay. Tripping of the only connected line resulted in a complete outage of 230kV Sprng_Pugalur wind generating station.	1. 400kV Karaikudi Pugalur Line-2 2. 230kV Pugalur Sprng_Pugalur Line
4	GD-1	Karnataka	09-Sep-23 09:10	09-Sep-23 09:59	49mins	90	0	0.18%	0.00%	49834	49871	Complete Outage of 220kV Tirumani SS-1 of KSPDCL: As per the reports submitted, the triggering incident was tripping of 220kV Pavagada Tirumani Line-1 only at Tirumani end on relay maloperation. Tripping of only connected line resulted in complete outage of 220kV Tirumani SS-1.	1. 220kV Pavagada Tirumani Line-1
5	GD-1	Tamil Nadu	13-Sep-23 18:06	13-Sep-23 19:03	57mins	1060	0	2.29%	0.00%	46388	46951	Complete Outage of 400kV CEPL Generating Station: During antecedent conditions, 400kV Tuticorin PS Coastal Line-1 was under outage. As per the reports submitted, the triggering incident was YB-N fault in 400kV Tuticorin PS Coastal Line-2 and the line tripped. Tripping of the only connected line resulted in a complete outage of the 400kV CEPL Generating station.	1. 400kV Tuticorin PS Coastal Line-2
6	GD-1	Tamil Nadu	19-Sep-23 00:12	19-Sep-23 02:18	2hrs 6mins	171	0	0.45%	0.00%	38400	42349	Complete Outage of 230kV OPG_CPP SS and Tripping of 230kV Bus-2 of 400kV/230kV Alamythy SS : During antecedent conditions, 230kV Alamythy OPG_CPP Line-1 was under LC. As per the reports submitted, the triggering incident was B N fault in 230kV Bus-2 of 400kV/230kV Alamythy SS, BBP operated and all the elements connected to the Bus including 230kV Alamythy OPG_CPP Line-2 tripped. Tripping of the only connected line resulted in the complete outage of 230kV OPG_CPP SS.	1. 230kV Alamythy OPG_CPP Line-2 2. 230kV Alamythy Mosur 3. 230kV Alamythy Manali Line-2 4. 400kV/230kV Alamythy ICT-3
7	GI-1	Karnataka	03-Sep-23 17:39	03-Sep-23 18:29	50mins	0	0	0.00%	0.00%	41461	41304	Tripping of 220kV Bus-1 of 400kV/220kV Munirabad SS of PGCIL SR-1: During antecedent conditions, at 400kV/220kV Munirabad SS, main circuit breaker of 220kV Munirabad Lingapur line-2 was bypassed and the line was charged through Bus Coupler by connecting to 220kV Bus-1. The triggering incident was tripping of 220kV Munirabad Lingapur line-2 only at the Munirabad end due to distance protection overreach. Since this is the only line connected to Bus-1 at Munirabad, this resulted in the de-energization of 220kV Bus-1 of 400kV/220kV Munirabad SS.	1. 220kV Munirabad Lingapur line-2
8	GI-1	Karnataka	16-Sep-23 09:46	16-Sep-23 10:08	22mins	0	0	0.00%	0.00%	53349	56065	Tripping of 220kV Bus-1 of 400kV/220kV Munirabad SS of PGCIL SR-1: During antecedent conditions, at 400kV/220kV Munirabad SS, main circuit breaker of 220kV Munirabad Lingapur line-2 was bypassed and the line was charged through Bus Coupler by connecting to 220kV Bus-1. The triggering incident was tripping of 220kV Munirabad Lingapur line-2 only at Munirabad end after the hand tripping 220kV Munirabad Lingapur line-1. Since this is the only line connected to Bus-1 at Munirabad, this resulted in the de-energization of 220kV Bus-1 of 400kV/220kV Munirabad SS.	1. 220kV Munirabad Lingapur line-2
9	GI-1	Karnataka	16-Sep-23 10:08	16-Sep-23 18:46	8hr 38mins	0	0	0.00%	0.00%	53300	55248	Tripping of 220kV Bus-1 of 400kV/220kV Munirabad SS of PGCIL SR-1: 400kV/220kV Munirabad SS, main circuit breaker of 220kV Munirabad Lingapur line-2 was bypassed and the line was charged through Bus Coupler by connecting to 220kV Bus-1. The triggering incident was tripping of 220kV Munirabad Lingapur line-2 only at Munirabad end on operation of OC protection. Since this is the only line connected to Bus-1 at Munirabad, this resulted in the de-energization of 220kV Bus-1 of 400kV/220kV Munirabad SS.	1. 220kV Munirabad Lingapur line-2

20.Details of Grid Events during the Month of September 2023
Southern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load v.s. Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Name of Elements (Tripped/Manually opened)
	(GI for 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
10	GI-1	Karnataka	19-Sep-23 17:17	19-Sep-23 20:07	2hr 53mins	0	0	0.00%	0.00%	35850	40669	Tripping of 220kV Bus-1 of 400kV/220kV Munirabad SS of PGCIL SR-1: During antecedent conditions, at 400kV/220kV Munirabad SS, main circuit breaker of 220kV Munirabad Lingapur line-2 was bypassed and the line was charged through Bus Coupler by connecting to 220kV Bus-1. The triggering incident was tripping of 220kV Munirabad Lingapur line-2 only at the Munirabad end due to distance protection overreach. Since this is the only line connected to Bus-1 at Munirabad, this resulted in the de-energization of 220kV Bus-1 of 400kV/220kV Munirabad SS.	1. 220kV Munirabad Lingapur line-2
11	GI-1	Andhra Pradesh	27-Sep-23 15:51	27-Sep-23 18:51	3hrs	0	0	0.00%	0.00%	48354	53146	Tripping of 220kV Bus-2 of 400kV/220kV/132kV Jammalamadugu SS: As per the reports submitted, the triggering incident was R-N fault in 220kV Bus-2 of 400kV/220kV Jammalamadugu SS. Immediately, BBP operated and all the elements conncted to the Bus-2 tripped. At the same time, 400kV Talaricheruvu Jammalamadugu line-1 which is on the same dia of 400kV/220kV ICT-3 got tripped due to tripping of Tie CB(Main CB of the line was under outage during antecedent conditions .	1. 400kV/220kV Jammalamadugu ICT-3 2. 400kV Talaricheruvu Jammalamadugu Line-1 3. 220kV Jammalamadugu Animala Line 4. 220kV Jammalamadugu Mylavaram Line 5. 220kV Jammalamadugu Porumamilla Line-1 6. 220kV/132kV 160MVA Transfromer-1

20.Details of Grid Events during the Month of September 2023**Eastern Region**

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(G1 or 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Tenughat, Govindpur, Dumka, Gooda, Jasidih, Giridih, Deoghar, Pakur, Rajmahal	08.09.2023 20:33	08.09.2023 20:57	00:34	140	437	0.43%	1.72%	32618	25440	At 20:30 Hrs, 220 kV Maithon-Dumka-1 tripped due to R_N fault leading to increased loading of 220 kV maithon-Dumka-2, which tripped at 20:33 Hrs due to R_N fault. After 12 seconds, 220 kV Tenughat-Biharsharif also tripped from Biharsharif. This led to islanding of entire Tenughat, Govindpur, Dumka, Godda, Deoghar complex with only one unit of Tenughat (140 MW generation) and 410 MW load, which didn't survive, and the island collapsed immediately.	220 kV Maithon-Dumka D/c 220 kV Biharsharif-Tenughat 210 MW U#2 at Tenughat

20.Details of Grid Events during the Month of September 2023

North-Eastern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(G1 for 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD 1	Golaghat area of Assam Power System	02-Sep-23 09:51	02-Sep-23 10:21	00:30:00	0	22	0.00%	0.83%	3075	2657	Golaghat area of Assam Power System was connected with the rest of NER grid through 132 kV Golaghat - Sarupathar line. 132 kV Mariani (AS)-Golaghat line was under planned shutdown prior to event. At 09:51 Hrs on 02.09.2023, 132 kV Golaghat - Sarupathar line tripped. Due to tripping of this element Golaghat area of Assam Power System got separated from the rest of NER Grid and collapsed due to no source available in this area. Power supply was extended to Golaghat area of Assam Power System by charging 132 kV Golaghat - Sarupathar line at 10:21 Hrs on 02.09.2023.	132 kV Golaghat - Sarupathar line
2	GD 1	Kakching, Elangkangpokpi, Chandel, Old Thoubal and 11 kV New Moreh areas of Manipur Power System along with Tamu load of Myanmar Power System	05-Sep-23 05:46	05-Sep-23 06:05	00:19:00	0	19	0.00%	0.66%	2928	2872	Kakching, Elangkangpokpi, Chandel, Old Thoubal and 11 kV New Moreh areas of Manipur Power System along with Tamu load of Myanmar Power System were connected with the rest of NER Grid through 132 kV Thoubal New-Thoubal Old & 132 kV Thoubal Old-Kakching lines. 132 kV Churachandpur - Kakching, 132 kV Elangkangpokpi - Churachandpur, 132 kV Ningthoukhong - Churachandpur 2 and 132 kV Thoubal New-Kakching lines were under forced outage prior to event. At 05:46 Hrs on 05.09.2023, 132 kV Thoubal New-Thoubal Old & 132 kV Thoubal Old-Kakching lines tripped. Due to tripping of these elements, Kakching, Elangkangpokpi, Chandel, Old Thoubal and 11 kV New Moreh areas of Manipur Power System along with Tamu load of Myanmar Power System got separated from the rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Kakching, Elangkangpokpi, Chandel, Old Thoubal and 11 kV New Moreh areas of Manipur Power System along with Tamu load of Myanmar Power System by charging 132 kV Thoubal New-Thoubal Old line at 06:05 Hrs on 05.09.2023.	132 kV Thoubal New-Thoubal Old & 132 kV Thoubal Old-Kakching lines
3	GD 1	Leshka Generating Station of Meghalaya Power System	05-Sep-23 11:38	05-Sep-23 12:00	00:22:00	77	0	2.69%	0.00%	2861	2975	Leshka Generating Station of Meghalaya Power System was connected with the rest of NER Grid through 132 kV Myntdu Leshka - Khleihriat D/C lines. At 11:38 Hrs on 05.09.2023, 132 kV Myntdu Leshka - Khleihriat D/C lines tripped. Due to tripping of these elements, Leshka Generating Station of Meghalaya Power System got separated from the rest of NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Leshka Generating Station of Meghalaya Power System by charging 132 kV Myntdu Leshka - Khleihriat 1 line at 12:00 Hrs on 05.09.2023.	132 kV Myntdu Leshka - Khleihriat D/C lines
4	GD 1	Dhaligaon, Barpeta, Kokrajhar, Gosaigaon, Bilasipara, Gauripur and part load of Bornagar areas of Assam Power System	05-Sep-23 18:28	05-Sep-23 18:53	00:25:00	0	272	0.00%	8.03%	3168	3388	Dhaligaon, Barpeta, Kokrajhar, Gosaigaon, Bilasipara, Gauripur and part load of Bornagar areas of Assam Power System were connected with the rest of NER Grid through 160 MVA, 220/132 kV ICT 1 & 2 at BTPS. 132 kV Nalbari-Barpeta line and 132kV Gosaigaon-Gauripur lines were under shutdown prior to event. At 18:28 Hrs on 05.09.2023, 160 MVA, 220/132 kV ICT 1 & 2 at BTPS tripped. Due to tripping of these elements, Dhaligaon, Barpeta, Kokrajhar, Gosaigaon, Bilasipara, Gauripur and part load of Bornagar areas of Assam Power System got separated from the rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Dhaligaon, Barpeta, Kokrajhar, Gosaigaon, Bilasipara, Gauripur and part load of Bornagar areas of Assam Power System by charging 160 MVA, 220/132 kV ICT 2 at BTPS at 18:53 Hrs on 05.09.2023.	160 MVA, 220/132 kV ICT 1 & 2 at BTPS
5	GD 1	Monarchak & Rabindranagar areas of Tripura Power System	07-Sep-23 14:08	07-Sep-23 14:32	00:24:00	75	11	2.43%	0.37%	3081	2998	Monarchak & Rabindranagar areas of Tripura Power System were connected with the rest of NER Grid through 132 kV Monarchak - Rokhia line. 132 kV Monarchak - Udaipur line was under planned shutdown prior to event. At 14:08 Hrs on 07.09.2023, 132 kV Monarchak - Rokhia line tripped. Due to tripping of this element, Monarchak & Rabindranagar areas of Tripura Power System got separated from the rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Monarchak & Rabindranagar areas of Tripura Power System by charging 132 kV Monarchak - Rokhia line at 14:32 Hrs on 07.09.2023.	132 kV Monarchak - Rokhia line
6	GD 1	Kohima, Chiephobozou, Wokha and Sanis areas of Nagaland Power System	07-Sep-23 14:50	07-Sep-23 15:07	00:17	0	14	0%	0%	3117	2936	Kohima, Chiephobozou, Wokha and Sanis areas of Nagaland Power System were connected with the rest of NER Grid through 132 kV Karong - Kohima line. 132 kV Dimapur(PG)-Kohima and 132 kV Doyang-Sanis lines were under forced outage and 132 kV Kohima-Meluri line was under state approved shutdown prior to event. At 14:50 Hrs on 07.09.2023, 132 kV Karong - Kohima line tripped. Due to tripping of this element, Kohima, Chiephobozou, Wokha and Sanis areas of Nagaland Power System got separated from the rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Kohima, Chiephobozou, Wokha and Sanis areas of Nagaland Power System by charging 132 kV Karong - Kohima line at 15:07 Hrs on 07.09.2023.	132 kV Karong - Kohima line

20.Details of Grid Events during the Month of September 2023

North-Eastern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(G1 for 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
7	GD 1	220 kV Rangia, 132 kV Rangia, Nalbari, Nathkuchi, Sipajhar and Kamalpur areas of Assam Power System	07-Sep-23 21:44	07-Sep-23 22:10	00:26	0	175	0%	6%	3156	3086	220 kV Rangia, 132 kV Rangia, Nalbari, Nathkuchi, Sipajhar and Kamalpur areas of Assam Power System were connected with the rest of NER Grid through 220 kV BTPS - Rangia D/C lines. 132 kV Rangia-Motonga, 132 kV Rangia - Tanga, 132 kV Sipajhar - Rowta and 132 kV Amingaon - Kamalpur I & II lines were under shutdown prior to event. At 21:44 Hrs on 07.09.2023, 220 kV BTPS - Rangia D/C lines tripped. Due to tripping of these elements, 220 kV Rangia, 132 kV Rangia, Nalbari, Nathkuchi, Sipajhar and Kamalpur areas of Assam Power System got separated from the rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to 220 kV Rangia, 132 kV Rangia, Nalbari, Nathkuchi, Sipajhar and Kamalpur areas of Assam Power System by charging 220 kV BTPS - Rangia 1 line at 22:10 Hrs on 07.09.2023.	220 kV BTPS - Rangia D/C lines
8	GD 1	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System	09-Sep-23 09:50	09-Sep-23 10:15	00:25:00	17	24	0.64%	0.89%	2657	2691	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System were connected with the rest of NER Grid through 132 kV Balipara - Tenga line. At 09:50 Hrs on 09.09.2023, 132 kV Balipara - Tenga line tripped. Due to tripping of this element, 132 kV Balipara - Tenga line got separated from the rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System by charging 132 kV Balipara - Tenga line at 10:15 Hrs on 09.09.2023.	132 kV Balipara - Tenga line
9	GD 1	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System	11-Sep-23 11:46	11-Sep-23 13:22	01:36:00	17	23	0.64%	0.90%	2657	2543	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System were connected with the rest of NER Grid through 132 kV Balipara - Tenga line. At 11:46 Hrs on 11.09.2023, 132 kV Balipara - Tenga line tripped. Due to tripping of this element, Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System got separated from the rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System by charging 132 kV Balipara - Tenga line at 13:22 Hrs on 11.09.2023.	132 kV Balipara - Tenga line
10	GD 1	Leshka Generating Station of Meghalaya Power System	11-Sep-23 15:08	11-Sep-23 15:15	00:07:00	77	0	2.90%	0.00%	2657	2802	Leshka Generating Station of Meghalaya Power System was connected with rest of NER grid through 132 kV Khleiriat(ME) - Leshka D/C lines. At 15:08 Hrs on 11.09.2023, 132 kV Myntdu Leshka - Khleiriat D/C lines tripped. Due to tripping of these elements, Leshka Generating Station of Meghalaya Power System got separated from rest of NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Leshka Generating Station of Meghalaya Power System by charging 132 kV Leska - Khleiriat (ME) 1 line at 15:15 Hrs on 11.09.2023.	132 kV Myntdu Leshka - Khleiriat D/C lines
11	GD 1	Dharmanagar area of Tripura Power System	15-Sep-23 10:40	15-Sep-23 11:06	00:26:00	0	18	0.00%	0.65%	3036	2759	Dharmanagar area of Tripura Power System was connected with rest of NER grid through 132 kV Dharmanagar - Dullavchera line. 132 kV PK Bari - Dharmanagar line was under forced outage prior to event. At 10:40 Hrs on 15.09.2023, 132 kV Dharmanagar - Dullavchera line tripped. Due to tripping of this element, Dharmanagar area of Tripura Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in this area. Power supply was extended to Dharmanagar area of Tripura Power System by charging 132 kV Dharmanagar - Dullavchera line at 11:06 Hrs on 15.09.2023.	132 kV Dharmanagar - Dullavchera line
12	GD 1	Dimapur(NL) area of Nagaland Power System	15-Sep-23 18:32	15-Sep-23 20:11	01:39	0	110	0%	3%	3502	3627	Dimapur(NL) area of Nagaland Power System was connected with rest of grid through 132 kV Dimapur(PG) - Dimapur(NL) D/C lines. At 18:32 Hrs on 15.09.2023, 132 kV Dimapur(PG) - Dimapur(NL) D/C lines tripped. Due to tripping of this element, Dimapur(NL) area of Nagaland Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in this area. 132 kV Dimapur(PG) - Dimapur(NL) D/C lines were declared faulty at 20:11 Hrs on 15.09.2023. Power supply was extened to Dimapur(NL) area of Nagaland Power System by charging 132 kV Dimapur(PG) - Dimapur(NL) 2 line at 23:08 Hrs on 15.09.2023.	132 kV Dimapur(PG) - Dimapur(NL) D/C lines
13	GD 1	Churachandpur and Thanlon areas of Manipur Power System	16-Sep-23 16:43	16-Sep-23 18:48	02:05	0	16	0%	1%	2872	3102	Churachandpur and Thanlon areas of Manipur Power System were connected with rest of NER grid through 132 kV Ningthoukhong - Churachandpur 1 line. 132 kV Churachandpur - Kakching, 132 kV Elangkangpokpi - Churachandpur and 132kV Ningthoukhong - Churachandpur 2 lines were under forced outage prior to event. At 16:43 Hrs on 16.09.2023, 132 kV Ningthoukhong - Churachandpur 1 line tripped. Due to tripping of this element, Churachandpur and Thanlon areas of Manipur Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extened to Churachandpur and Thanlon areas of Manipur Power System by charging 132 kV Ningthoukhong - Churachandpur 1 line at 18:48 Hrs of 16.09.2023.	132 kV Ningthoukhong - Churachandpur 1 line

20.Details of Grid Events during the Month of September 2023

North-Eastern Region



Sl No.	Category of Grid Event (G1 for 2/ GD-1 to GD-5)	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	Generation Loss(MW)					Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)			
14	GD 1	Dimapur(NL) area of Nagaland Power System	16-Sep-23 17:38	16-Sep-23 18:59	01:21	0	96	0%	3%	2947	3194	Dimapur(NL) area of Nagaland Power System was connected with rest of grid through 132 kV Dimapur(PG) - Dimapur(NL) D/C lines. At 17:38 Hrs on 16.09.2023, 132 kV Dimapur(PG) - Dimapur(NL) D/C lines tripped. Due to tripping of this element, Dimapur(NL) area of Nagaland Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in this area. Power supply was extended to Dimapur(NL) area of Nagaland Power System by charging 132 kV Dimapur(PG) - Dimapur(NL) 2 line at 18:59 Hrs on 16.09.2023.	132 kV Dimapur(PG) - Dimapur(NL) D/C lines
15	GD 1	Dhaligaon, Barpeta, Gosaigaon and part load of Bornagar areas of Assam Power System	18-Sep-23 16:43	18-Sep-23 17:05	00:22	0	83	1%	1%	2783	2547	Dhaligaon, Barpeta, Gosaigaon and part load of Bornagar areas of Assam Power System were connected with rest of NER grid through 132 kV BTPS - Dhaligaon D/C lines. 132 kV Barpeta – Nalbari line was kept opened due to overloading of 132kV Rangia – Nalbari line and 132kV Gauripur – Gosaigaon line was kept opened due to overloading of 132kV Bilasipara- Gauripur- Gosaigaon-Dhaligaon link prior to event. At 16:43 Hrs on 18.09.2023, 132 kV BTPS - Dhaligaon D/C lines tripped. Due to tripping of these elements, Dhaligaon, Barpeta, Gosaigaon and part load of Bornagar areas of Assam Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Dhaligaon, Barpeta, Gosaigaon and part load of Bornagar areas of Assam Power System by charging 132kV BTPS - Dhaligaon 1 line at 17:05 Hrs on 18.09.2023.	132 kV BTPS - Dhaligaon D/C lines
16	GD 1	Monarchak Generating Station and Rabindranagar area of Tripura Power System	19-Sep-23 12:08	19-Sep-23 12:18	00:10	70	9	3%	0%	2485	2442	Monarchak Generating Station and Rabindranagar area of Tripura Power System were connected with rest of NER Grid through 132kV Monarchak - Udaipur line. 132 kV Monarchak - Rokhia was under planned shutdown prior to event. At 12:08 Hrs on 19.09.2023, 132 kV Monarchak - Udaipur line tripped. Due to tripping of this element, Monarchak Generating Station and Rabindranagar area of Tripura Power System got separated from rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Monarchak Generating Station and Rabindranagar area of Tripura Power System by charging 132 kV Monarchak - Udaipur line at 12:18 Hrs on 19.09.2023.	132 kV Monarchak - Udaipur line
17	GD 1	Margherita(Ledo), Rupai and Chapakhowa areas of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System	19-Sep-23 17:20	19-Sep-23 17:31	00:11	0	72	0%	3%	3095	2734	Margherita(Ledo), Rupai and Chapakhowa areas of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System were connected with rest of NER Grid through 132 kV Tinsukia - Margherita line. 132kV Along - Pasighat line was under emergency shutdown, 132 kV Tinsukia - Rupai line was under forced outage prior to event. At 17:20 Hrs on 19.09.2023, 132kV Tinsukia - Margherita line tripped. Due to tripping of this element, Margherita(Ledo), Rupai and Chapakhowa areas of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Margherita(Ledo), Rupai and Chapakhowa areas of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System by charging 132 kV Tinsukia - Margherita line at 17:31 Hrs on 19.09.2023.	132 kV Tinsukia - Margherita line
18	GD 1	Chapakhowa area of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System	20-Sep-23 02:11	20-Sep-23 03:40	01:29	0	12.8	0%	1%	2809	2213	Chapakhowa area of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System were connected with rest of the NER Grid through 132kV Rupai - Chapakhowa line. 132 kV Along - Pasighat was under planned shutdown prior to event. At 02:11 Hrs on 20.09.2023, 132 kV Rupai - Chapakhowa line tripped. Due to tripping of this element, Chapakhowa area of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Chapakhowa area of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System by charging 132 kV Rupai - Chapakhowa line at 03:40 Hrs on 20.09.2023.	132 kV Rupai - Chapakhowa line
19	GD 2	Kameng Hydro Power Station (NEEPCO)	20-Sep-23 02:55	20-Sep-23 03:50	00:55	450	0	18%	0%	2530	2093	Kameng Hydro Power Station (NEEPCO) was connected with rest of the NER Grid through 400 kV Balipara - Kameng D/C lines. At 02:55 Hrs on 20.09.2023, 400 kV Balipara - Kameng D/C lines tripped. Due to tripping of these lines, Kameng Hydro Power Station (NEEPCO) got separated from rest of NER Grid and subsequently collapsed due to loss of evacuation path. Power supply was extended to Kameng Hydro Power Station (NEEPCO) by charging 400kV Kameng - Balipara 1 line at 03:50 Hrs on 20.09.2023.	400 kV Balipara - Kameng D/C lines

20.Details of Grid Events during the Month of September 2023

North-Eastern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM:SS)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the		Antecedent Generation/Load in the Regional Grid		Brief details of the event (pre-fault and post-fault system conditions)	Elements Tripped
	(G1 for 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss(MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
20	GD 1	Along, Basar and Daporizo areas of Arunachal Pradesh Power System	22-Sep-23 13:01	22-Sep-23 15:02	02:01	0	15	0%	1%	2549	2422	Along, Basar and Daporizo areas of Arunachal Pradesh Power System were connected with rest of the NER Grid through 132 kV Along - Pasighat line. 132 kV Daporijo - Ziro line was under forced outage since 00:35 Hrs on 22.09.2023. At 13:01 Hrs on 22.09.2023, 132 kV Along - Pasighat line tripped. Due to tripping of this element, Along, Basar and Daporizo areas of Arunachal Pradesh Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in these areas. Power supply was extended to Along, Basar and Daporizo areas of Arunachal Pradesh Power System by charging 132 kV Along - Pasighat line at 15:02 Hrs on 22.09.2023.	132 kV Along - Pasighat line
21	GD 1	Karong area of Manipur Power System	25-Sep-23 17:24	25-Sep-23 17:39	00:15	0	15	0%	0%	3039	3156	Karong area of Manipur Power System was connected with rest of the NER Grid through 132 kV Imphal (MSPCL) - Karong line. 132 kV Kohima-Karong line was under forced outage prior to event. At 17:24 Hrs on 25.09.2023, 132 kV Imphal (MSPCL) - Karong line tripped. Due to tripping of this element, Karong area of Manipur Power System got separated from rest of NER Grid and subsequently collapsed due to no source available in this area. Power supply was extended to Karong area of Manipur Power System by charging 132 kV Imphal (MSPCL) - Karong line at 17:39 Hrs on 25.09.2023	132 kV Imphal (MSPCL) - Karong line
22	GD 1	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System	27-Sep-23 13:35	27-Sep-23 13:53	00:18	13	27	1%	1%	2529	2721	Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System were connected with the rest of NER Grid through 132 kV Balipara - Tenga line. At 13:35 Hrs on 27.09.2023, 132 kV Balipara - Tenga line tripped. Due to tripping of this element, Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System were got separated from the rest of NER Grid and subsequently collapsed due to load generation mismatch in these areas. Power supply was extended to Tenga, Khupi areas and Dikshi HEP of Arunachal Pradesh Power System by charging 132 kV Balipara - Tenga line at 13:53 Hrs on 27.09.2023.	132 kV Balipara - Tenga line
23	GD 1	Chapakhowa area of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System	28-Sep-23 12:17	28-Sep-23 12:43	00:26	0	13	0%	0%	2520	2718	Chapakhowa area of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System were connected with rest of the NER Grid through 132 kV Rupai - Chapakhowa line. 132 kV Along - Pasighat was under forced outage prior to event. At 12:17 Hrs on 28.09.2023, 132 kV Rupai - Chapakhowa line tripped. Due to tripping of this element Chapakhowa area of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System got separated due to no source available in these areas. Power supply was extended to Chapakhowa area of Assam Power System and Roing, Tezu, Namsai and Pasighat areas of Arunachal Pradesh Power System by charging 132 kV Rupai - Chapakhowa line at 12:43 Hrs on 28.09.2023.	132 kV Rupai - Chapakhowa line
24	GD 1	Daporijo, Basar and Along areas of Arunachal Pradesh Power System	28-Sep-23 12:32	28-Sep-23 14:19	01:47	0	11	0%	0%	2561	2721	Daporijo, Basar and Along areas of Arunachal Pradesh Power System were connected with rest of the NER Grid through 132 kV Ziro-Daporijo line. 132 kV Along - Pasighat line was under forced outage prior to event. At 12:32 Hrs on 28.09.2023, 132 kV Ziro-Daporijo line tripped. Due to tripping of this element Daporijo, Basar and Along areas of Arunachal Pradesh Power System got separated due to no source available in these areas. Power supply was extended to Daporijo, Basar and Along areas of Arunachal Pradesh Power System by charging 132 kV Along - Pasighat line at 14:19 Hrs on 28.09.2023.	132 kV Ziro-Daporijo line
25	GI-II	Assam	06-Sep-23 10:41	06-Sep-23 12:30	01:49	230	0	8%	0%	3038	2955	BgTPP Unit 3 tripped at 10:41 Hrs on 06.09.2023 due to Turbine Shaft Vibration High. Revision done from Block No. 51 on 06.09.2023.	BgTPP Unit 3
26	GI-II	Assam	06-Sep-23 16:54	06-Sep-23 18:30	01:36	200	0	6%	0%	3122	2987	BgTPP Unit 3 tripped at 16:54 Hrs on 06.09.2023 due to Feed Water System Problem. Revision done from Block No. 75 on 06.09.2023.	BgTPP Unit 3
27	GI-I	Tripura	07-Sep-23 02:51	07-Sep-23 04:30	01:39	27	0	1%	0%	2907	2849	AGTCCPP Unit 4 & 6 tripped at 02:51 Hrs on 07.09.2023 due to Lube Oil Temperature High Trip and due to tripping of GTG - 4 respectively. Revision done from Block No. 19 on 07.09.2023.	AGTCCPP Unit 4 & 6
28	GI-II	Assam	14-Sep-23 21:20	14-Sep-23 23:00	01:40	46	0	1%	0%	3556	3545	AGBPP Unit 2 tripped at 21:20 Hrs on 14.09.2023 due to high vibration. Revision done from Block No. 93 on 14.09.2023.	AGBPP Unit 2
29	GI-II	Tripura	30-Sep-23 16:04	30-Sep-23 17:30	01:26	218	0	9%	0%	2394	2703	Palatana Module II tripped at 16:04 Hrs on 30.09.2023 due to Stator Earth Fault. Revision done from Block No. 71 on 30.09.2023	Palatana Module II

21. NEW ELEMENTS COMMISSIONED* IN SEPTEMBER 2023

COMMISSIONING OF NEW GENERATING UNITS

REGION	S.NO.	ENTITY	UNIT NAME / LOCATION	ADD. CAPACITY (MW)	UNIT NO.	INS. CAPACITY (MW)	DATE®
NR	1	SBSR POWER CLEANTECH ELEVEN PRIVATE LIMITED	SBSR Power Cleantech Eleven Private Limited(Solar)/Rajasthan	12.5	-	300	11-09-2023
WR	2	MASAYA	Masaya(Solar)/Madhya Pradesh	100	I	250	08-09-2023
	3	APRAAVA	Apraava Energy Private Limited(Wind)/Gujarat	21	I	128.1	16-09-2023
	4	APRAAVA	Apraava Energy Private Limited(Wind)/Gujarat	10.5	I	138.6	27-09-2023
	5	ADANI WIND ENERGY KUTCHH FOUR LTD	Adani Wind Energy Kutchh Four Ltd(Wind)/Gujarat	16.8	I	174.3	12-09-2023
SR	6	JSW RENEW TWO ENERGY LIMITED	JSW Renew Two(Wind)/Tamilnadu	18.9	-	234.9	28-09-2023

^(a)RE (Solar & Wind generation added capacity) is in phases. Date mentioned is the date by which complete (Added) capacity

COMMISSIONING OF NEW ICTs

REGION	S.NO.	ENTITY	SUB-STATION	VOLTAGE (kV)	ICT NO.	CAPACITY (MVA)	DATE
NR	1	PRTL	Fatehgarh_III	400/220	III	500	28-09-2023
ER	2	WBSETCL	Gokarna	400/220/33	III	315	15-09-2023
NER	3	NERTS	Salakati	220/132	III	100	09-09-2023
	4	AEGL	Agia	220/132	II	160	22-09-2023
	5	MePTCL	New Shillong	220/132	I	160	15-09-2023
	6	MePTCL	New Shillong	220/132	II	160	15-09-2023

COMMISSIONING OF NEW TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME	CKT NO.	VOLTAGE (kV)	DATE
NR	1	PRTL	400kV Jaisalmer-Fatehgarh_III	II	400	22-09-2023
	2	PRTL	400kV Jaisalmer-Fatehgarh_III	I	400	22-09-2023
WR	3	MSETCL	400kV Kharghar-Vikhroli(GIS)	I	400	26-09-2023
	4	MSETCL	400kV Kharghar-Vikhroli(GIS)	II	400	26-09-2023
SR	5	TSTRANSCO	400kV Yedula - Narlapur	I	400	27-09-2023
	6	TSTRANSCO	400kV Yedula - Dindi	I	400	27-09-2023
	7	WCTL	765kV Warangal_New - Chilakaluripeta	I	765	07-09-2023
	8	WCTL	765kV Warangal_New - Chilakaluripeta	II	765	07-09-2023
NER	9	MePTCL	220kV Mawphlang (Mawngap)-New Shillong	I	220	15-09-2023
	10	MePTCL	220kV Mawphlang (Mawngap)-New Shillong	II	220	15-09-2023

LILO OF TRANSMISSION LINES

REGION	S.NO.	ENTITY	LINE NAME (LILOed LINE)	LILO AT	VOLTAGE (kV)	DATE
NR	1	OCBTL,UPPTCL,UPRVUNL	400kV Jaunpur-Obra_B-I (400kV Obra_B- Jaunpur -II)	Jaunpur	400	03-09-2023
	2	OCBTL,UPPTCL	400kV Obra_C_TPS-Jaunpur-I (400kV Obra_B- Obra_C)	Jaunpur	400	06-09-2023
SR	3	KSEB	400kV Tirunelveli-Kottayam-II	Kottayam	400	21-09-2023
	4	KSEB	400kV Kottayam-Kochi-II	Kottayam	400	21-09-2023

COMMISSIONING OF BUS/LINE REACTORS

REGION	S.NO.	ENTITY	SUB-STATION / LINE (@SUB-STATION)	VOLTAGE (kV)	CAPACITY (MVAR)	DATE
NR	1	UPRVUNL	Bus Reactor-I@Jawaharpur_TPS	765	330	05-09-2023
	2	PRTL	Bus Reactor-I@Fatehgarh_III	400	125	28-09-2023
	3	OCBTL,UPPTCL,UPRVUNL	400kV Jaunpur- Obra_B@Jaunpur	400	50	03-09-2023
	4	OCBTL,UPPTCL,UPRVUNL	400kV Jaunpur- Obra_B@Obra_B	400	63	03-09-2023
	5	PASTL	765kV Aligarh-Sikar-2 Ckt-I@Aligarh	765	330	14-09-2023
	6	PASTL	765kV Aligarh-Sikar-2 Ckt-II@Aligarh	765	330	15-09-2023
WR	7	MSETCL	Bus Reactor-I@Vikhroli	400	125	28-09-2023
NER	8	NERTS	Bus Reactor@Bongaigaon	420	50	21-09-2023

COMMISSIONING OF HVDC/FACTS DEVICE/SYSTEM

REGION	S.NO.	NAME	SUB-STATION / CONSTITUENT	RATINGS	VOLTAGE (kV)	DATE

*Includes first time charging (FTC) as well

22. Voltage Profile

विद्युत विभव रूपरेखा माह - सितंबर 2023

VOLTAGE PROFILE - SEPTEMBER 2023

Date of reporting: 05-Oct-23

क्र.स. Sl. No.	क्षेत्र REGION	उपकेंद्र SUBSTATION	VOLTAGE < V(lower)* (V=380,728 kV)	V(lower) < VOLTAGE < V(upper)*	VOLTAGE > V(upper)* (V=420,800 kV)	Voltage Deviation Index (%age of time voltage is outside range)	Number of hours voltage was outside IEGC band during month	उच्चतम (कि.वो.) Maximum(kV)	निम्नतम(कि.वो.) Minimum(kV)	औसत (कि.वो.) Average(kV)
1	पूर्वी क्षेत्र ER	अंगुल ANGUL	0%	100%	0%	0%	0	800	753	771
2		दालिपली DARPALI	0%	100%	0%	0%	0	795	764	777
3		गया GAYA	0%	100%	0%	0%	0	795	752	776
4		जेरत JERAT	0%	100%	0%	0%	0	793	738	763
5		झारसुगडा JHARSUGUDA	0%	100%	0%	0%	3	805	772	786
6		मिदनापुर MEDINIPUR	0%	100%	0%	0%	0	792	746	770
7		रांची RANCHI	0%	100%	0%	0%	0	796	764	782
8		सासाराम SASARAM	0%	100%	0%	0%	0	789	749	771
1	उत्तरी क्षेत्र NR	आगरा AGRA	0%	100%	0%	0%	0	798	747	777
2		आगरा (फतेहाबाद) AGRA (FATEHABAD)	0%	100%	0%	0%	0	795	737	767
3		अजमेर AJMER	0%	100%	0%	0%	1	801	756	780
4		अलीगढ़ ALIGARH	0%	100%	0%	0%	0	795	733	770
5		अनपरा सी ANPORA-C	0%	100%	0%	0%	0	783	758	769
6		अनपरा डी ANPORA-D	0%	100%	0%	0%	0	782	754	767
7		अंता ANTA	0%	100%	0%	0%	0	794	757	779
8		बलिया BALLIA	0%	100%	0%	0%	0	785	740	766
9		बारा BARA	0%	100%	0%	0%	0	783	754	769
10		भादला BHADLA	0%	100%	0%	0%	0	790	751	772
11		भादला-2 BHADLA-2	0%	100%	0%	0%	0	796	755	775
12		भिवानी BHIWANI	0%	100%	0%	0%	0	799	750	777
13		बीकानेर BIKANER	0%	100%	0%	0%	0	796	753	774
14		बरैली BAREILLY	0%	99%	1%	1%	4	807	741	773
15		चित्तौड़गढ़ CHITTORGARH	0%	100%	0%	0%	3	803	760	782
16		फतेहागढ़-2 FATEHGARH-2	0%	100%	0%	0%	0	796	746	773
17		फतेहपुर FATEHPUR	0%	100%	0%	0%	0	799	740	772
18		ग्रेंटर नोएडा GREATER NOIDA	0%	100%	0%	0%	0	799	738	772
19		हापड़ HAPUR	0%	100%	0%	0%	3	810	730	770
20		जवाहरपुर JAWAHARPUR	0%	100%	0%	0%	0	809	742	770
21		झाटिकरा JHATIKARA	0%	100%	0%	0%	0	792	743	770
22		कानपुर जीआईएस KANPUR GIS	0%	100%	0%	0%	0	794	748	772
23		खेतरी KHETRI	0%	100%	0%	0%	0	795	748	774
24		कोटेश्वर KOTESHWAR	0%	100%	0%	0%	0	792	741	769
25		लालितपुर LALITPUR	0%	100%	0%	0%	0	781	740	756
26		लखनऊ LUCKNOW	0%	100%	0%	0%	0	798	742	771
27		मैनपुरी MAINPURI	0%	100%	0%	0%	0	794	739	768
28		मेरठ MEERUT	0%	100%	0%	0%	0	798	743	774
29		मोहा MOHA	0%	100%	0%	0%	0	803	741	770
30		ओबरा सी OBRA C	0%	100%	0%	0%	0	782	753	765
31		उरई ORAI	0%	100%	0%	0%	2	804	758	782
32		फागी PHAGI	0%	100%	0%	0%	0	799	751	779
33		उन्नाव UNNAO	0%	100%	0%	0%	0	788	738	763
34		वाराणसी VARANASI	0%	100%	0%	0%	0	794	759	777
1	पश्चिमी क्षेत्र WR	अकोला AKOLA	0%	100%	0%	0%	0	791	755	774
2		औरंगाबाद AURANGABAD	0%	100%	0%	0%	0	796	753	778
3		भोपाल (बीडीटीसीएल) BHOPAL (BDTCL)	0%	100%	0%	0%	0	790	747	772
4		भुज BHUJ	0%	100%	0%	0%	0	799	755	780
5		भुज-2 BHUJ-II	0%	100%	0%	0%	0	790	751	775
6		बिलासपुर BILASPUR	0%	100%	0%	0%	0	789	762	776
7		बिना BINA	0%	100%	0%	0%	0	796	757	778
8		बनारस BANARSKANTHA	0%	99%	1%	1%	10	804	764	787
9		चाण CHAMPA	0%	97%	1%	3%	20	807	757	788
10		धुले (बीडीटीसीएल) DHULE (BDTCL)	0%	100%	0%	0%	0	798	752	777
11		धर्मजगढ़ DHARAMJAGARH	0%	100%	0%	0%	0	790	761	775
12		रायपुर पूलिंग RAIPUR POOLING	0%	100%	0%	0%	0	793	754	782
13		एकतुनी EKTUNI	0%	100%	0%	0%	0	792	749	772
14		गदरवाड़ा GADARWARA	0%	100%	0%	0%	0	799	772	786
15		ग्वालियर GWALIOR	0%	100%	0%	0%	0	800	754	780
16		इंदौर INDORE	0%	100%	0%	0%	0	793	749	775
17		जबलपुर JABALPUR	0%	100%	0%	0%	0	802	768	786
18		खंडवा KHANDWA	0%	100%	0%	0%	0	796	752	777
19		कोरडी KORADI	0%	100%	0%	0%	0	782	752	767
20		लकाडिया LAKADIYA	0%	96%	4%	4%	27	807	760	787
21		रायगढ़ पूलिंग RAIGARH POOLING	0%	100%	0%	0%	0	798	772	786
22		पड़घे PADGHE	0%	100%	0%	0%	2	803	748	780
23		परली PARLI	0%	98%	2%	2%	14	805	761	786
24		पुणे PUNE	0%	99%	1%	1%	7	805	744	779
25		राजनन्दागांव RAJNANDGAON	0%	99%	1%	1%	6	804	772	790
26		सासन SASAN	0%	100%	0%	0%	0	792	768	780
27		सतना SATNA	0%	100%	0%	0%	1	800	722	784
28		सिवनी SEONI	0%	100%	0%	0%	0	795	762	779
29		सिपत SIPAT	0%	100%	0%	0%	0	790	765	777
30		सोलापुर SOLAPUR	0%	97%	3%	3%	24	805	763	787
31		तिरुठा TIRORA	0%	100%	0%	0%	0	776	754	766
32		तामना TAMNAR	0%	100%	0%	0%	0	797	772	785
33		वडोदरा VADODARA	0%	100%	0%	0%	3	802	746	780
34		विंध्याचल पूलिंग VINDHYACHAL PS	0%	100%	0%	0%	0	795	771	783
35	दक्षिणी क्षेत्र SR	वर्धा WARDHA	0%	100%	0%	0%	0	797	758	781
36		वरारो WARORA	0%	100%	0%	0%	0	800	765	785
1		कडपा CUDDAH	0%	95%	5%	5%	35	812	755	785
2		चिक्लरीचिक्ल CHILAKALURIPETA	0%	100%	0%	0%	0	794	740	769
3		कुरुन कुरुन KURNOOL	0%	100%	0%	0%	0	798	746	776
4		महेश्वरम MAHESWARAM	0%	95%	5%	5%	37	808	746	783
5		निजामाबाद NIZAMABAD	0%	87%	13%	13%	93	814	750	788
6		नेल्लूर पूलिंग NELLORE PS	0%	0%	0%	0%	0	788	745	770
7		रायचूर RAICHUR	0%	100%	0%	0%	0	799	751	780
8		श्रीकाकुलम SRIKAKULAM	0%	99%	1%	1%	9	809	753	780
9	पूर्वोत्तर क्षेत्र NER	तिरुवल्लम THIRUVALEM	0%	100%	0%	0%	0	795	735	768
10		वेमगिरी VEMAGIRI	0%	98%	2%	2%	15	808	744	774
1		बिस्वनाथ चौराहा BISWANATH CHARIALI (400 kV)	0%	100%	0%	0%	0	419	398	409
2		बोंगाईगांव BONGAIGAON (400 kV)	0%	100%	0%	0%	0	411	397	405
3		बोंगाईगांव टीपीएस BONGAIGAON TPS (400 kV)	0%	100%	0%	0%	0	415	401	409
4		इम्फाल IMPHAL (400 kV)	0%	100%	0%	0%	0	416	395	405
5		बर्निहाट BYRNHAT (400 kV)	0%	100%	0%	0%	0	418	400	411
6		कामेंग KAMENG (400 kV)	0%	100%	0%	0%	0	412	395	403
7		अज़ारा AZARA (400 kV)	0%	100%	0%	0%	0	399	389	395
8		मिसा MISA (400 kV)	0%	100%	0%	0%	0	413	391	403
9		न्यू मारियानी NEW MARIANI (400 kV)	0%	100%	0%	0%	0	417	397	405
10		न्यू कोहिमा NEW KOHIMA (400 kV)	0%	100%	0%	0%	0	411	380	394
11		पालाताना PALATANA (400 kV)	0%	100%	0%	0%	0	413	401	407
12		पुबेकचनबारी PK BARI (400 kV)	0%	100%	0%	0%	0	411	393	403
13		रंगानादी RANGANADI (400 kV)	0%	98%	2%	2%	12	424	401	412
14		सिलचर SILCHAR (400 kV)	0%	100%	0%	0%	0	415	397	407
15		सूर्ययामोनिनगर SURJYAMANINAGAR (400 kV)	0%	100%	0%	0%	0	411	393	402
16		थोबाल THOUBAL (400 kV)	0%	100%	0%	0%	0	412	390	398

All listed stations are 765 kV stations unless otherwise mentioned

*Percentage is calculated w.r.t. Time of one month.

24. ALL TIME HIGHEST

30-09-2023

	Maximum Demand Met during the day (MW)	Demand Met during Evening Peak hrs(MW)	Energy Met (MU)	Hydro Gen. (MU)	Wind Gen. (MU)	Solar Gen. (MU)
NR	81220 04-09-2023 @14:55	76274 04-09-2023	1796 04-09-2023	443 01-08-2023	86 07-08-2023	147 10-05-2023
WR	74035 01-09-2023 @14:29	64984 04-09-2023	1618 02-09-2023	167 18-12-2014	273 07-08-2023	70.69 12-05-2023
SR	64314 15-03-2023	52366 18-04-2023	1329 19-04-2023	208 31-08-2018	298 21-07-2023	142.7 21-03-2023
ER	29641 28-07-2023	28579 18-04-2023	639 08-06-2023	157 14-09-2022	-	7 23-03-2023
NER	3712 04-09-2023 @18:57	3510 12-08-2022	72 15-09-2023	41 29-08-2023	-	2 22-06-2022
All India	239978 01-09-2023 @12:22	215592 28-08-2023	5224 02-09-2023	877 30-08-2022	611 02-08-2023	351 10-05-2023

Regions	States	Max. Demand Met during the day (MW)		Energy Consumption (MU)	
			As on date		As on date
NR	Punjab	15237	23-06-2023	343.5	24-06-2023
	Haryana	12692	18-08-2023	273.1	18-08-2023
	Rajasthan	17772	02-09-2023 @14:00	371.6	04-09-2023
	Delhi	7770	29-06-2022	153.5	28-06-2022
	UP	28553	25-07-2023	580.0	03-09-2023
	Uttarakhand	2594	14-06-2022	55.8	17-06-2023
	HP	2071	06-01-2023	40.7	22-08-2023
	J&K(UT) and Ladakh(UT)	3636	29-06-2023	65.4	20-01-2023
	Chandigarh	426	08-07-2021	8.4	08-07-2021
	Railways NR ISTS	-	-	-	-
WR	Chhattisgarh	5832	02-09-2023 @15:00	133.5	28-08-2023
	Gujarat	24544	01-09-2023 @14:00	511.6	02-09-2023
	MP	17238	13-01-2023	329.3	10-01-2023
	Maharashtra	28846	14-04-2022	636.1	21-06-2023
	Goa	776	07-06-2023	17.1	08-06-2023
	DD & DNH	1327	17-08-2023	30.9	10-08-2023
	AMNSIL	977	18-08-2021	21.0	31-05-2022
	Balco	-	-	-	-
SR	Andhra Pradesh*	12900	17-06-2023	263.8	16-06-2023
	Telangana*	15497	30-03-2023	303.4	03-03-2023
	Karnataka	16950	25-08-2023	309.5	20-04-2023
	Kerala	5024	18-04-2023	103.3	19-04-2023
	Tamil Nadu	19387	20-04-2023	415.4	20-04-2023
	Pondy	512	17-05-2023	11.6	18-05-2023
ER	Bihar	7578	27-07-2023	161.3	28-07-2023
	DVC	3751	18-04-2023	81.2	22-04-2022
	Jharkhand	1923	16-06-2023	43.3	17-06-2023
	Odisha	7192	19-04-2023	145.5	25-09-2022
	West Bengal	11868	18-04-2023	245.7	18-04-2023
	Sikkim	137	03-02-2022	2.5	28-01-2020
	Railways ER ISTS	-	-	-	-
NER	Arunachal Pradesh	179	16-09-2023 @19:24	3.2	04-09-2023
	Assam	2498	16-09-2023 @21:13	50.0	15-09-2023
	Manipur	258	31-01-2022	3.7	10-02-2022
	Meghalaya	415	09-02-2022	7.8	31-01-2022
	Mizoram	157	16-01-2023	2.3	18-01-2023
	Nagaland	179	28-09-2023 @17:50	3.29	05-09-2023
	Tripura	370	07-06-2023	7.0	06-06-2023
*SR	Andhra Pradesh (Undivided)	13162	23-03-2014	284.8	22-03-2014

25.System Reliability Indices Report for the month of September 2023

Percentage (%) of times ATC was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	92	23.00	3.19
2	ER-NR	3	0.75	0.10
3	Import of NR	17	4.25	0.59
4	NEW-SR	2	0.50	0.07
5	NER Import	0	0.00	0.00

Percentage(%) of times (N-1) Criteria was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	35	8.75	1.22
2	ER-NR	4	1.00	0.14
3	Import of NR	4	1.00	0.14
4	NEW-SR	3	0.75	0.10
4	NER Import	0	0.00	0.00

Remarks: Flows crossing Total Transfer Capability (TTC) on interregional corridors has been worked out as a proxy for (N-1) violation.



POWER SYSTEM OPERATION CORPORATION LIMITED

पावर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड

NATIONAL LOAD DESPATCH CENTRE – राष्ट्रीय भार प्रेषण केन्द्र

First floor, B-9, Qutab Institutional Area,
Katwaria Sarai, New Delhi-110016.